

# GLP-1 Author Journey Extraction Pipeline

```
In [ ]: %pip install --upgrade pip
        %pip install pandas requests torch transformers accelerate tiktoken openai rouge-score bert-score matplotlib
```

Requirement already satisfied: pip in /opt/homebrew/lib/python3.11/site-packages (26.1.2)  
Note: you may need to restart the kernel to use updated packages.  
Requirement already satisfied: pandas in /opt/homebrew/lib/python3.11/site-packages (3.0.3)  
Requirement already satisfied: requests in /opt/homebrew/lib/python3.11/site-packages (2.34.2)  
Requirement already satisfied: torch in /opt/homebrew/lib/python3.11/site-packages (2.13.0)  
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Requirement already satisfied: urllib3<3,>=1.26 in /opt/homebrew/lib/python3.11/site-packages (from requests) (2.7.0)  
Requirement already satisfied: certifi>=2023.5.7 in /opt/homebrew/lib/python3.11/site-packages (from requests) (2023.5.20)  
Requirement already satisfied: filelock in /opt/homebrew/lib/python3.11/site-packages (from torch) (3.29.1)  
Requirement already satisfied: typing-extensions>=4.10.0 in /Users/telissatitley/Library/Python/3.11/lib/python/site-packages (from torch) (4.15.0)  
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Requirement already satisfied: fsspec>=0.8.5 in /opt/homebrew/lib/python3.11/site-packages (from torch) (2026.4.0)  
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /opt/homebrew/lib/python3.11/site-packages (from transformers) (1.18.0)  
Requirement already satisfied: packaging>=20.0 in /Users/telissatitley/Library/Python/3.11/lib/python/site-packages (from transformers) (26.2)  
Requirement already satisfied: pyyaml>=5.1 in /opt/homebrew/lib/python3.11/site-packages (from transformers) (6.0.3)  
Requirement already satisfied: regex>=2025.10.22 in /opt/homebrew/lib/python3.11/site-packages (from transformers) (2026.5.9)  
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /opt/homebrew/lib/python3.11/site-packages (from transformers) (0.22.2)  
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Requirement already satisfied: hf-xet<2.0.0, >=1.4.3 in /opt/homebrew/lib/python3.11/site-packages (from huggingface-hub<2.0, >=1.5.0->transformers) (1.5.0)

Requirement already satisfied: httpx<1, >=0.23.0 in /opt/homebrew/lib/python3.11/site-packages (from huggingface-hub<2.0, >=1.5.0->transformers) (0.28.1)

Requirement already satisfied: anyio in /opt/homebrew/lib/python3.11/site-packages (from httpx<1, >=0.23.0->huggingface-hub<2.0, >=1.5.0->transformers) (4.13.0)

Requirement already satisfied: httpcore==1.\* in /opt/homebrew/lib/python3.11/site-packages (from httpx<1, >=0.23.0->huggingface-hub<2.0, >=1.5.0->transformers) (1.0.9)

Requirement already satisfied: h11>=0.16 in /opt/homebrew/lib/python3.11/site-packages (from httpcore==1.\*->httpx<1, >=0.23.0->huggingface-hub<2.0, >=1.5.0->transformers) (0.16.0)

Requirement already satisfied: shellingham>=1.3.0 in /opt/homebrew/lib/python3.11/site-packages (from typer->transformers) (1.5.4)

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Requirement already satisfied: annotated-types>=0.6.0 in /opt/homebrew/lib/python3.11/site-packages (from pydantic<3, >=1.9.0->openai) (0.7.0)

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Requirement already satisfied: absl-py in /opt/homebrew/lib/python3.11/site-packages (from rouge-score) (2.5.0)

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Requirement already satisfied: pillow>=9 in /opt/homebrew/lib/python3.11/site-packages (from matplotlib) (12.2.0)  
Requirement already satisfied: pyparsing>=3 in /opt/homebrew/lib/python3.11/site-packages (from matplotlib) (3.3.2)  
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /Users/telissatitley/Library/Python/3.11/lib/python/site-packages (from rich>=13.8.0->typer->transformers) (2.20.0)  
Requirement already satisfied: mdurl~=0.1 in /opt/homebrew/lib/python3.11/site-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer->transformers) (0.1.2)  
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/homebrew/lib/python3.11/site-packages (from sympy>=1.13.3->torch) (1.3.0)  
Requirement already satisfied: MarkupSafe>=2.0 in /opt/homebrew/lib/python3.11/site-packages (from jinja2->torch) (3.0.3)  
Requirement already satisfied: defusedxml in /opt/homebrew/lib/python3.11/site-packages (from nltk->rouge-score) (0.7.1)  
Requirement already satisfied: joblib in /opt/homebrew/lib/python3.11/site-packages (from nltk->rouge-score) (1.5.3)  
Note: you may need to restart the kernel to use updated packages.

```
In [ ]: #####  
# Imports #  
#####  
  
# Standard library  
import os  
import re  
import sys  
import glob  
import json  
import time  
import getpass  
import hashlib  
from pathlib import Path  
from collections import defaultdict  
  
# Data / plotting  
import pandas as pd  
import matplotlib.pyplot as plt  
import matplotlib.image as mpimg
```

```

from matplotlib.gridspec import GridSpec
from matplotlib.colors import TwoSlopeNorm
from matplotlib.cm import ScalarMappable
from matplotlib.lines import Line2D
from scipy.stats import fisher_exact, mannwhitneyu
from statsmodels.stats.multitest import multipletests

# Modelling / tokenisation (kept from the original environment)
import requests
import torch
import accelerate
import transformers
from transformers import pipeline
import tiktoken

# OpenAI client
from openai import OpenAI

# Evaluation metrics
import rouge_score
import bert_score
from rouge_score import rouge_scorer
from bert_score import score as bert_score

```

/opt/homebrew/lib/python3.11/site-packages/tqdm/auto.py:21: TqdmWarning: IPProgress not found. Please update jupyter and ipywidgets. See [https://ipywidgets.readthedocs.io/en/stable/user\\_install.html](https://ipywidgets.readthedocs.io/en/stable/user_install.html)

```
from .autonotebook import tqdm as notebook_tqdm
```

```

In [ ]: #####
# Defining paths and constants #
#####

base_dir = "/Users/telissatitley/llm_env/GLP1_Thesis"
subreddits_root = f"{base_dir}/Subreddits"

subsample_dir = f"{subreddits_root}/Subsamples"
test_output_dir = f"{subreddits_root}/Test"
gpt_result_dir = f"{subreddits_root}/GPT_Results"

for d in [subsample_dir, test_output_dir, gpt_result_dir]:
    os.makedirs(d, exist_ok=True)

```

```
# Manually annotated files
few_shot_annotated_path = f"{test_output_dir}/few_shot_author_annotated_examples.json"
gold_standard_annotated_path = f"{test_output_dir}/gold_standard_evaluation_annotated.json"

# API key request line
if not os.environ.get("OPENAI_API_KEY"):
    os.environ["OPENAI_API_KEY"] = getpass.getpass("Enter OpenAI API key: ")
```

# 1. Load raw subreddit json files

## Semaglutide

```
In [6]: sema_oz_c = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/r_Ozempic_comments.jsonl')
sema_oz_p = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/r_Ozempic_posts.jsonl', l
sema_weg_c = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/r_Wegovy_comments.jsonl')
sema_weg_p = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/r_Wegovy_posts.jsonl', l
sema_gen_c = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/r_Semaglutide_comments.j
sema_gen_p = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/r_Semaglutide_posts.json
```

## Retatrutide

```
In [7]: reta_c = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Retatrutide/r_Retatrutide_comments.jsonl')
reta_p = pd.read_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Retatrutide/r_Retatrutide_posts.jsonl', l
```

# 2. Clean and standardise

```
In [8]: #Check all posts and comments alignment across each indication.

keep_cols = ['author', 'archived', 'body', 'created_utc', 'id'] #defining relevant information following file inspe

def align_posts(df):
    df = df.rename(columns={'selftext': 'body'}) #updating posts 'selftext' to 'body', for standardised naming conv
```

```

        return df[keep_cols]

def align_comments(df):
    return df[keep_cols]

#Aligning relevant tags from posts and comments section for each drug indication.
# Semaglutide
# Ozempic
sema_oz_p = align_posts(sema_oz_p)
sema_oz_c = align_comments(sema_oz_c)

# Wegovy
sema_weg_p = align_posts(sema_weg_p)
sema_weg_c = align_comments(sema_weg_c)

# Generic
sema_gen_p = align_posts(sema_gen_p)
sema_gen_c = align_comments(sema_gen_c)

# Retatrutide
reta_p = align_posts(reta_p)
reta_c = align_comments(reta_c)

#Sense checking only relevant fields remain.
# Retatrutide
print(sema_oz_p.columns.tolist())
print(sema_oz_c.columns.tolist())

```

```

['author', 'archived', 'body', 'created_utc', 'id']
['author', 'archived', 'body', 'created_utc', 'id']

```

```

In [9]: sema_oz_p["source"] = "post"
        sema_oz_c["source"] = "comment"

        sema_weg_p["source"] = "post"
        sema_weg_c["source"] = "comment"

        sema_gen_p["source"] = "post"
        sema_gen_c["source"] = "comment"

```

```
reta_p["source"] = "post"
reta_c["source"] = "comment"
```

In [10]: *#Removing automoderator, deleted and removed authors.*

```
def clean_data(df):
    df['body'] = df['body'].astype(str)

    return df[
        (df['author'].notna()) &
        (~df['author'].isin(['[deleted]', '[removed]', 'AutoModerator'])) &
        (~df['body'].isin(['[removed]', '[deleted]'])) &
        (df['body'].str.strip() != '')
    ]

# Ozempic
sema_oz_c = clean_data(sema_oz_c)
sema_oz_p = clean_data(sema_oz_p)

# Wegovy
sema_weg_c = clean_data(sema_weg_c)
sema_weg_p = clean_data(sema_weg_p)

# Generic Semaglutide
sema_gen_c = clean_data(sema_gen_c)
sema_gen_p = clean_data(sema_gen_p)

# Retatrutide
reta_c = clean_data(reta_c)
reta_p = clean_data(reta_p)
```

In [11]: *## Merging posts and comments for ozempic, wegovy, generic semaglutide and now that only relevant columns remain an*

```
# Ozempic
sema_oz = pd.concat([sema_oz_p, sema_oz_c], ignore_index=True)
# Wegovy
sema_weg = pd.concat([sema_weg_p, sema_weg_c], ignore_index=True)
# Generic semaglutide
sema_gen = pd.concat([sema_gen_p, sema_gen_c], ignore_index=True)
```



```
# Retatrutide
reta = pd.concat([reta_p, reta_c], ignore_index=True)

sema_oz["drug"] = "ozempic"
sema_weg["drug"] = "wegovy"
sema_gen["drug"] = "semaglutide"
reta["drug"] = "retatrutide"
```

In [12]: *#Checking for archived users.*

```
print(sema_oz['archived'].value_counts(dropna=False))
print(sema_weg['archived'].value_counts(dropna=False))
print(sema_gen['archived'].value_counts(dropna=False))
print(reta['archived'].value_counts(dropna=False))
```

```
archived
False    943779
True       1706
Name: count, dtype: int64
archived
False    34697
Name: count, dtype: int64
archived
False    973128
Name: count, dtype: int64
archived
False    803706
Name: count, dtype: int64
```

In [13]: *#Sense checking for any empty posts/comments in body*

```
sema_oz['body'].isna().sum()
#Output: np.int64(0)
sema_weg['body'].isna().sum()
#Output: np.int64(0)
sema_gen['body'].isna().sum()
#Output: np.int64(0)
reta['body'].isna().sum()
#Output: np.int64(0)

#Removing duplicate bodies of texts
```

```

sema_oz = sema_oz.drop_duplicates(subset=["body"])
sema_weg = sema_weg.drop_duplicates(subset=["body"])
sema_gen = sema_gen.drop_duplicates(subset=["body"])
reta = reta.drop_duplicates(subset=["body"])

print("oz:", len(sema_oz))
print("weg:", len(sema_weg))
print("gen:", len(sema_gen))
print("reta:", len(reta))

```

```

oz: 898471
weg: 34174
gen: 923001
reta: 764833

```

```

In [14]: # Ozempic
print("Ozempic:", sema_oz.shape)
#Output: Ozempic: (898471, 6)

# Wegovy
print("Wegovy:", sema_weg.shape)
#Output: Wegovy: (34174, 6)

# Generic Semaglutide
print("Generic Semaglutide:", sema_gen.shape)
#Output: Generic Semaglutide: (923001, 6)

# Retatrutide
print("Retatrutide:", reta.shape)
#Output: Retatrutide: (764833, 6)

```

```

Ozempic: (898471, 7)
Wegovy: (34174, 7)
Generic Semaglutide: (923001, 7)
Retatrutide: (764833, 7)

```

```

In [ ]: sema_oz.head()

```

```

In [16]: #Handling NaNs to prevent dropping relevant columns.
sema_oz = sema_oz[sema_oz['archived'].fillna(False) == False]

```

```
sema_weg = sema_weg[sema_weg['archived'].fillna(False) == False]
sema_gen = sema_gen[sema_gen['archived'].fillna(False) == False]
reta = reta[reta['archived'].fillna(False) == False]
```

```
In [17]: print("oz:", len(sema_oz))
print("weg:", len(sema_weg))
print("gen:", len(sema_gen))
print("reta:", len(reta))
```

```
oz: 896774
weg: 34174
gen: 923001
reta: 764833
```

```
In [ ]: #Dropping archived column completely.
```

```
sema_oz = sema_oz.drop(columns=['archived'])
sema_weg = sema_weg.drop(columns=['archived'])
sema_gen = sema_gen.drop(columns=['archived'])
reta = reta.drop(columns=['archived'])
```

```
#Sense checking each indication.
```

```
print(sema_oz.columns)
print(sema_weg.columns)
print(sema_gen.columns)
print(reta.columns)
```

```
#Output: Index(['author', 'body', 'created_utc', 'id', 'source', 'drug'], dtype='str')
```

```
Index(['author', 'body', 'created_utc', 'id', 'source', 'drug'], dtype='str')
Index(['author', 'body', 'created_utc', 'id', 'source', 'drug'], dtype='str')
Index(['author', 'body', 'created_utc', 'id', 'source', 'drug'], dtype='str')
Index(['author', 'body', 'created_utc', 'id', 'source', 'drug'], dtype='str')
```

## Convert UTC timestamps to datetime

```
In [19]: #Convert UTC to standard date format
```

```
sema_oz['created_utc'] = pd.to_datetime(sema_oz['created_utc'], unit='s')
```

```

sema_weg['created_utc'] = pd.to_datetime(sema_weg['created_utc'], unit='s')
sema_gen['created_utc'] = pd.to_datetime(sema_gen['created_utc'], unit='s')
reta['created_utc'] = pd.to_datetime(reta['created_utc'], unit='s')

#Sense-checking change complete
sema_oz[['created_utc']].head()

```

Out [19]:

|   | created_utc         |
|---|---------------------|
| 0 | 2019-12-04 20:01:54 |
| 1 | 2021-05-09 03:51:59 |
| 2 | 2021-05-09 13:41:26 |
| 3 | 2021-05-11 22:29:54 |
| 4 | 2021-05-12 12:49:08 |

### 3. Filtering for longitudinal unique authors

```

In [ ]: output_dir = "/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Subsamples"
os.makedirs(output_dir, exist_ok=True)

start_date = pd.Timestamp("2019-12-04")
end_date = pd.Timestamp("2026-06-10 22:50:09")

def make_longitudinal_subsample(df, name):

    # 1. Filter date range
    sub_df = df.loc[df["created_utc"].between(start_date, end_date)].copy()

    total_rows = len(sub_df)
    total_authors = sub_df["author"].nunique()

    # 2. Keep only repeat authors
    repeat_authors = (sub_df["author"].value_counts().loc[lambda x: x > 1].index)

    sub_df = sub_df[sub_df["author"].isin(repeat_authors)].copy()

```

```

repeat_author_count = len(repeat_authors)

# 3. Sort chronologically within author
sub_df = sub_df.sort_values(["author", "created_utc"])
grouped = sub_df.groupby("author")

# 4. Add longitudinal metadata
sub_df["n_author_posts"] = (sub_df.groupby("author")["author"].transform("count"))
sub_df["journey_step"] = (sub_df.groupby("author").cumcount() + 1)
sub_df["first_post_date"] = grouped["created_utc"].transform("min")
sub_df["days_since_first_post"] = (sub_df["created_utc"] - sub_df["first_post_date"]).dt.days

retained_rows = len(sub_df)

# Save
base_path = f"{output_dir}/sub_{name}_longitudinal"

sub_df.to_json(f"{base_path}.jsonl", orient="records", lines=True, date_format="iso")
sub_df.to_csv(f"{base_path}.csv", index=False)

# Summary
print(f"\n{name.upper()}")
print(f"Total rows: {total_rows:,}")
print(f"Unique authors: {total_authors:,}")
print(f"Repeat authors: {repeat_author_count:,}")
print(f"Retained rows: {retained_rows:,}")

return sub_df

```

```

In [21]: sub_sem_oz = make_longitudinal_subsample(sema_oz, "oz")
sub_sem_weg = make_longitudinal_subsample(sema_weg, "weg")
sub_sem_gen = make_longitudinal_subsample(sema_gen, "gen")
sub_reta = make_longitudinal_subsample(reta, "reta")

```

OZ

Total rows: 896,774

Unique authors: 98,544

Repeat authors: 58,498

Retained rows: 856,728

WEG

Total rows: 34,174

Unique authors: 8,277

Repeat authors: 4,479

Retained rows: 30,376

GEN

Total rows: 923,001

Unique authors: 114,308

Repeat authors: 67,054

Retained rows: 875,747

RETA

Total rows: 764,833

Unique authors: 67,483

Repeat authors: 41,643

Retained rows: 738,993

In [22]: *#Saving for reproducibility*

```
sema_oz.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/Clean/sema_oz.jsonl',orient='records')
sema_weg.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/Clean/sema_weg.jsonl',orient='records')
sema_gen.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/Clean/sema_gen.jsonl',orient='records')
reta.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Retatrutide/Clean/reta.jsonl',orient='records', l
```

```

/var/folders/5l/fhws2zt56lg1ns3d7llkfmkw0000gn/T/ipykernel_32450/3317793212.py:3: Pandas4Warning: The default 'epoch' date format is deprecated and will be removed in a future version, please use 'iso' date format instead.
    sema_oz.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/Clean/sema_oz.jsonl',orient='records', lines=True)
/var/folders/5l/fhws2zt56lg1ns3d7llkfmkw0000gn/T/ipykernel_32450/3317793212.py:4: Pandas4Warning: The default 'epoch' date format is deprecated and will be removed in a future version, please use 'iso' date format instead.
    sema_weg.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/Clean/sema_weg.jsonl',orient='records', lines=True)
/var/folders/5l/fhws2zt56lg1ns3d7llkfmkw0000gn/T/ipykernel_32450/3317793212.py:5: Pandas4Warning: The default 'epoch' date format is deprecated and will be removed in a future version, please use 'iso' date format instead.
    sema_gen.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Semaglutide/Clean/sema_gen.jsonl',orient='records', lines=True)
/var/folders/5l/fhws2zt56lg1ns3d7llkfmkw0000gn/T/ipykernel_32450/3317793212.py:6: Pandas4Warning: The default 'epoch' date format is deprecated and will be removed in a future version, please use 'iso' date format instead.
    reta.to_json('/Users/telissatitley/llm_env/GLP1_Thesis/Subreddits/Retatrutide/Clean/reta.jsonl',orient='records', lines=True)

```

## 4. Keywords and terms search: lexical screening

```

In [ ]: seed_terms = {"journey": ["week", "month", "day", "days", "year", "started", "starting", "since", "after", "before",
    "weight": ["lost", "down", "gained", "lbs", "kg", "weight", "increased", "improved", "decreased", "change", "fluctu
    "dose": ["dose", "dosage", "mg", "0.25", "0.5", "1mg", "2.4mg", "upped dose", "maintenance dose", "starter dose", "
    "experience": ["side effect", "nausea", "appetite", "food noise", "binge", "vomiting", "diarrhea", "constipation",
    "height": ["ft", "cm", "inches", "centimetres"],
    "age": ["years old", "age", "yo", "y/o"],
    "sex": ["female", "male", "woman", "man", "cis", "sex", "gender"],
    "exercise": ["exercise", "workout", "gym", "training", "sedentary", "active", "activity", "walking", "running", "ca
    }

#Removing seed search sensitivity
seed_terms_lower = {
    category: [term.lower() for term in terms]
    for category, terms in seed_terms.items()
}

def flag_terms(text, terms):
    text = str(text).lower()
    return any(term in text for term in terms)

```

```

def add_seed_flags(df, seed_terms_lower):
    df = df.copy()

    # Create category flags
    for category, terms in seed_terms_lower.items():
        df[category] = df["body"].apply(lambda x: flag_terms(x, terms))

    # Calculate relevance score
    df["relevance_score"] = df[list(seed_terms_lower.keys())].sum(axis=1)

    # Apply relevance threshold
    df["relevance"] = df["relevance_score"] >= 6

    return df

datasets = {
    "oz": sub_sem_oz,
    "weg": sub_sem_weg,
    "sema": sub_sem_gen,
    "reta": sub_reta
}

# Apply seed term flags to each dataset
for name, df in datasets.items():
    datasets[name] = add_seed_flags(df, seed_terms_lower)

# Summary output
for name, df in datasets.items():

    print(f"\n{'=' * 50}")
    print(f"{name.upper()}")
    print(f"{'=' * 50}")

    print("\nCounts")
    print(df[list(seed_terms_lower.keys())].sum())

    print("\nProportions")
    print(df[list(seed_terms_lower.keys())].mean())

```



```
print(  
    f"\nRelevant Posts: "  
    f"{df['relevance'].sum():,} / {len(df):,} "  
    f"({df['relevance'].mean():.1%})"  
)
```

=====

0Z

=====

Counts

|            |        |
|------------|--------|
| journey    | 372424 |
| weight     | 264878 |
| dose       | 148128 |
| experience | 141397 |
| height     | 91815  |
| age        | 426112 |
| sex        | 84724  |
| exercise   | 51342  |

dtype: int64

Proportions

|            |          |
|------------|----------|
| journey    | 0.434705 |
| weight     | 0.309174 |
| dose       | 0.172900 |
| experience | 0.165043 |
| height     | 0.107169 |
| age        | 0.497371 |
| sex        | 0.098893 |
| exercise   | 0.059928 |

dtype: float64

Relevant Posts: 21,958 / 856,728 (2.6%)

=====

WEG

=====

Counts

|            |       |
|------------|-------|
| journey    | 15955 |
| weight     | 10281 |
| dose       | 7288  |
| experience | 7390  |
| height     | 4083  |
| age        | 15278 |
| sex        | 2874  |
| exercise   | 1732  |

dtype: int64

Proportions

|            |          |
|------------|----------|
| journey    | 0.525250 |
| weight     | 0.338458 |
| dose       | 0.239926 |
| experience | 0.243284 |
| height     | 0.134415 |
| age        | 0.502963 |
| sex        | 0.094614 |
| exercise   | 0.057019 |

dtype: float64

Relevant Posts: 1,143 / 30,376 (3.8%)

=====

SEMA

=====

Counts

|            |        |
|------------|--------|
| journey    | 391296 |
| weight     | 277282 |
| dose       | 130477 |
| experience | 139072 |
| height     | 93497  |
| age        | 433538 |
| sex        | 77949  |
| exercise   | 55069  |

dtype: int64

Proportions

|            |          |
|------------|----------|
| journey    | 0.446814 |
| weight     | 0.316623 |
| dose       | 0.148989 |
| experience | 0.158804 |
| height     | 0.106763 |
| age        | 0.495049 |
| sex        | 0.089009 |
| exercise   | 0.062882 |

dtype: float64

Relevant Posts: 19,957 / 875,747 (2.3%)

```
=====
RETA
=====
```

```
Counts
journey      265962
weight       162428
dose         172547
experience    86179
height       62870
age          350079
sex          56179
exercise     49101
dtype: int64
```

```
Proportions
journey      0.359898
weight       0.219796
dose         0.233489
experience    0.116617
height       0.085075
age          0.473724
sex          0.076021
exercise     0.066443
dtype: float64
```

Relevant Posts: 15,991 / 738,993 (2.2%)

## 5. Combined longitudinal pool, anonymised author IDs, and candidate authors

```
In [ ]: # Combine all longitudinal drug-group datasets
longitudinal_pool = pd.concat([df.assign(drug_group=drug_group)
                               for drug_group, df in datasets.items()],
                              ignore_index=True)
```

```

# Ensure the date column is datetime
longitudinal_pool["created_utc"] = pd.to_datetime(
    longitudinal_pool["created_utc"],
    errors="coerce"
)

# Sort posts chronologically within each author and drug group
longitudinal_pool = (
    longitudinal_pool
    .sort_values(
        [
            "drug_group",
            "author",
            "created_utc"
        ]
    ).reset_index(drop=True))

print(f"Combined longitudinal posts: " f"{len(longitudinal_pool):,}")

```

Combined longitudinal posts: 2,501,844

```

In [ ]: def create_author_id(drug_group, author):
        """
        Anonymising author IDs using SHA256.
        """
        author_key = f"{drug_group}|{author}"
        author_hash = hashlib.sha256(author_key.encode("utf-8")).hexdigest()[:12]
        return f"{drug_group}_{author_hash}"

private_key_path = f"{test_output_dir}/private_author_key.csv"

# Checking for old author_id exists
longitudinal_pool = longitudinal_pool.drop(columns=["author_id"], errors="ignore")

if os.path.exists(private_key_path):
    # ---- Existing key found: LOAD it, do not regenerate ----
    private_author_key = pd.read_csv(private_key_path)

    longitudinal_pool = longitudinal_pool.merge(
        private_author_key,

```

```

        on=["drug_group", "author"],
        how="left",
    )

    # Only hash authors that are genuinely new (absent from the saved key)
    missing = longitudinal_pool["author_id"].isna()
    if missing.any():
        longitudinal_pool.loc[missing, "author_id"] = (
            longitudinal_pool.loc[missing]
            .apply(lambda r: create_author_id(r["drug_group"], r["author"]), axis=1)
        )
        private_author_key = (
            longitudinal_pool[["drug_group", "author", "author_id"]]
            .drop_duplicates()
        )
        private_author_key.to_csv(private_key_path, index=False)
        print(f"Loaded existing key and added {int(missing.sum())} new author ids")
    else:
        print(f"Loaded existing author key ({len(private_author_key):}, authors) "
              "- no new ids generated")

else:
    # ---- Generate and save the key ----
    print("No private_author_key.csv found - generating author ids for the first time")
    longitudinal_pool["author_id"] = longitudinal_pool.apply(
        lambda r: create_author_id(r["drug_group"], r["author"]), axis=1
    )
    private_author_key = (
        longitudinal_pool[["drug_group", "author", "author_id"]]
        .drop_duplicates()
    )
    private_author_key.to_csv(private_key_path, index=False)
    print(f"Saved new author key: {private_author_key['author_id'].nunique():}, authors")

longitudinal_pool = longitudinal_pool.sort_values(
    ["drug_group", "author_id", "created_utc"]
).copy()

longitudinal_pool["post_number"] = (
    longitudinal_pool.groupby(["drug_group", "author_id"]).cumcount().add(1)
)

```

```
print(f"Unique author journeys: {longitudinal_pool['author_id'].nunique():,}")
```

Loaded existing author key (171,674 authors) – no new ids generated

Unique author journeys: 171,674

```
In [ ]: author_summary = (
    longitudinal_pool
    .groupby(["drug_group", "author_id"], as_index=False)
    .agg(n_posts=("id", "size"),
         first_post_date=("created_utc", "min"),
         last_post_date=("created_utc", "max"),
         has_candidate_post=("relevance", "any"),
         max_relevance_score=( "relevance_score", "max")
    )

    author_summary["days_between_first_and_last_post"] =
    (author_summary["last_post_date"] - author_summary["first_post_date"]).dt.days

    candidate_authors =
    author_summary[(author_summary["n_posts"] >= 2) &
                    (author_summary["days_between_first_and_last_post"] >= 14) &
                    (author_summary["has_candidate_post"])]
                    .copy()

    print(candidate_authors.groupby("drug_group")
          .agg(candidate_authors=("author_id", "nunique"),

               total_posts=("n_posts", "sum"),
               median_posts=("n_posts", "median"),
               maximum_posts=("n_posts", "max")
          )
    )
```

|            | candidate_authors | total_posts | median_posts | maximum_posts |
|------------|-------------------|-------------|--------------|---------------|
| drug_group |                   |             |              |               |
| oz         | 8530              | 432078      | 18.0         | 8794          |
| reta       | 5878              | 390628      | 24.0         | 6296          |
| sema       | 9079              | 391685      | 17.0         | 5192          |
| weg        | 527               | 9723        | 10.0         | 242           |

## Journey text builder

```
In [ ]: def build_annotation_input(author_posts):  
        """  
        Convert one author journey into prompt compatible text.  
        """  
        author_posts = author_posts.sort_values("created_utc")  
  
        sections = []  
        for row in author_posts.itertuples():  
            sections.append(  
                f"""POST {row.post_number}  
  
DATE:  
{row.created_utc}  
  
TEXT:  
{row.body}  
"""  
            )  
  
        return "\n".join(sections)
```

## 6. Load fixed few-shot and gold-standard annotation sets

```
In [28]: # Load manually annotated few-shot examples (fixed – identical on every run)  
  
assert os.path.exists(few_shot_annotated_path), (  
    f"Missing annotated few-shot file: {few_shot_annotated_path}"  
)  
  
with open(few_shot_annotated_path, "r", encoding="utf-8") as f:  
    few_shot_examples = json.load(f)  
  
print(f"Loaded {len(few_shot_examples)} annotated few-shot author journeys")
```



Loaded 8 annotated few-shot author journeys

```
In [29]: # Load manually annotated gold-standard evaluation set (fixed - identical on every run)
```

```
assert os.path.exists(gold_standard_annotated_path), (  
    f"Missing annotated gold-standard file: {gold_standard_annotated_path}"  
)  
  
with open(gold_standard_annotated_path, "r", encoding="utf-8") as f:  
    gold_standard_examples = json.load(f)  
  
print(f"Loaded {len(gold_standard_examples)} annotated gold-standard journeys")
```

Loaded 8 annotated gold-standard journeys

```
In [30]: # Derive the author splits directly from the annotated files  
# (instead of re-sampling with random_state, which regenerated new sets whenever  
# upstream data changed)
```

```
few_shot_author_ids = {ex["author_id"] for ex in few_shot_examples}  
  
evaluation_authors = pd.DataFrame(  
    [  
        {"drug_group": ex["drug_group"], "author_id": ex["author_id"]}  
        for ex in gold_standard_examples  
    ]  
)  
.drop_duplicates()  
  
print(f"Few-shot authors: {len(few_shot_author_ids)}")  
print(f"Evaluation authors: {len(evaluation_authors)}")
```

Few-shot authors: 8

Evaluation authors: 8

```
In [ ]: # Remove few-shot authors from the candidate author pool
```

```
remaining_candidate_authors = candidate_authors[  
    ~candidate_authors["author_id"].isin(  
        few_shot_author_ids  
    )  
].copy()
```

```

print(
    f"Candidate authors before exclusion: "
    f"{len(candidate_authors):,}"
)

print(
    f"Candidate authors after exclusion: "
    f"{len(remaining_candidate_authors):,}"
)

print(
    f"Few-shot authors removed: "
    f"{len(candidate_authors) - len(remaining_candidate_authors):,}"
)

```

Candidate authors before exclusion: 24,014  
 Candidate authors after exclusion: 24,006  
 Few-shot authors removed: 8

```

In [ ]: evaluation_author_keys = (
    evaluation_authors[["drug_group", "author_id"]]
    .drop_duplicates()
    .assign(evaluation_author=True)
)

extraction_candidate_authors = (
    remaining_candidate_authors
    .merge(evaluation_author_keys, on=["drug_group", "author_id"], how="left")
)

extraction_candidate_authors = (
    extraction_candidate_authors[
        extraction_candidate_authors
        ["evaluation_author"].isna()
    ].drop(columns="evaluation_author").copy()
)

print(
    f"Authors after few-shot exclusion: "
    f"{len(remaining_candidate_authors):,}"
)

```

```

print(
    f"Evaluation authors excluded: "
    f"{len(evaluation_authors):,}"
)

print(
    f"Authors remaining for final extraction: "
    f"{len(extraction_candidate_authors):,}"
)

```

Authors after few-shot exclusion: 24,006

Evaluation authors excluded: 8

Authors remaining for final extraction: 23,998

```

In [ ]: few_shot_keys = {
    (example["drug_group"], example["author_id"])
    for example in few_shot_examples
}

evaluation_keys = set(
    evaluation_authors[["drug_group", "author_id"]].itertuples(
        index=False, name=None
    )
)

extraction_keys = set(
    extraction_candidate_authors[
        ["drug_group", "author_id"]].drop_duplicates().itertuples(
            index=False, name=None
        )
)

assert not (few_shot_keys & evaluation_keys), ("Few-shot and evaluation sets overlap.")
assert not (few_shot_keys & extraction_keys), ("Few-shot and extraction sets overlap.")
assert not (evaluation_keys & extraction_keys), ("Evaluation and extraction sets overlap.")

print(
    "Confirmed: few-shot, evaluation and "
    "final extraction authors are independent."
)

```

Confirmed: few-shot, evaluation and final extraction authors are independent.

## 7. Prompt construction

Schema, consistency rules, inclusion rules, task prompt, and the few-shot block built from the **manually annotated** examples.

```
In [165... extraction_schema = """
{
  "author_id": null,
  "drug_group": null,
  "route_of_administration": null,

  "include": null,
  "exclusion_reason": null,

  "longitudinal_criteria": {
    "n_posts": null,
    "days_between_first_and_last_post": null,
    "meets_minimum_requirement": null
  },

  "treatment_history": {
    "treatment_start_date": null,
    "treatment_start_source": null,
    "reported_treatment_duration": null
  },

  "drug_switching": {
    "previous_glp1": [],
    "current_glp1": [],
    "reason_for_switch": []
  },

  "patient_profile": {
    "age_years": null,
    "sex_gender": null,
    "race_ethnicity": null,
    "height_cm": null,
```

```
    "baseline_conditions": [
      {
        "specific_label": null,
        "visualisation_group": null
      }
    ],
  },

  "baseline_metrics": {
    "starting_weight_kg": null,
    "starting_weight_lb": null,
    "starting_bmi": null,
    "goal_weight_kg": null,
    "goal_weight_lb": null
  },

  "timeline_events": [
    {
      "post_number": null,
      "date": null,

      "treatment_event": {
        "specific_label": null,
        "visualisation_group": null
      },

      "drug": null,
      "dose_mg": null,
      "weight_kg": null,
      "weight_lb": null,
      "bmi": null,

      "desired_effects": [
        {
          "specific_label": null,
          "visualisation_group": null
        }
      ],
    },

    "adverse_effects": [
      {
```

```

        "specific_label": null,
        "visualisation_group": null,
        "severity": null
    }
],

    "confounding_variables": [
        {
            "specific_label": null,
            "visualisation_group": null
        }
    ],

    "adherence_or_access_issues": [
        {
            "specific_label": null,
            "visualisation_group": null
        }
    ],

    "explicit_notes": [],
    "inferred_notes": []
}
],

    "journey_summary": null
}
"""

```

In [166... `schema_consistency_rules = """`

Schema consistency rules:

- Always return every top-level field in the extraction schema.
- Use null for unavailable scalar values.
- Use [] for unavailable list values.
- Never place empty placeholder objects inside arrays.
- `previous_glp1`, `current_glp1` and `reason_for_switch` must always be arrays of strings.
- Use an empty array when no GLP-1 switch occurred.
- `treatment_event` must be one object, not an array.
- `baseline_conditions`, `desired_effects`, `adverse_effects`, `confounding_variables` and `adherence_or_access_issues` must always be arrays.

```
- severity must be a string: mild, moderate or severe.
- route_of_administration must be injection, oral, other or null.
- Dates must use YYYY-MM-DD format.
"""
```

```
In [167... longitudinal_inclusion_rules = """
FIRST-PERSON RULE (applies to every post and every extracted item):
- Extract ONLY the author's OWN GLP-1 treatment experience.
- IGNORE any content where the author is advising, replying to, or describing
  SOMEONE ELSE - e.g. "you should try...", "my wife is on...", "for most people
  nausea fades". Do not create timeline_events, effects, doses, or weights from
  such posts, even when they mention GLP-1 terms.
- A dose, effect, or weight counts ONLY if the author states it about themselves
  (first person: "I", "my", "me"). If it is unclear whose experience it is, do
  NOT extract it.

INCLUSION (whether the journey as a whole is usable):
- include = true only if, using FIRST-PERSON content only, the journey has:
  * at least 2 of the author's own posts spanning at least 14 days, AND
  * at least one first-person post with treatment-relevant content
    (dose, weight, effect, adherence, or timeline information).
- include = false if the author never describes their own GLP-1 use (e.g. they
  only give advice, ask questions, or discuss others), or the first-person
  content is too sparse to meet the thresholds above.

When include = false:
- set exclusion_reason to a short explanation.
- If excluded because the content is NOT the author's own experience
  (advice-only, about others, questions-only): extract NOTHING - set effects,
  events and metrics to empty. Do not record other people's effects.
- If excluded only because the author's own journey is too short/sparse:
  you MAY still extract their own desired/adverse effects where clearly present.
When include = true: set exclusion_reason = null.
"""
```

```
In [ ]: # Visualisation grouping instructions.
# The vocabulary is derived from the FEW-SHOT annotations only (not the gold set),
# so it never leaks gold category names into the gold-standard evaluation.
# The model is prompted to prefer these groups, coin a consistent new one when none
# fits, and only use "other" only as a last resort.
```

```

effect_insights = ["desired_effects", "adverse_effects", "confounding_variables",
                  "adherence_or_access_issues"]
semantic_categories = {
    "dose_intolerability": "dose intolerance", "intolerable dosage": "dose intolerance",
    "dysomnia": "sleep disturbance", "dietary improvements": "dietary improvement",
    "cholesterol": "high cholesterol", "delayed scheduled dose": "delayed dose",
    "delayed dose escalation": "delayed dose",
}
undefined = {"not stated", "side effects", ""}

def build_visualisation_grouping(examples):
    groups = defaultdict(set)

    def add(field, vg):
        if isinstance(vg, str):
            v = vg.strip().lower()
            if v in undefined:
                return
            groups[field].add(semantic_categories.get(v, v))

    for ex in examples:
        eo = ex["expected_output"]
        for c in eo.get("patient_profile", {}).get("baseline_conditions") or []:
            if isinstance(c, dict):
                add("baseline_conditions", c.get("visualisation_group"))
        for te in eo.get("timeline_events") or []:
            tev = te.get("treatment_event")
            if isinstance(tev, dict):
                add("treatment_event", tev.get("visualisation_group"))
        for fld in effect_insights:
            #fields: desired_effects, adverse_effects, confounding_vari
            for it in te.get(fld) or []:
                if isinstance(it, dict):
                    add(fld, it.get("visualisation_group"))

    order = ["desired_effects", "adverse_effects", "confounding_variables",
            "adherence_or_access_issues", "treatment_event", "baseline_conditions"]
    vocab_block = "\n".join(
        f"{cat}:\n " + "; ".join(sorted(groups[cat]))
        for cat in order if groups[cat]
    )

```



```

)

section = """VISUALISATION GROUPING
Every desired effect, adverse effect, confounder, treatment_event, adherence/access
issue and baseline condition must be returned as a SEPARATE item with BOTH:
- specific_label: the exact issue as reported, concise and lower-case. Keep one
  distinct issue per item – never merge two issues (e.g. reduced food noise AND
  reduced cravings) into one item or one label.
- visualisation_group: a broad, chart-ready category, lower-case.

How to choose visualisation_group:
1. PREFER the vocabulary below. If an existing group reasonably fits the concept,
  use it verbatim – even when the wording in the post differs. Match on meaning,
  not exact words.
2. If NO existing group fits, CREATE a new group, following the naming rules below,
  so the same concept gets the same label in any other journey. Do not fall back
  to "other" just to avoid naming something.
3. Use "other" ONLY as a last resort – when no clinically or pharmacovigilance-
  meaningful label can be applied consistently across journeys (e.g. vague,
  off-topic or uninterpretable content). Prefer a real category whenever one exists.

Naming rules for a NEW group (so identical concepts get identical strings across
independent journeys):
- lower-case; a short noun phrase naming the concept, not a sentence.
- keep all specific detail (symptom, dose, number, body part) in specific_label,
  NOT in the group name. The group is the reusable bucket.
- do NOT encode severity, dose, frequency, dates or quantifiers in the group name.
- follow the directional patterns already in use for effects:
  "reduced X" / "improved X" / "increased X" / "worsened X".
- for a medical condition or comorbidity, use the recognised clinical term,
  singular (e.g. "hypothyroidism", "gallbladder disease").
- self-administered supplements -> "supplement use"; prescribed medications ->
  "co-medication". Never combine the two.
- reuse a group you have already created earlier in THIS journey for the same
  concept, so a journey never contains two spellings of one category.

PREFERRED VOCABULARY (by field):
{vocab}"""

return section.replace("{vocab}", vocab_block)

```

```
visualisation_grouping = build_visualisation_grouping(few_shot_examples)
print(f"visualisation_grouping section: ~{len(visualisation_grouping.split())} words")
```

visualisation\_grouping section: ~562 words

```
In [169... author_journey_prompt = f"""
You extract a structured, longitudinal treatment history for GLP-1 medications from one Reddit author's posts.

OUTPUT FORMAT
- Return exactly one valid JSON object and nothing else (no markdown, no commentary).
- Follow the schema below exactly: same field names, same nesting. Do not add, drop or rename fields.
- Unavailable scalar -> null. Unavailable list -> [].
- Do not invent details. Extract only what the text supports, and keep explicit statements (explicit_notes) separat

INPUT
- All posts are from ONE author, in chronological order, each with a POST number and date.
- drug_group is provided as metadata; use it to label the drug ONLY when the specific drug is not named in the text

WHAT TO CAPTURE
- The journey over time: drug, dose, weight/BMI, desired effects, adverse effects, adherence/access issues and conf
- Timing, taken from the post dates.
- An overall read of whether the evidence suggests efficacy, safety concerns, both or neither (journey_summary).

INCLUSION
{longitudinal_inclusion_rules}
Always extract desired and adverse effects even when include = false.

TIMELINE COMPLETENESS
- Produce ONE timeline_events entry for EVERY post that contains any dose, weight,
  effect, adverse event, confounder, or adherence/access information.
- Do NOT merge multiple posts into one event, and do NOT summarise the journey into
  a few representative events. A long journey must produce many events.
- Copy post_number EXACTLY from the input for each event; never invent post numbers.
- Omit a post only when it has no treatment-relevant information.

DRUG & DOSE
- Record drug names as written; map obvious misspellings to the intended GLP-1, but do not guess when unclear.
- Label every post with a GLP-1; if only drug_group is known, use that.
- Record non-GLP-1 medications as confounding_variables, never as the treatment drug.
- Capture dose transitions as concrete values (e.g. "0.25mg -> 0.5mg"), not vague phrases ("titrated up").
```

- drug\_switching applies ONLY between two GLP-1s (e.g. Ozempic -> Wegovy). GLP-1 -> non-GLP-1 (metformin, an antidiabetic).

DEMOGRAPHICS, HEIGHT, BMI

- sex\_gender: parse forms like 34F / 28M and explicit words (male, female, woman, man).
- height\_cm: only if explicitly stated; convert (1 in = 2.54 cm, 1 ft = 30.48 cm), else null.
- Compute bmi = weight\_kg / (height\_m squared) whenever both are known (1 lb = 0.453592 kg).

DATES & DURATION

- reported\_treatment\_duration only when explicitly quantified ("3 weeks", "2 years").
- You may infer duration from a stated start plus the post date (e.g. "started in November" + post on 2026-03-25 -> 3 months).
- All dates as YYYY-MM-DD.

CONFOUNDERS & CO-MEDICATIONS

- Extract explicitly mentioned confounders (e.g. PCOS, diabetes, menopause, thyroid disease, ADHD, depression, steroid use).
- Do NOT exclude an author merely for a comorbidity or a co-medication; record these as confounders. Exclude only when explicitly stated.

EFFECTS & SEVERITY

- Desired effects include weight loss, appetite/satiety change, reduced food noise/cravings, glucose control, improved mood.
- Adverse effects include nausea, vomiting, constipation, diarrhea, reflux, bloating, headache, fatigue, plateau, weight regain.
- Whenever an adverse effect is present, set severity to the highest burden reported during treatment: mild (tolerable), moderate, severe.

ACTIVITY

- exercise/activity level as low / moderate / high when stated, else null.

```
{visualisation_grouping}

{schema_consistency_rules}
```

SCHEMA:

```
{extraction_schema}
```

POSTS:

```
{{text}}
```

OUTPUT:

```
""""
```

```
In [ ]: # Build the few-shot block from the MANUALLY ANNOTATED examples.
```

```
def build_fewshot_from_annotations(examples):
```

```

    blocks = []

    for ex in examples:
        blocks.append(
            f""""POSTS:
{ex['input']}

OUTPUT:
{json.dumps(ex['expected_output'], indent=2, ensure_ascii=False)}""""

        )

    return "\n\n---\n\n".join(blocks)

few_shot = build_fewshot_from_annotations(few_shot_examples)

author_journey_prompt_fewshot = author_journey_prompt.replace(
    "POSTS:\n{text}",
    f"Examples:\n{few_shot}\n\nPOSTS:\n{{text}}"
)

enc = tiktoken.get_encoding("o200k_base")
print(f"Few-shot prompt length: ~{len(enc.encode(author_journey_prompt_fewshot))}, tokens (before journey text)")

```

Few-shot prompt length: ~51,070 tokens (before journey text)

## 8. GPT extraction setup

```

In [ ]: # =====
# GPT setup
# =====

model = "gpt-5.2"
max_output_tokens = 50000 # must also cover reasoning tokens
sleep_time = 0.5
max_retries = 3

# GPT-5.2 accepts temperature/top_p/reasoning_effort; update one, leave the other at default.
# temperature and top_p disabled when reasoning_effort is set.

```

```

# temperature = 0
# top_p = 1.0
reasoning_effort = "medium"

client = OpenAI(api_key=os.environ["OPENAI_API_KEY"])

def parse_model_output(x):
    if x is None:
        return None
    try:
        return json.loads(x)
    except Exception:
        return None

def call_openai(prompt):
    """Single extraction call. JSON output is enforced by the API,
    so parse failures should be rare rather than routine."""

    last_error = None

    for attempt in range(max_retries):
        try:
            response = client.responses.create(
                model=model,
                input=prompt,
                max_output_tokens=max_output_tokens,
                #temperature=TEMPERATURE,
                #top_p=TOP_P,
                #seed=SEED,
                reasoning={"effort": reasoning_effort},
                text={"format": {"type": "json_object"}},
            )
            return response.output_text

        except Exception as e:
            last_error = e
            wait = 2 ** attempt * 5
            print(f"  retry {attempt + 1}/{max_retries} after error: {e} (waiting {wait}s)")
            time.sleep(wait)

```

```
raise last_error
```

```
In [ ]: MAX_POSTS_PER_CALL = 60  # journeys longer than this are extracted in chunks

def _chunk_group(group, max_posts=MAX_POSTS_PER_CALL):
    group = group.sort_values("created_utc")
    for start in range(0, len(group), max_posts):
        yield group.iloc[start:start + max_posts]

def _merge_results(parts):
    """Merge per-chunk outputs: concatenate timeline_events, coalesce the rest."""
    merged, events = None, []
    for p in parts:
        if not isinstance(p, dict):
            continue
        events.extend(p.get("timeline_events") or [])
        if merged is None:
            merged = {k: v for k, v in p.items() if k != "timeline_events"}
        else: # fill any empty journey fields using chunks
            for k, v in p.items():
                if k != "timeline_events" and merged.get(k) in (None, "", [], {}):
                    and v not in (None, "", [], {}):
                        merged[k] = v
    merged = merged or {}
    # remove duplicate events by post_number, keep chronological order
    seen, uniq = set(), []
    for e in sorted(events, key=lambda e: (e.get("post_number") is None, e.get("post_number"))):
        key = e.get("post_number")
        key = json.dumps(e, sort_keys=True) if key is None else key
        if key not in seen:
            seen.add(key); uniq.append(e)
    merged["timeline_events"] = uniq
    return merged

def extract_journey(group, prompt_template, max_posts=MAX_POSTS_PER_CALL):
    """Single call for short journeys; chunked + merged for long ones.
    Returns a JSON string (drop-in for the runner's model_output)."""
    group = group.sort_values("created_utc")
    if len(group) <= max_posts:
```

```

        return call_openai(prompt_template.replace("{text}", build_annotation_input(group)))
    parts = [parse_model_output(
        call_openai(prompt_template.replace("{text}", build_annotation_input(chunk))))
        for chunk in _chunk_group(group, max_posts)]
    return json.dumps(_merge_results(parts), ensure_ascii=False)

```

```

In [173]: def post_coverage(group, model_output):
    parsed = parse_model_output(model_output) if isinstance(model_output, str) else model_output
    if not isinstance(parsed, dict):
        return None
    emitted = {e.get("post_number") for e in (parsed.get("timeline_events") or [])}
    total = set(group.sort_values("created_utc")["post_number"])
    covered = len(emitted & total)
    return {"n_posts": len(total),
            "n_events": len(parsed.get("timeline_events") or []),
            "posts_covered": covered,
            "coverage": round(covered / len(total), 3) if total else None}

```

```

In [ ]: #####
# Extraction runner #
#####

def run_journey_extraction(
    inputs,
    output_path,
    prompt_template,
    dry_run=True,
):
    """
    inputs: list of dicts with keys author_id, drug_group, input (+ metadata).
           If an item also has a "posts" dataframe, the journey is chunked;
           otherwise journey is extracted with one pass from the "input" string.
    output_path: .jsonl file; one line appended per completed author.
    """

    # Skip authors already present in the output file
    processed = set()
    if os.path.exists(output_path):
        done = pd.read_json(output_path, lines=True)
        if len(done) > 0:

```

```

        processed = set(done["author"])
        print(f"Resuming: {len(processed)} authors already processed")

    todo = [x for x in inputs if x["author_id"] not in processed]
    print(f"Journeys to process: {len(todo)} (of {len(inputs)})")

    if dry_run:
        print("DRY RUN - no API calls made")
        return None

    with open(output_path, "a", encoding="utf-8") as out_file:

        for n, item in enumerate(todo, start=1):

            # Build the record FIRST, then fill model_output.
            record = {
                # 'author' holds the anonymised author_id; 'drug_group'
                # matches the extraction schema field name.
                "author": item["author_id"],
                "drug_group": item["drug_group"],
                "n_author_posts": item.get("n_author_posts"),
                "first_post": str(item.get("first_post")),
                "last_post": str(item.get("last_post")),
                "days_between_first_and_last_post": item.get("days_between_first_and_last_post"),
                "weeks_between_first_and_last_post": item.get("weeks_between_first_and_last_post"),
                "months_between_first_and_last_post": item.get("months_between_first_and_last_post"),
                "journey_text": item.get("input"),
                "model": model,
                "model_output": None,
                "error": None,
            }

            try:
                if isinstance(item.get("posts"), pd.DataFrame):
                    # full extraction: chunk long journeys
                    record["model_output"] = extract_journey(item["posts"], prompt_template)
                else:
                    record["model_output"] = call_openai(
                        prompt_template.replace("{text}", str(item["input"]))
                    )
            except:
                record["error"] = str(sys.exc_info()[0])

            print(f"[{n}/{len(todo)}] {item['drug_group']} {item['author_id']}: done")

```



```

    except Exception as e:
        record["error"] = str(e)
        print(f"[{n}/{len(todo)}] {item['drug_group']} {item['author_id']}: ERROR - {e}")

    out_file.write(json.dumps(record, ensure_ascii=False) + "\n")
    out_file.flush()

    time.sleep(sleep_time)

results = pd.read_json(output_path, lines=True)
print(f"\nSaved {len(results)} results to {output_path}")
return results

```

## 9. Gold-standard Model Evaluation

In [178... *# Run the model on the gold-standard journeys*

```

evaluation_output_path = f"{gpt_result_dir}/gold_standard_predictions.jsonl"

dry_run = False # flip to False to make API calls

evaluation_inputs = [
    {
        "author_id": ex["author_id"],
        "drug_group": ex["drug_group"],
        "input": ex["input"],
    }
    for ex in gold_standard_examples
]

run_journey_extraction(
    inputs=evaluation_inputs,
    output_path=evaluation_output_path,
    prompt_template=author_journey_prompt_fewshot,
    dry_run=dry_run,
)

```

Resuming: 0 authors already processed

Journeys to process: 8 (of 8)

[1/8] oz oz\_4a9ba8ff5d43: done

[2/8] oz oz\_f06fb96ca50f: done

[3/8] reta reta\_bb45854c891d: done

[4/8] reta reta\_d0c18b2637e9: done

[5/8] sema sema\_46dfbf71f08c: done

[6/8] sema sema\_4d9944dbe2bb: done

[7/8] weg weg\_2cd38c74d0ef: done

[8/8] weg weg\_62ee4e4ed320: done

Saved 8 results to /Users/telissatitley/llm\_env/GLP1\_Thesis/Subreddits/GPT\_Results/gold\_standard\_predictions.jsonl

Out[178...

|   | author            | drug_group | n_author_posts | first_post | last_post | days_between_first_and_last_post | weeks_between_firs |
|---|-------------------|------------|----------------|------------|-----------|----------------------------------|--------------------|
| 0 | oz_4a9ba8ff5d43   | oz         | NaN            | None       | None      |                                  | NaN                |
| 1 | oz_f06fb96ca50f   | oz         | NaN            | None       | None      |                                  | NaN                |
| 2 | reta_bb45854c891d | reta       | NaN            | None       | None      |                                  | NaN                |
| 3 | reta_d0c18b2637e9 | reta       | NaN            | None       | None      |                                  | NaN                |
| 4 | sema_46dfbf71f08c | sema       | NaN            | None       | None      |                                  | NaN                |
| 5 | sema_4d9944dbe2bb | sema       | NaN            | None       | None      |                                  | NaN                |
| 6 | weg_2cd38c74d0ef  | weg        | NaN            | None       | None      |                                  | NaN                |
| 7 | weg_62ee4e4ed320  | weg        | NaN            | None       | None      |                                  | NaN                |

```

In [ ]: #####
# Split evaluation: structured fields (exact-match P/R/F1)#
# vs free-text fields (ROUGE-L, BERTScore)                #
#####

FREETEXT_KEYS = {"journey_summary", "exclusion_reason", "treatment_start_source",
                 "explicit_notes", "inferred_notes", "specific_label"}
META_KEYS = {"author_id", "drug_group"} # inputs, not predictions -> excluded

def _norm(v):
    return re.sub(r"\s+", " ", str(v).strip().lower())

def _flatten(v, parent=""):
    """(leaf_key, path, value) for every non-empty scalar leaf; fixes the route key typo."""
    out = []
    if isinstance(v, dict):
        for k, it in v.items():
            key = k.replace("route_of administration", "route_of_administration")
            out += _flatten(it, f"{parent}.{key}" if parent else key)
    elif isinstance(v, list):
        for it in v:
            out += _flatten(it, parent)
    else:
        nv = _norm(v)
        if nv not in ("", "none", "null", "not reported", "unknown"):
            out.append((parent.split(".")[1], parent.lower(), nv))
    return out

def _structured_facts(obj):
    return {(p, v) for leaf, p, v in _flatten(obj)
            if leaf not in FREETEXT_KEYS and leaf not in META_KEYS}

def _freetext_string(obj):
    return " ".join(v for leaf, p, v in _flatten(obj) if leaf in FREETEXT_KEYS)

def _prf(exp, pred):
    tp = len(exp & pred); fp = len(pred - exp); fn = len(exp - pred)
    p = tp / (tp + fp) if tp + fp else 0.0
    r = tp / (tp + fn) if tp + fn else 0.0
    f = 2 * p * r / (p + r) if p + r else 0.0

```

```

    return tp, fp, fn, p, r, f

# ---- load predictions from the checkpoint ----
_preds = pd.read_json(evaluation_output_path, lines=True)
_pred_by = {row["author"]: parse_model_output(row["model_output"])
            for _, row in _preds.iterrows()}

# ---- per-author structured facts + collect prose for batched scoring ----
rows, gold_texts, pred_texts = [], [], []
for ex in gold_standard_examples:
    aid = ex["author_id"]; exp = ex["expected_output"]; pred = _pred_by.get(aid) or {}
    E, P = _structured_facts(exp), _structured_facts(pred)
    tp, fp, fn, p, r, f = _prf(E, P)
    rows.append({"author_id": aid, "drug_group": ex["drug_group"],
                "struct_tp": tp, "struct_fp": fp, "struct_fn": fn,
                "struct_precision": round(p, 3), "struct_recall": round(r, 3),
                "struct_f1": round(f, 3)})
    gold_texts.append(_freetext_string(exp) or " ")
    pred_texts.append(_freetext_string(pred) or " ") # avoid empty string

# ---- free-text: ROUGE-L per author ----
_rs = rouge_scorer.RougeScorer(["rougeL"], use_stemmer=True)
for i, row in enumerate(rows):
    row["freetext_rougeL_f1"] = round(
        _rs.score(gold_texts[i], pred_texts[i])["rougeL"].fmeasure, 3)

# ---- free-text: BERTScore (one call over all authors) ----
_bp, _br, _bf = bert_score(pred_texts, gold_texts, lang="en", rescale_with_baseline=True, verbose=False)
for i, row in enumerate(rows):
    row["freetext_bertscore_f1"] = round(float(_bf[i]), 3)

split_metrics = pd.DataFrame(rows)

# ---- aggregates: macro (mean over authors) + micro (pooled) for structured ----
_cols = ["struct_precision", "struct_recall", "struct_f1",
         "freetext_rougeL_f1", "freetext_bertscore_f1"]
macro = split_metrics[_cols].mean().round(3)
TP, FP, FN = (split_metrics[c].sum() for c in ["struct_tp", "struct_fp", "struct_fn"])
micro_p = TP / (TP + FP) if TP + FP else 0.0
micro_r = TP / (TP + FN) if TP + FN else 0.0
micro_f = 2 * micro_p * micro_r / (micro_p + micro_r) if micro_p + micro_r else 0.0

```

```

# macro mean + std (std shows journey-to-journey variability)
macro_tbl = split_metrics[_cols].agg(["mean", "std"]).round(3).T
macro_tbl.columns = ["macro_mean", "std"]
print("\nMACRO mean +/- std:\n" + macro_tbl.to_string())

# per-drug-group
per_group = split_metrics.groupby("drug_group")[_cols].mean().round(3)
print("\nPer drug_group (indicative, n=2):\n" + per_group.to_string())

print(split_metrics.to_string(index=False))
print("\nMACRO (mean over authors):\n" + macro.to_string())
print(f"\nMICRO structured precision={micro_p:.3f} recall={micro_r:.3f} "
      f"f1={micro_f:.3f} (TP={TP} FP={FP} FN={FN})")

```

Loading weights: 100%|██████████| 389/389 [00:00<00:00, 6077.93it/s]  
[transformers] **RobertaModel LOAD REPORT** from: roberta-large

| Key                       | Status     |
|---------------------------|------------|
| lm_head.dense.bias        | UNEXPECTED |
| lm_head.layer_norm.bias   | UNEXPECTED |
| lm_head.dense.weight      | UNEXPECTED |
| lm_head.layer_norm.weight | UNEXPECTED |
| lm_head.bias              | UNEXPECTED |
| pooler.dense.bias         | MISSING    |
| pooler.dense.weight       | MISSING    |

#### Notes:

- UNEXPECTED: can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING: those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

MACRO mean +/- std:

|                       | macro_mean | std   |
|-----------------------|------------|-------|
| struct_precision      | 0.664      | 0.127 |
| struct_recall         | 0.693      | 0.079 |
| struct_f1             | 0.672      | 0.087 |
| freetext_rougeL_f1    | 0.243      | 0.051 |
| freetext_bertscore_f1 | 0.086      | 0.089 |

Per drug\_group (indicative, n=2):

|            | struct_precision | struct_recall | struct_f1 | freetext_rougeL_f1 | freetext_bertscore_f1 |
|------------|------------------|---------------|-----------|--------------------|-----------------------|
| drug_group |                  |               |           |                    |                       |
| oz         | 0.718            | 0.816         | 0.764     | 0.276              | 0.151                 |
| reta       | 0.631            | 0.643         | 0.620     | 0.214              | 0.079                 |
| sema       | 0.667            | 0.664         | 0.662     | 0.254              | 0.074                 |
| weg        | 0.640            | 0.649         | 0.644     | 0.226              | 0.041                 |

|                   | author_id | drug_group | struct_tp | struct_fp | struct_fn | struct_precision | struct_recall | struct_f1 | freetext_rougeL_f1 | freetext_bertscore_f1 |
|-------------------|-----------|------------|-----------|-----------|-----------|------------------|---------------|-----------|--------------------|-----------------------|
| oz_4a9ba8ff5d43   | oz        | 28         | 9         | 5         |           | 0.757            | 0.848         | 0.800     |                    |                       |
| 0.263             |           | 0.155      |           |           |           |                  |               |           |                    |                       |
| oz_f06fb96ca50f   | oz        | 40         | 19        | 11        |           | 0.678            | 0.784         | 0.727     |                    |                       |
| 0.289             |           | 0.147      |           |           |           |                  |               |           |                    |                       |
| reta_bb45854c891d | reta      | 75         | 97        | 40        |           | 0.436            | 0.652         | 0.523     |                    |                       |
| 0.136             |           | -0.048     |           |           |           |                  |               |           |                    |                       |
| reta_d0c18b2637e9 | reta      | 19         | 4         | 11        |           | 0.826            | 0.633         | 0.717     |                    |                       |
| 0.291             |           | 0.206      |           |           |           |                  |               |           |                    |                       |
| sema_46dfbf71f08c | sema      | 20         | 6         | 10        |           | 0.769            | 0.667         | 0.714     |                    |                       |
| 0.277             |           | 0.145      |           |           |           |                  |               |           |                    |                       |
| sema_4d9944dbe2bb | sema      | 35         | 27        | 18        |           | 0.565            | 0.660         | 0.609     |                    |                       |
| 0.232             |           | 0.004      |           |           |           |                  |               |           |                    |                       |
| weg_2cd38c74d0ef  | weg       | 29         | 20        | 17        |           | 0.592            | 0.630         | 0.611     |                    |                       |
| 0.239             |           | 0.039      |           |           |           |                  |               |           |                    |                       |
| weg_62ee4e4ed320  | weg       | 22         | 10        | 11        |           | 0.688            | 0.667         | 0.677     |                    |                       |
| 0.214             |           | 0.043      |           |           |           |                  |               |           |                    |                       |

MACRO (mean over authors):

|                       |       |
|-----------------------|-------|
| struct_precision      | 0.664 |
| struct_recall         | 0.693 |
| struct_f1             | 0.672 |
| freetext_rougeL_f1    | 0.243 |
| freetext_bertscore_f1 | 0.086 |

MICRO structured precision=0.583 recall=0.685 f1=0.630 (TP=268 FP=192 FN=123)

```
In [ ]: split_metrics.to_csv(
        f"{test_output_dir}/evaluation_metrics_by_author.csv",
        index=False,
    )

    # macro_tbl is indexed by metric name
    macro_tbl.to_csv(
        f"{test_output_dir}/evaluation_metrics_macro.csv"
    )

    # micro scores are scalars in the split cell
    micro_results = pd.DataFrame({
        "metric": ["micro_precision", "micro_recall", "micro_f1"],
        "score": [round(micro_p, 3), round(micro_r, 3), round(micro_f, 3)],
    })
    micro_results.to_csv(
        f"{test_output_dir}/evaluation_metrics_micro.csv",
        index=False,
    )

    # per_group is indexed by drug_group
    per_group.to_csv(
        f"{test_output_dir}/evaluation_metrics_by_drug_group.csv"
    )

    print("Evaluation metrics saved.")
```

Evaluation metrics saved.

## 10. Checking full extraction outputs

```
In [ ]: final_extraction_journeys = (
        longitudinal_pool
        .merge(extraction_candidate_authors[["drug_group", "author_id"]].drop_duplicates(),
              on=["drug_group", "author_id"], how="inner")
        .sort_values(["drug_group", "author_id", "created_utc"]).reset_index(drop=True)
```



```

    )

print(
    f"Final extraction authors: "
    f"{final_extraction_journeys[['drug_group', 'author_id']].drop_duplicates().shape[0]:,}"
)

print(
    f"Final extraction posts: "
    f"{len(final_extraction_journeys):,}"
)

```

Final extraction authors: 23,998  
Final extraction posts: 1,223,215

```
In [151... print(pd.DataFrame(llm_inputs)["n_author_posts"].describe())
```

```

count      23998.000000
mean         50.971539
std         160.088769
min           2.000000
25%           8.000000
50%          19.000000
75%          47.000000
max         8794.000000
Name: n_author_posts, dtype: float64

```

```
In [152... n = pd.DataFrame(llm_inputs)["n_author_posts"]
print(n.describe(percentiles=[.9, .95, .99, .999]))
print("journeys > 100 posts:", (n > 100).sum())
print("journeys > 300 posts:", (n > 300).sum())
```

```
count      23998.000000
mean        50.971539
std         160.088769
min          2.000000
90%         107.000000
95%         178.000000
99%         467.060000
99.9%       2197.045000
max         8794.000000
Name: n_author_posts, dtype: float64
journeys > 100 posts: 2593
journeys > 300 posts: 515
```

```
In [ ]: llm_inputs = []

for (drug_group, author_id), group in final_extraction_journeys.groupby(
    ["drug_group", "author_id"], sort=False
):
    group = group.sort_values("created_utc")
    first_post = group["created_utc"].min()
    last_post = group["created_utc"].max()
    days_between = (last_post - first_post).days

    llm_inputs.append({
        "author_id": author_id,
        "drug_group": drug_group,
        "n_author_posts": len(group),
        "first_post": first_post,
        "last_post": last_post,
        "days_between_first_and_last_post": days_between,
        "weeks_between_first_and_last_post": round(days_between / 7, 1),
        "months_between_first_and_last_post": round(days_between / 30.44, 1),
        "posts": group,  # dataframe for chunking
        "input": build_annotation_input(group), # for dry-run/preview only
    })

print(f"Prepared {len(llm_inputs):,} author journeys for LLM extraction")
```

Prepared 23,998 author journeys for LLM extraction

```

In [ ]: #####
# Select 450 randomly sampled authors per drug group for extraction #
#####

TARGET_PER_GROUP = 450
SEED = 42

inputs_df = pd.DataFrame(llm_inputs)

counts = inputs_df["drug_group"].value_counts()
print("available authors per drug_group:")
print(counts.to_string())

# sample per group WITHOUT losing drug_group: select columns explicitly after groupby
sampled_df = (
    inputs_df
    .groupby("drug_group", group_keys=False)[inputs_df.columns]
    .apply(lambda g: g.sample(n=min(TARGET_PER_GROUP, len(g)), random_state=SEED))
    .reset_index(drop=True)
)
assert "drug_group" in sampled_df.columns, "drug_group lost in sampling"

extraction_inputs = sampled_df.to_dict("records")

print(f"\nselected {len(extraction_inputs)} authors:")
print(sampled_df["drug_group"].value_counts().sort_index().to_string())
print("drug_group on first item:", "drug_group" in extraction_inputs[0])

# ---- run the extraction ----
scaled_output_path = f"{gpt_result_dir}/extraction_450_per_group.jsonl"

dry_run = False # flip True for dry run or False to make API calls

gpt_results = run_journey_extraction(
    inputs=extraction_inputs,
    output_path=scaled_output_path,
    prompt_template=author_journey_prompt_fewshot,
    dry_run=dry_run,
)

```

available authors per drug\_group:

| drug_group |      |
|------------|------|
| sema       | 9075 |
| oz         | 8526 |
| reta       | 5874 |
| weg        | 523  |

selected 1800 authors:

| drug_group |     |
|------------|-----|
| oz         | 450 |
| reta       | 450 |
| sema       | 450 |
| weg        | 450 |

drug\_group on first item: True

Journeys to process: 1800 (of 1800)

[1/1800] oz oz\_e1033af173bb: done  
[2/1800] oz oz\_583a6a9c7693: done  
[3/1800] oz oz\_fcd929b9e873: done  
[4/1800] oz oz\_20971e1af8e1: done  
[5/1800] oz oz\_19d743b3d8c1: done  
[6/1800] oz oz\_7eba97e3fb92: done  
[7/1800] oz oz\_3fd577c8fe60: done  
[8/1800] oz oz\_2905e0a36a43: done  
[9/1800] oz oz\_b840a6d9afbe: done  
[10/1800] oz oz\_c2b9f79634d1: done  
[11/1800] oz oz\_85567763fae0: done  
[12/1800] oz oz\_313296e5ce9c: done  
[13/1800] oz oz\_7ef8922b84ef: done  
[14/1800] oz oz\_2fb1535e27e0: done  
[15/1800] oz oz\_9c0c5b5e25ed: done  
[16/1800] oz oz\_953acd5a6cda: done  
[17/1800] oz oz\_0922c4918015: done  
[18/1800] oz oz\_69210be19541: done  
[19/1800] oz oz\_e766cb0cebd2: done  
[20/1800] oz oz\_3e720ffeabc38: done  
[21/1800] oz oz\_83ef59c0f2ea: done  
[22/1800] oz oz\_825a1c50a00c: done  
[23/1800] oz oz\_3f269bdc6ffd: done  
[24/1800] oz oz\_1f9566869088: done  
[25/1800] oz oz\_1a60e6f2ec27: done  
[26/1800] oz oz\_1503cf0ba73d: done

[27/1800] oz oz\_ce2986ee878b: done  
[28/1800] oz oz\_78817c74ffae: done  
[29/1800] oz oz\_87056cedbff2: done  
[30/1800] oz oz\_93331e466bff: done  
[31/1800] oz oz\_e61acbefd6f4: done  
[32/1800] oz oz\_fb96e8d3a853: done  
[33/1800] oz oz\_e32a7edcce67: done  
[34/1800] oz oz\_9623016e6e63: done  
[35/1800] oz oz\_e720fc9d5987: done  
[36/1800] oz oz\_22ebd34019f1: done  
[37/1800] oz oz\_17de2892391f: done  
[38/1800] oz oz\_b0685ad33349: done  
[39/1800] oz oz\_a9fdc5cf13c0: done  
[40/1800] oz oz\_d627b42b2a10: done  
[41/1800] oz oz\_d8b5c5a94eee: done  
[42/1800] oz oz\_d56dc040c3d1: done  
[43/1800] oz oz\_4a332e590ce1: done  
[44/1800] oz oz\_17b768ec2ab4: done  
[45/1800] oz oz\_06b9106858f4: done  
[46/1800] oz oz\_f736a39b2d13: done  
[47/1800] oz oz\_0cb1031a2055: done  
[48/1800] oz oz\_3add32dc76db: done  
[49/1800] oz oz\_5c0592743fef: done  
[50/1800] oz oz\_25a1271cacb2: done  
[51/1800] oz oz\_3f2cd1256584: done  
[52/1800] oz oz\_e57f6e58d577: done  
[53/1800] oz oz\_0074d6bd7c51: done  
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[82/1800] oz oz\_5ff203ee2097: done  
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[229/1800] oz oz\_ac3b8cc00753: done  
[230/1800] oz oz\_ad32eb615fd1: done  
[231/1800] oz oz\_880fac13081a: done

[232/1800] oz oz\_8d90c68c2570: done  
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[239/1800] oz oz\_adeb34092d09: done  
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[241/1800] oz oz\_49613871784e: done  
[242/1800] oz oz\_9ebd5f0ee3d1: done  
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[265/1800] oz oz\_d5def0262e7e: done  
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[267/1800] oz oz\_ad9f02098d5b: done  
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[269/1800] oz oz\_3c2aee8959bd: done  
[270/1800] oz oz\_da05a221a6ba: done  
[271/1800] oz oz\_f3e3a4cfe611: done  
[272/1800] oz oz\_99f6ac7ea11a: done

[273/1800] oz oz\_e6c9baabcbd5: done  
[274/1800] oz oz\_8018a60c5106: done  
[275/1800] oz oz\_8813ba445bd4: done  
[276/1800] oz oz\_c1a614ae8d81: done  
[277/1800] oz oz\_cb101910dd22: done  
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[279/1800] oz oz\_0a1b423507f2: done  
[280/1800] oz oz\_8957668db235: done  
[281/1800] oz oz\_4801dfc1f925: done  
[282/1800] oz oz\_217566178d3d: done  
[283/1800] oz oz\_14436c65c28a: done  
[284/1800] oz oz\_53169cf5e52d: done  
[285/1800] oz oz\_308fc695ad34: done  
[286/1800] oz oz\_e62adc053e8e: done  
[287/1800] oz oz\_53ce5d116361: done  
[288/1800] oz oz\_238b5c839a96: done  
[289/1800] oz oz\_35d6ba97255b: done  
[290/1800] oz oz\_ab5d38721b16: done  
[291/1800] oz oz\_3516a082ed20: done  
[292/1800] oz oz\_97220835ed69: done  
[293/1800] oz oz\_1de58ce82871: done  
[294/1800] oz oz\_55c3e3a8d608: done  
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[296/1800] oz oz\_155b2971bd03: done  
[297/1800] oz oz\_dffcb410068a: done  
[298/1800] oz oz\_2d9e5adf68c2: done  
[299/1800] oz oz\_009e82e87b5f: done  
[300/1800] oz oz\_5aa1315d243b: done  
[301/1800] oz oz\_e46c8e47b03c: done  
[302/1800] oz oz\_873f08fe29ce: done  
[303/1800] oz oz\_a815c9ad7e07: done  
[304/1800] oz oz\_e5a81e9766f5: done  
[305/1800] oz oz\_c34559f4ae47: done  
[306/1800] oz oz\_d33434fe8d88: done  
[307/1800] oz oz\_e9b6ca932d13: done  
[308/1800] oz oz\_7bb73af1d729: done  
[309/1800] oz oz\_ca5e3d9fafe0: done  
[310/1800] oz oz\_d02d226de7ea: done  
[311/1800] oz oz\_c29989b1e758: done  
[312/1800] oz oz\_3ae5fa98b9: done  
[313/1800] oz oz\_70653cf5615e: done

[314/1800] oz oz\_aaa88f801a6f: done  
[315/1800] oz oz\_43125f21c57e: done  
[316/1800] oz oz\_2b89690914f6: done  
[317/1800] oz oz\_783621f87ba6: done  
[318/1800] oz oz\_338815cfec7f: done  
[319/1800] oz oz\_d31e4c0133d2: done  
[320/1800] oz oz\_582d1b4e1eea: done  
[321/1800] oz oz\_47ee349c4696: done  
[322/1800] oz oz\_b11edd5683bf: done  
[323/1800] oz oz\_59e2f7a679a2: done  
[324/1800] oz oz\_135f87387b23: done  
[325/1800] oz oz\_7926c95620b8: done  
[326/1800] oz oz\_ca95da0cc53f: done  
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[1754/1800] weg weg\_ee51e12d0487: done  
[1755/1800] weg weg\_224d52fd9d7c: done  
[1756/1800] weg weg\_949759f9a61b: done  
[1757/1800] weg weg\_095a1ef051e7: done  
[1758/1800] weg weg\_4a9591292482: done  
[1759/1800] weg weg\_130c7e88a058: done  
[1760/1800] weg weg\_b6a20e22a51d: done  
[1761/1800] weg weg\_e9fa599530c8: done  
[1762/1800] weg weg\_cb9717a601d7: done  
[1763/1800] weg weg\_6a0e854e1865: done  
[1764/1800] weg weg\_c5c721d1eb42: done  
[1765/1800] weg weg\_8ac51a12c3d8: done  
[1766/1800] weg weg\_dd7350f50aba: done  
[1767/1800] weg weg\_dbd0687ba9cb: done  
[1768/1800] weg weg\_a7d69edccccc: done  
[1769/1800] weg weg\_76abb515387d: done  
[1770/1800] weg weg\_6582e3698956: done  
[1771/1800] weg weg\_67eaa818d903: done  
[1772/1800] weg weg\_91ec14d0e2d8: done  
[1773/1800] weg weg\_e79620d60632: done  
[1774/1800] weg weg\_7d7e7aef669f: done  
[1775/1800] weg weg\_fd414ef90bf4: done  
[1776/1800] weg weg\_e8e985039569: done  
[1777/1800] weg weg\_95d3deaf1c1d: done  
[1778/1800] weg weg\_adc59dfc6989: done  
[1779/1800] weg weg\_84bcbf3781ee: done  
[1780/1800] weg weg\_628596deca24: done  
[1781/1800] weg weg\_4d3b7f5faac8: done  
[1782/1800] weg weg\_1408d25bbeb0: done  
[1783/1800] weg weg\_6a9269e77b04: done  
[1784/1800] weg weg\_c393b8eb7b65: done  
[1785/1800] weg weg\_5f22a6ced880: done  
[1786/1800] weg weg\_9850b9d26798: done  
[1787/1800] weg weg\_816131565457: done  
[1788/1800] weg weg\_31d047f6878c: done  
[1789/1800] weg weg\_1ee06fefa5d6: done

```

[1790/1800] weg weg_bbcd70b100ce: done
[1791/1800] weg weg_014a2999ca9e: done
[1792/1800] weg weg_e9039680bc9f: done
[1793/1800] weg weg_face14096b43: done
[1794/1800] weg weg_18f3a857d510: done
[1795/1800] weg weg_c95585468d86: done
[1796/1800] weg weg_289a8a3fcb61: done
[1797/1800] weg weg_643de9045798: done
[1798/1800] weg weg_0fe3c95e7874: done
[1799/1800] weg weg_ceef7df2245e: done
[1800/1800] weg weg_8297299c3b5e: done

```

Saved 1800 results to /Users/telissatitley/llm\_env/GLP1\_Thesis/Subreddits/GPT\_Results/extraction\_450\_per\_group.jsonl

```

In [ ]: #####
# Post-coverage monitor for the full extraction #
#####

COVERAGE_FLAG = 0.70 # journeys below this are flagged for review

# content posts = posts that passed the seed-relevance screen
_content = (
    final_extraction_journeys[final_extraction_journeys["relevance"]]
    .groupby(["drug_group", "author_id"])["post_number"]
    .apply(set)
    .to_dict()
)
# fall back to all posts if an author has no relevant rows
# coverage against ALL posts (relevance under-counts content, esp. for reta)
_allposts = (
    final_extraction_journeys
    .groupby(["drug_group", "author_id"])["post_number"]
    .apply(set).to_dict()
)

cov_rows = []
for _, r in _scaled.iterrows():
    parsed = parse_model_output(r["model_output"])
    aid, dg = r["author"], r["drug_group"]
    allposts = _allposts.get((dg, aid), set())
    events = (parsed or {}).get("timeline_events") or []

```



```

emitted = {e.get("post_number") for e in events if isinstance(e, dict)}
covered = len(emitted & allposts)
cov_rows.append({
    "author_id": aid, "drug_group": dg,
    "n_posts": len(allposts),
    "n_events": len(events),
    "posts_covered": covered,
    "coverage": round(covered / len(allposts), 3) if allposts else None,
    "valid_json": parsed is not None,
})
coverage_df = pd.DataFrame(cov_rows)

# summary
print(f"journeys: {len(coverage_df)}")
print(f"invalid/empty JSON: {(~coverage_df['valid_json']).sum()}")
print(f"median coverage: {coverage_df['coverage'].median():.2f}")
print(f"flagged (< {COVERAGE_FLAG:.0%}): {(coverage_df['coverage'] < COVERAGE_FLAG).sum()}")

# is coverage worse on longer journeys? (the length-bias check)
print("\ncoverage by journey length:")
print(
    coverage_df.assign(
        length_bucket=pd.cut(coverage_df["n_posts"],
                               [0, 10, 30, 60, 10_000],
                               labels=["1-10", "11-30", "31-60", "60+"])
    )
    .groupby("length_bucket", observed=True)["coverage"]
    .agg(["count", "median"]).round(3).to_string()
)

# the worst offenders to eyeball / re-extract
flagged = coverage_df[coverage_df["coverage"] < COVERAGE_FLAG].sort_values("coverage")
print(f"\nlowest-coverage journeys:\n{flagged.head(15).to_string(index=False)}")

coverage_df.to_csv(f"{gpt_result_dir}/full_extraction_coverage.csv", index=False)

```

```
journeys: 1800
invalid/empty JSON: 0
median coverage: 0.53
flagged (< 70%): 1278
```

```
coverage by journey length:
      count  median
length_bucket
1-10         659    0.750
11-30        561    0.571
31-60        262    0.423
60+         318    0.333
```

```
lowest-coverage journeys:
      author_id drug_group  n_posts  n_events  posts_covered  coverage  valid_json
reta_ad87e7ce0df7      reta         3         0             0         0.0         True
reta_65e21ddd07df      reta        14         0             0         0.0         True
weg_6139aa4d1126      weg         13         0             0         0.0         True
weg_c8f167c3bf23      weg          8         0             0         0.0         True
sema_cf1b8ab997fa      sema        31         0             0         0.0         True
sema_ca4a185f2a0d      sema        11         0             0         0.0         True
sema_1689902c87a4      sema         9         0             0         0.0         True
oz_2c1e6a4f9b6a       oz         14         0             0         0.0         True
sema_67ffff2ca0cf      sema         9         0             0         0.0         True
reta_6c123957e936      reta        16         0             0         0.0         True
reta_0f8993cab34b      reta         4         0             0         0.0         True
reta_ea4d40a1fcf7      reta        13         0             0         0.0         True
reta_ac83ded1809e      reta        10         0             0         0.0         True
reta_dc09b6acbcf8      reta         5         0             0         0.0         True
reta_01163194ed94      reta         3         0             0         0.0         True
```

```
In [ ]: # Checking why events show empty on both include=false (correct) and genuine miss (re-run)
zero_ev = coverage_df[coverage_df["n_events"] == 0]
print(f"{len(zero_ev)} journeys with 0 events")
_by = {r["author"]: parse_model_output(r["model_output"]) for _, r in _scaled.iterrows()}
for aid in zero_ev["author_id"].head(15):
    p = _by.get(aid) or {}
    print(f"{aid}: include={p.get('include')} | reason={p.get('exclusion_reason')}")
```

64 journeys with 0 events

oz\_d56dc040c3d1: include=False | reason=Posts discuss Medicare/policy coverage for Wegovy and other medications but do not describe the author's own GLP-1 treatment use (no first-person dosing, effects, weight, or adherence history).

oz\_382b3f3d51c8: include=False | reason=No clear first-person GLP-1 treatment use, dosing, effects, weight, or adherence information; posts are primarily advice/commentary about others.

oz\_c3f33f946118: include=False | reason=author does not describe their own GLP-1 use; posts are advice/general information only

oz\_1cf9dd669a2d: include=False | reason=author does not describe their own glp-1 use; posts are advice/questions about others

oz\_aeea4e9f7c1e: include=False | reason=No first-person GLP-1 medication use, dosing, effects, weight/BMI, or access/adherence details were described; posts are general advice about exercise/body composition.

oz\_ac3b8cc00753: include=False | reason=Author does not describe their own GLP-1 medication use; posts are questions/advice and they explicitly state they have never been on ozempic.

oz\_2c1e6a4f9b6a: include=False | reason=No first-person GLP-1 treatment use described; posts are about considering treatment, access, or advising others.

reta\_ea4d40a1fcf7: include=True | reason=None

reta\_01163194ed94: include=False | reason=Author mentions retatrutide but provides no first-person treatment details (no dose, weight, effects, adherence, or timing).

reta\_91f65777b366: include=False | reason=No first-person GLP-1 treatment use described; posts are about planned dosing/questions before starting.

reta\_34523cff2998: include=False | reason=author does not describe their own glp-1 medication use; posts discuss trt and general commentary

reta\_b8a8c962b8cd: include=False | reason=no clear first-person GLP-1 treatment experience; posts are primarily general discussion/advice about retatrutide and other GLP-1s

reta\_2745e174ecd3: include=False | reason=No first-person GLP-1 medication use described; posts are primarily advice/opinions about dosing and safety.

reta\_c442af0dad98: include=False | reason=author does not describe their own GLP-1 medication use; posts are advice/discussion without first-person GLP-1 treatment details

reta\_ada88e7988ef: include=False | reason=Posts discuss GLP-1 dosing safety in general/for others; author does not describe their own GLP-1 use.

```
In [209... inc = coverage_df.merge(
    pd.DataFrame([{"author_id": a, "include": (_by[a] or {}).get("include")} for a in coverage_df["author_id"]]),
    on="author_id"
)
inc = inc[inc["include"] == True].copy()
inc["events_per_post"] = inc["n_events"] / inc["n_posts"].clip(lower=1)
print(inc.groupby(pd.cut(inc["n_posts"], [0,10,30,60,10000]))["events_per_post"].median())
```

```
n_posts
(0, 10]      0.777778
(10, 30]     0.583333
(30, 60]     0.428571
(60, 10000]  0.343750
Name: events_per_post, dtype: float64
```

```
In [210... long_low = inc[(inc["n_posts"] > 60)].sort_values("events_per_post").iloc[0]
aid = long_low["author_id"]; dg = long_low["drug_group"]
posts = final_extraction_journeys[
    (final_extraction_journeys["author_id"]==aid) & (final_extraction_journeys["drug_group"]==dg)
].sort_values("created_utc")
print(f"{aid}: {len(posts)} posts -> {long_low['n_events']} events\n")
for _, row in posts.head(30).iterrows():
    print(".", row["body"][:150].replace("\n", " "))
# then compare against:
print("\nEVENTS:")
for e in (_by[aid].get("timeline_events") or []):
    print(" -", [d.get("specific_label") for d in (e.get("desired_effects") or [])],
          [a.get("specific_label") for a in (e.get("adverse_effects") or [])])
```

reta\_e69eb1b18052: 67 posts -> 1 events

- Holding more fat in your chest isn't a sign of high estrogen it's juust where he holds a lot of fat. it's no diffe rent from holding a lot of fat in th
- Sounds like you're nutrient deficient
- Tren ace EOD and fewer sides and mood stability shouldn't be in the same sentence, ace should be done ED
- bro what, you adjust the dose so you do less ED. Its not cleaner on paper its literally cleaner in your blood. The half life is 24-48 hours. If you ca
- 2 days isn't enough time to show sustained weight loss, less food will result in more weight loss.
- >stubborn abdomen and hip Im sorry but its my understanding that tesamorelin only helps reduce visceral fat. Ab domen and hip fat would fall under
- You need to decrease your calories if you want more weight loss, stalls aren't dose related. Reta will slightly i ncrease your tdee but if you lose en
- You either have a metabolic disease like inactive thyroid or you are counting wrong. Do you weigh everything you e at ? Every time you cook with oil do
- Bro the only reason you don't know what the long term side effects from cjc/ipa is because no study has made it pa st few months before getting cancell
- Literally none of this is evidence based and is just assumptions and feelings. Hyperplasia only happens with crazy dosages 20iu+. All of the side effe
- Hgh is a peptide why do you keep saying peptides stimulate production. It's like you are confusing hgh with mk or something. You take 1 shot of hgh
- You don't cycle tb500, you use it while healing whatever injury you are trying to heal then you don't use it. Th ere is no reason to use bpc or tb if
- You probably shouldn't use Klow if you don't have an injury. Klow makes no sense these peptides have different use cases. Klow/glow is just clever mar
- There isn't any evidence to the contrary either, but because it increases VEGFR2 → AKT/eNOS signaling. More blood vessels can theoretically feed tumor
- Like I said I just said it theoretically could, not saying it's 100% definitive. I'd like it if you could answer m y other questions though, I genuinel
- Why do you think it will peak on day 5/6, when at that time it should've halfed not peaked. half life of reta is around 6 days as in half of it has b
- It doesn't make sense pharmacologically though, even if someone were a slow metaboliser of the drug it wouldn't la st 2 or 3x as long in the body.
- Yeah maybe for you but I highly doubt it because there literally isn't a documented case of this happening in any of the studies. You shouldn't pres
- Giving false information is worse than no information, also I corrected your false information. There are publis hed studies on reta, just because wh
- Shutting down natural test production to cut on rad is crazy work
- Triz raises rhr, Reta rises it a bit more, but it still does
- 500cals above maintenance is crazy overkill, 200 will achieve the same with less fat gain
- you don't need to gain 90% fat for every 10% muscle you gain

- This is what Janoshik recommends. • Room temp: Years for properly made lyophilized vials • Fridge: Years of viability • Freezer: Potentially DECADE
- Yeah like a 100–150 more calories maybe, a natural doesn't need to eat above 100–150 calories above maintenance. 500 calories above maintenance is 2.2
- Sorry I wasn't clear an enhanced lifter doesn't need to eat in a surplus that's more than 300 calories, a natural lifter shouldn't eat more than a 10
- Used to take tesa 2mg every night for around 3 months didn't really see any improvement just more water retention. Switched to Hgh and it's been way b
- I think it's safe to assume most people aren't getting a bad reaction from their injections. I pin daily both IM and subq and it doesn't feel like I'm
- Animal based diet is high in saturated fats which is counterproductive if you want to keep ldl, and general heart markers in range
- I'll listen to cardiologist over random people on Reddit or psychiatrist trying to sell me on some alternative diet.

EVENTS:

– ['not hungry while on it'] []

```
In [ ]: #####
# ANALYSIS SET #
#####

MIN_FIRST_PERSON_EVENTS = 2      # number of extracted events required
MIN_DESIRED_OR_ADVERSE   = 1      # number of adverse and desired effects required

def summarise(aid, p):
    p = p or {}
    events = p.get("timeline_events") or []
    n_events = sum(1 for e in events if isinstance(e, dict))
    n_desired = sum(len(e.get("desired_effects") or []) for e in events if isinstance(e, dict))
    n_adverse = sum(len(e.get("adverse_effects") or []) for e in events if isinstance(e, dict))
    return {
        "author_id": aid,
        "include_model": p.get("include"),
        "n_events": n_events,
        "n_desired": n_desired,
        "n_adverse": n_adverse,
        "n_effects": n_desired + n_adverse,
    }

analysis = pd.DataFrame([summarise(a, _by[a]) for a in _by])
```

```

analysis = analysis.merge(
    _scaled[["author", "drug_group", "n_author_posts"]].rename(columns={"author": "author_id"}),
    on="author_id", how="left",
)

keep = (
    (analysis["include_model"] == True) &
    (analysis["n_events"] >= MIN_FIRST_PERSON_EVENTS) &
    (analysis["n_effects"] >= MIN_DESIRED_OR_ADVERSE)
)
analysis["in_analysis_set"] = keep

print("full extraction:", len(analysis))
print(f"model include=True: {(analysis['include_model']==True).sum()}")
print(f"in analysis set (>= {MIN_FIRST_PERSON_EVENTS} events, >= {MIN_DESIRED_OR_ADVERSE} effect): {keep.sum()}")
print("\nanalysis set by drug_group:")
print(analysis[keep].groupby("drug_group").size().to_string())
print("\ndropped by reason:")
print(f"  model-excluded:      {(analysis['include_model'] != True).sum()}")
print(f"  too few events:      {((analysis['include_model']==True) & (analysis['n_events'] < MIN_FIRST_PERSON_EVENTS)).sum()}")
print(f"  no effects extracted: {((analysis['include_model']==True) & (analysis['n_events'] >= MIN_FIRST_PERSON_EVENTS) & (analysis['n_effects'] < MIN_DESIRED_OR_ADVERSE)).sum()}")

analysis_set_ids = set(analysis[keep]["author_id"])
analysis.to_csv(f"{gpt_result_dir}/analysis_set_membership.csv", index=False)
print(f"\nanalysis set: {len(analysis_set_ids)} authors -> use these for aggregate stats")

```

```
full extraction: 1800
model include=True: 1688
in analysis set (>= 2 events, >= 1 effect): 1633
```

analysis set by drug\_group:

```
drug_group
oz      422
reta    392
sema    409
weg     410
```

dropped by reason:

```
model-excluded:      112
too few events:       39
no effects extracted:  16
```

analysis set: 1633 authors -> use these for aggregate stats

## 11. Interpretation and visualisation

### Cohort EDA

```
In [ ]: #####
# Cohort Selection EDA Setup #
#####

# ---- drug display constants (shared by every figure) ----
DRUG_COLOUR = {"reta": "#e41a1c", "sema": "#377eb8", "weg": "#7c4aaf", "oz": "#ff7f00"}
DRUG_NAME    = {"reta": "Retatrutide", "sema": "Semaglutide", "weg": "Wegovy", "oz": "Ozempic"}
DRUG_ORDER   = ["reta", "sema", "weg", "oz"]
SEMA_FAMILY  = ("oz", "sema", "weg")          # pooled semaglutides

def wilson(reporters, total, z=1.96):
    """
    Wilson score 95% confidence interval for a proportion.
    reporters = number who reported the effect (k)
    total     = number of authors in the group (sample size, n)
```



```

    z = 1.96 gives a 95% interval
    Returns (rate, ci_low, ci_high). The Wilson interval is used instead
    of the simple normal approximation because it stays inside [0,1] and
    performs well in small samples and rare effects.
    """
    if total == 0:
        return (np.nan, np.nan, np.nan)
    rate = reporters / total # observed proportion p = k/n
    denominator = 1 + z**2 / total # Wilson denominator
    centre = rate + z**2 / (2 * total) # shifted centre of the interval
    half_width = z * np.sqrt(rate*(1-rate)/total # spread from sampling variance
                    + z**2 / (4*total*2)) # + small-sample correction
    return rate, (centre - half_width)/denominator, (centre + half_width)/denominator

# ---- PED (performance-enhancing-drug) flag ----
STEROID_TERMS = ["trt", "testosterone", "hgh", "steroid", "anavar", "tren", "sarms",
                  "nandrolone", "deca", "winstrol", "primo", "masteron", "boldenone",
                  "mk-677", "mk677", "clen"]

def uses_ped(aid):
    """True if any confounder label for this author matches a PED/steroid term."""
    profile = _by.get(aid) or {}
    for event in (profile.get("timeline_events") or []):
        if isinstance(event, dict):
            for conf in (event.get("confounding_variables") or []):
                if isinstance(conf, dict) and any(
                    term in (conf.get("specific_label") or "").lower() for term in STEROID_TERMS):
                    return True
    return False

```

```

In [ ]: #####
# Cohort selection, injection route-confirmed filter, and author-level demographics #
#####
_scaled = pd.read_json(scaled_output_path, lines=True)
_by = {r["author"]: parse_model_output(r["model_output"]) for _, r in _scaled.iterrows()}
_drug = dict(zip(_scaled["author"], _scaled["drug_group"]))

# --- administration by injection route-confirmed filter: drop explicit oral everywhere; require an
# injection route for oz/sema/weg ---
analysis_set_ids = [
    aid for aid in analysis_set_ids

```

```

        if "oral" not in str((_by.get(aid) or {}).get("route_of_administration") or "").lower()
        and (str((_by.get(aid) or {}).get("route_of_administration") or "").strip()
            or _drug.get(aid) == "reta")
    ]
print("route-confirmed per-group sizes:",
      dict(sorted(Counter(_drug.get(a) for a in analysis_set_ids).items())))
print(f"route-confirmed total: {len(analysis_set_ids)}")

# --- demographic extractors – return raw values recorded. The only exception
# for DERIVED BMI, which keeps a minimal height>50, to limit noise/nonsense BMI via kg/m^2. ---
def get_age(aid):
    v = ((_by.get(aid) or {}).get("patient_profile") or {}).get("age_years")
    return float(v) if isinstance(v, (int, float)) else None
def get_sex(aid):
    s = (((_by.get(aid) or {}).get("patient_profile") or {}).get("sex_gender") or "").lower().strip()
    if s in ("m", "man", "male"): return "male"
    if s in ("f", "woman", "female"): return "female"
    return None
def get_height(aid):
    h = ((_by.get(aid) or {}).get("patient_profile") or {}).get("height_cm")
    return float(h) if isinstance(h, (int, float)) else None
def get_bmi(aid):
    """BMI if directly recorded, else DERIVED from height + first valid weight.
    Recorded BMI is returned as-is (type-checked only). For the DERIVED value we
    require height>50cm – not a plausibility filter on real people, but a guard so
    a garbage height doesn't blow up the kg/m^2 division into a nonsense BMI."""
    prof = (_by.get(aid) or {}).get("patient_profile") or {}
    bmi = prof.get("bmi")
    if isinstance(bmi, (int, float)): # recorded BMI
        return round(float(bmi), 1)
    height = get_height(aid)
    if height and height > 50:
        for e in ((_by.get(aid) or {}).get("timeline_events") or []):
            if isinstance(e, dict):
                weight = e.get("weight_kg")
                if weight is None and e.get("weight_lb") is not None:
                    weight = float(e["weight_lb"]) * 0.453592 # convert lb to kg
                if isinstance(weight, (int, float)):
                    return round(weight / ((height/100) ** 2), 1) # BMI = kg / m^2
    return None
def get_race(aid):

```

```

    r = ((_by.get(aid) or {}).get("patient_profile") or {}).get("race_ethnicity")
    if isinstance(r, str) and r.strip() and r.strip().lower() not in ("none", "unknown", "n/a"):
        return r.strip()
    return None

age_map    = {a: get_age(a)    for a in analysis_set_ids}
sex_map    = {a: get_sex(a)    for a in analysis_set_ids}
height_map = {a: get_height(a) for a in analysis_set_ids}
bmi_map    = {a: get_bmi(a)    for a in analysis_set_ids}
race_map   = {a: get_race(a)   for a in analysis_set_ids}

# --- long-format effects table for the FULL injection confirmed cohort (n=1506),
#     with RAW adverse and desired effect labels. ---
rows = []
for aid in analysis_set_ids:
    p = _by.get(aid) or {}; drug = _drug.get(aid)
    for e in p.get("timeline_events") or []:
        if not isinstance(e, dict):
            continue
        for event, field in [("desired", "desired_effects"), ("adverse", "adverse_effects")]:
            for item in e.get(field) or []:
                if not isinstance(item, dict):
                    continue
                rows.append({
                    "author_id": aid, "drug_group": drug, "event": event,
                    "visualisation_group": (item.get("visualisation_group") or "other").lower(),
                    "specific_label": item.get("specific_label"),
                    "severity": item.get("severity"), "date": e.get("date"),
                    "age": age_map.get(aid), "sex": sex_map.get(aid),
                })
effects_long = pd.DataFrame(rows)
print(f"effect mentions: {len(effects_long):,} "
      f"(desired {sum(effects_long['event']=='desired'):,} / adverse {sum(effects_long['event']=='adverse'):,})")

```

route-confirmed per-group sizes: {'oz': 403, 'reta': 392, 'sema': 342, 'weg': 369}

route-confirmed total: 1506

effect mentions: 36,555 (desired 19,092 / adverse 17,463)

```

In [ ]: #####
# Explore recovery + completeness for each demographic field #
# to drive the cohort selection decision.                      #

```

```
#####
N = len(analysis_set_ids)
SATURATION_FLOOR = 30    #minimum authors required for cohort selection

FIELDS = {"age": age_map, "sex": sex_map, "height": height_map,
          "BMI": bmi_map, "race/ethnicity": race_map}
print(f"=== demographic recovery (route-confirmed, n={N}) ===")
print(f"{'field':16s} {'n':>5s} {'%':>6s}    per drug (oz/reta/sema/weg)")
for name, m in FIELDS.items():
    have = {a for a in analysis_set_ids if m.get(a) is not None}
    by_drug = [sum(1 for a in have if _drug.get(a) == dg) for dg in DRUG_ORDER]
    print(f"{'name':16s} {'len(have):>5d} {'100*len(have)/N:>5.1f}%    {by_drug}")

def cohort_with(*maps):
    """Authors having a non-None value in EVERY listed demographic map."""
    return {a for a in analysis_set_ids if all(m.get(a) is not None for m in maps)}

# candidate cohorts of increasing completeness
has_sex = cohort_with(sex_map)
has_age = cohort_with(age_map)
has_both = cohort_with(sex_map, age_map) # sex + age
has_sex_age_bmi = cohort_with(sex_map, age_map, bmi_map) # + BMI
has_sex_age_height = cohort_with(sex_map, age_map, height_map) # + height
full_demo = cohort_with(sex_map, age_map, height_map, bmi_map) # + height + BMI
# (race excluded from every cohort – recovered for almost no one, would collapse n)

print("\n=== cohort options (per-drug counts) ===")
cohort_options = [
    ("sex", has_sex), ("age", has_age),
    ("sex+age", has_both), ("sex+age+BMI", has_sex_age_bmi),
    ("sex+age+height", has_sex_age_height), ("sex+age+ht+BMI", full_demo),
]
for label, s in cohort_options:
    by_drug = dict(sorted(Counter(_drug.get(a) for a in s).items()))
    weg_ok = "OK" if by_drug.get("weg", 0) >= SATURATION_FLOOR else f"< {SATURATION_FLOOR} FLOOR"
    print(f"{'label':16s} n={len(s):4d}    {by_drug}    weg: {weg_ok}")

# --- distributions within the chosen cohort ---
print(f"\n=== distributions within sex+age (n={len(has_both)}) ===")
for name, m in [("age", age_map), ("height", height_map), ("BMI", bmi_map)]:
    vals = [m[a] for a in has_both if m.get(a) is not None]
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    if vals:
        print(f" {name:8s} n={len(vals):3d} median {np.median(vals):.1f} "
              f"IQR {np.percentile(vals,25):.1f}-{np.percentile(vals,75):.1f} "
              f"range {min(vals):.1f}-{max(vals):.1f}")
    print(" sex:", dict(Counter(sex_map[a] for a in has_both if sex_map.get(a))))
    race_vals = [race_map[a] for a in has_both if race_map.get(a)]
    print(f" race/ethnicity: {len(race_vals)} recorded",
          dict(Counter(race_vals)) if race_vals else "(none – generalisability limitation)")

```

=== demographic recovery (route-confirmed, n=1506) ===

| field          | n   | %     | per drug (oz/reta/sema/weg) |
|----------------|-----|-------|-----------------------------|
| age            | 450 | 29.9% | [143, 116, 63, 128]         |
| sex            | 485 | 32.2% | [140, 127, 72, 146]         |
| height         | 438 | 29.1% | [134, 127, 57, 120]         |
| BMI            | 407 | 27.0% | [127, 118, 50, 112]         |
| race/ethnicity | 5   | 0.3%  | [1, 2, 0, 2]                |

=== cohort options (per-drug counts) ===

|                |        |                                                  |                 |
|----------------|--------|--------------------------------------------------|-----------------|
| sex            | n= 485 | {'oz': 146, 'reta': 140, 'sema': 127, 'weg': 72} | weg: OK         |
| age            | n= 450 | {'oz': 128, 'reta': 143, 'sema': 116, 'weg': 63} | weg: OK         |
| sex+age        | n= 339 | {'oz': 101, 'reta': 104, 'sema': 88, 'weg': 46}  | weg: OK         |
| sex+age+BMI    | n= 212 | {'oz': 62, 'reta': 67, 'sema': 59, 'weg': 24}    | weg: < 30 FLOOR |
| sex+age+height | n= 226 | {'oz': 66, 'reta': 71, 'sema': 63, 'weg': 26}    | weg: < 30 FLOOR |
| sex+age+ht+BMI | n= 212 | {'oz': 62, 'reta': 67, 'sema': 59, 'weg': 24}    | weg: < 30 FLOOR |

=== distributions within sex+age (n=339) ===

|                 |                                       |              |                 |                   |
|-----------------|---------------------------------------|--------------|-----------------|-------------------|
| age             | n=339                                 | median 42.0  | IQR 32.0–52.0   | range 15.0–76.0   |
| height          | n=226                                 | median 170.2 | IQR 162.6–180.3 | range 147.3–198.1 |
| BMI             | n=212                                 | median 30.4  | IQR 26.6–35.1   | range 18.6–63.1   |
| sex:            | { 'female': 222, 'male': 117 }        |              |                 |                   |
| race/ethnicity: | 2 recorded { 'black': 1, 'white': 1 } |              |                 |                   |

```

In [ ]: #####
# Inspect the raw adverse and desired effect labels.#
# Define & apply the canonicalised events          #
#####

# distinct raw authors reporting a RAW adverse and desired effects
def raw_authors_reporting(effect, event, ids):
    ids = set(ids)
    return effects_long[(effects_long["event"] == event) &

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        (effects_long["visualisation_group"] == effect) &
        (effects_long["author_id"].isin(ids))["author_id"].nunique()

# --- list the raw labels + their author-level prevalence ---
print("=== RAW effect labels (sex+age), author-level prevalence – pre-canonicalisation ===")
for event in ["desired", "adverse"]:
    pool = effects_long[(effects_long["event"] == event) &
                        (effects_long["author_id"].isin(has_both))]
    counts = pool.groupby("visualisation_group")["author_id"].nunique().sort_values(ascending=False)
    print(f"\n {event.upper()} – {len(counts)} distinct raw labels:")
    for label, k in counts.items():
        print(f"    {label:38s} {k:4d}  ({100*k/len(has_both):.1f}%)")

# --- define and APPLY the canonical map ---
DESIRED_CANONICAL = {
    "improved satiety": "reduced appetite",
    "portion control": "reduced appetite",
    "psychological wellbeing": "improved quality of life",
    "improved mood": "improved quality of life",
    "improved confidence": "improved quality of life",
}
ADVERSE_CANONICAL = {
    "loss of appetite": "reduced appetite",
}
def canonical_label(label, event):
    """Map a raw effect label to its canonical form, per event type."""
    key = (label or "").lower().strip()
    if event == "desired": return DESIRED_CANONICAL.get(key, key)
    if event == "adverse": return ADVERSE_CANONICAL.get(key, key)
    return key

effects_long["visualisation_group_canonical"] = effects_long.apply(
    lambda r: canonical_label(r["visualisation_group"], r["event"]), axis=1)
effects_long.to_csv(f"{gpt_result_dir}/effects_long_analysis_set.csv", index=False)

# --- verify exactly which labels changed ---
print("\n=== canonicalised labels that changed ===")
for event in ["desired", "adverse"]:
    changed = effects_long[(effects_long["event"] == event) &
                          (effects_long["visualisation_group"] != effects_long["visualisation_group_canonical"])]
    print(f" {event.upper()}:")

```

```

    print(changed.groupby(["visualisation_group", "visualisation_group_canonical"]).size().to_string())

# --- canonicalised reporting ---
def authors_reporting(effect, event, ids):
    """Distinct authors in `ids` reporting this CANONICAL effect for this event type."""
    ids = set(ids)
    return effects_long[(effects_long["event"] == event) &
                        (effects_long["visualisation_group_canonical"] == effect) &
                        (effects_long["author_id"].isin(ids))]["author_id"].nunique()
def prevalence(effect, event, ids):
    """Fraction of a group who reported this canonical effect at least once."""
    return authors_reporting(effect, event, ids) / len(ids) if ids else np.nan

```

=== RAW effect labels (sex+age), author-level prevalence – pre-canonicalisation ===

DESIRED – 110 distinct raw labels:

|                                    |     |         |
|------------------------------------|-----|---------|
| weight loss                        | 314 | (92.6%) |
| reduced appetite                   | 206 | (60.8%) |
| reduced food noise                 | 170 | (50.1%) |
| improved quality of life           | 150 | (44.2%) |
| dietary improvement                | 118 | (34.8%) |
| portion control                    | 101 | (29.8%) |
| improved impulse control           | 93  | (27.4%) |
| relieved gastrointestinal symptoms | 89  | (26.3%) |
| improved satiety                   | 82  | (24.2%) |
| weight maintenance                 | 79  | (23.3%) |
| psychological wellbeing            | 76  | (22.4%) |
| increased physical activity        | 68  | (20.1%) |
| improved blood glucose control     | 63  | (18.6%) |
| reduced inflammation               | 55  | (16.2%) |
| improved mood                      | 47  | (13.9%) |
| maintained energy                  | 45  | (13.3%) |
| improved confidence                | 40  | (11.8%) |
| improved fatigue                   | 37  | (10.9%) |
| improved blood pressure            | 35  | (10.3%) |
| reduced cholesterol                | 22  | (6.5%)  |
| improved sleep                     | 14  | (4.1%)  |
| improved relationship with food    | 13  | (3.8%)  |
| reduced alcohol cravings           | 11  | (3.2%)  |
| reduced alcohol use                | 10  | (2.9%)  |
| improved diabetes symptoms         | 9   | (2.7%)  |
| menstrual changes                  | 8   | (2.4%)  |
| sleep disturbance                  | 7   | (2.1%)  |
| other                              | 7   | (2.1%)  |
| improved sleep apnea               | 6   | (1.8%)  |
| improved lab results               | 6   | (1.8%)  |
| improved menstrual regularity      | 5   | (1.5%)  |
| reduced alcohol desire             | 5   | (1.5%)  |
| muscle loss                        | 4   | (1.2%)  |
| improved sleep quality             | 4   | (1.2%)  |
| reduced alcohol craving            | 4   | (1.2%)  |
| improved menstrual symptoms        | 4   | (1.2%)  |
| improved resting heart rate        | 3   | (0.9%)  |
| improved pcos symptoms             | 3   | (0.9%)  |



|                                |   |        |
|--------------------------------|---|--------|
| increased libido               | 3 | (0.9%) |
| improved mobility              | 3 | (0.9%) |
| weight plateau                 | 3 | (0.9%) |
| treatment continuation         | 3 | (0.9%) |
| improved fatty liver           | 2 | (0.6%) |
| reduced musculoskeletal pain   | 2 | (0.6%) |
| craving changes                | 2 | (0.6%) |
| improved acne                  | 2 | (0.6%) |
| loss of appetite               | 2 | (0.6%) |
| improved sexual function       | 2 | (0.6%) |
| improved liver enzymes         | 2 | (0.6%) |
| improved iron status           | 2 | (0.6%) |
| improved liver function        | 2 | (0.6%) |
| muscle preservation            | 2 | (0.6%) |
| improved libido                | 2 | (0.6%) |
| reduced infections             | 1 | (0.3%) |
| preserved muscle mass          | 1 | (0.3%) |
| no appetite suppression        | 1 | (0.3%) |
| reduced fluid retention        | 1 | (0.3%) |
| transplant eligibility         | 1 | (0.3%) |
| stable vision                  | 1 | (0.3%) |
| pcos                           | 1 | (0.3%) |
| smoking cessation              | 1 | (0.3%) |
| relieved dermatologic symptoms | 1 | (0.3%) |
| reduced medication use         | 1 | (0.3%) |
| reduced water retention        | 1 | (0.3%) |
| reduced triglycerides          | 1 | (0.3%) |
| reduced substance use          | 1 | (0.3%) |
| reduced cravings               | 1 | (0.3%) |
| reduced smoking                | 1 | (0.3%) |
| reduced hair loss              | 1 | (0.3%) |
| reduced resting heart rate     | 1 | (0.3%) |
| reduced night sweats           | 1 | (0.3%) |
| reduced edema                  | 1 | (0.3%) |
| body recomposition             | 1 | (0.3%) |
| muscle gain                    | 1 | (0.3%) |
| improved hair growth           | 1 | (0.3%) |
| improved focus                 | 1 | (0.3%) |
| improved fertility             | 1 | (0.3%) |
| improved fatty liver disease   | 1 | (0.3%) |
| improved colorectal health     | 1 | (0.3%) |

|                                  |   |        |
|----------------------------------|---|--------|
| improved colon health            | 1 | (0.3%) |
| improved cognitive clarity       | 1 | (0.3%) |
| improved cognition               | 1 | (0.3%) |
| improved cardiovascular health   | 1 | (0.3%) |
| improved cardiometabolic markers | 1 | (0.3%) |
| improved breathing               | 1 | (0.3%) |
| improved body composition        | 1 | (0.3%) |
| improved adhd symptoms           | 1 | (0.3%) |
| headaches                        | 1 | (0.3%) |
| hair regrowth                    | 1 | (0.3%) |
| dysgeusia                        | 1 | (0.3%) |
| improved gout symptoms           | 1 | (0.3%) |
| improved headaches               | 1 | (0.3%) |
| maintained strength              | 1 | (0.3%) |
| improved insulin resistance      | 1 | (0.3%) |
| increased lean mass              | 1 | (0.3%) |
| improved vision                  | 1 | (0.3%) |
| improved urinary symptoms        | 1 | (0.3%) |
| improved tolerability            | 1 | (0.3%) |
| improved stroke symptoms         | 1 | (0.3%) |
| body size reduction              | 1 | (0.3%) |
| improved retinopathy             | 1 | (0.3%) |
| improved respiratory symptoms    | 1 | (0.3%) |
| improved plantar fasciitis       | 1 | (0.3%) |
| improved neuropathy symptoms     | 1 | (0.3%) |
| improved migraine symptoms       | 1 | (0.3%) |
| improved menopause symptoms      | 1 | (0.3%) |
| improved macular edema           | 1 | (0.3%) |
| improved liver disease           | 1 | (0.3%) |
| improved laboratory markers      | 1 | (0.3%) |
| improved skin                    | 1 | (0.3%) |

ADVERSE – 166 distinct raw labels:

|                         |     |         |
|-------------------------|-----|---------|
| gastrointestinal issues | 248 | (73.2%) |
| fatigue                 | 144 | (42.5%) |
| no appetite suppression | 106 | (31.3%) |
| loss of appetite        | 77  | (22.7%) |
| weight gain             | 77  | (22.7%) |
| dose intolerance        | 64  | (18.9%) |
| headaches               | 60  | (17.7%) |
| low mood                | 55  | (16.2%) |

|                              |    |         |
|------------------------------|----|---------|
| weight plateau               | 54 | (15.9%) |
| sleep disturbance            | 48 | (14.2%) |
| dehydration                  | 46 | (13.6%) |
| dysgeusia                    | 43 | (12.7%) |
| muscle loss                  | 41 | (12.1%) |
| increased sugar appetite     | 39 | (11.5%) |
| body dysmorphia              | 37 | (10.9%) |
| skin sensitivity             | 37 | (10.9%) |
| injection site inflammation  | 32 | (9.4%)  |
| reduced appetite             | 30 | (8.8%)  |
| elevated heart rate          | 30 | (8.8%)  |
| other                        | 29 | (8.6%)  |
| cold temperature sensitivity | 27 | (8.0%)  |
| menstrual changes            | 27 | (8.0%)  |
| treatment failure            | 21 | (6.2%)  |
| hair loss                    | 21 | (6.2%)  |
| dizziness                    | 19 | (5.6%)  |
| weight maintenance           | 14 | (4.1%)  |
| anxiety                      | 12 | (3.5%)  |
| increased food noise         | 12 | (3.5%)  |
| loose skin                   | 12 | (3.5%)  |
| hypoglycemia                 | 11 | (3.2%)  |
| reduced food noise           | 11 | (3.2%)  |
| musculoskeletal pain         | 7  | (2.1%)  |
| hypoglycaemia                | 7  | (2.1%)  |
| increased appetite           | 6  | (1.8%)  |
| hypoglycemia symptoms        | 6  | (1.8%)  |
| reduced libido               | 6  | (1.8%)  |
| alcohol intolerance          | 6  | (1.8%)  |
| sexual dysfunction           | 6  | (1.8%)  |
| increased cravings           | 4  | (1.2%)  |
| vision changes               | 4  | (1.2%)  |
| hypotension                  | 4  | (1.2%)  |
| increased calorie intake     | 4  | (1.2%)  |
| urinary frequency            | 3  | (0.9%)  |
| cognitive impairment         | 3  | (0.9%)  |
| heat intolerance             | 3  | (0.9%)  |
| lightheadedness              | 3  | (0.9%)  |
| fluid retention              | 3  | (0.9%)  |
| syncope                      | 3  | (0.9%)  |
| anhedonia                    | 3  | (0.9%)  |

|                                    |   |        |
|------------------------------------|---|--------|
| improved blood glucose control     | 3 | (0.9%) |
| orthostatic hypotension            | 3 | (0.9%) |
| worsened blood pressure            | 2 | (0.6%) |
| worsened food noise                | 2 | (0.6%) |
| worsened blood glucose control     | 2 | (0.6%) |
| persistent food noise              | 2 | (0.6%) |
| weight loss                        | 2 | (0.6%) |
| visual disturbance                 | 2 | (0.6%) |
| jaw pain                           | 2 | (0.6%) |
| urinary changes                    | 2 | (0.6%) |
| skin changes                       | 2 | (0.6%) |
| muscle discomfort                  | 2 | (0.6%) |
| relieved gastrointestinal symptoms | 2 | (0.6%) |
| reduced mobility                   | 2 | (0.6%) |
| acne                               | 2 | (0.6%) |
| irritability                       | 2 | (0.6%) |
| hypoglycaemia symptoms             | 2 | (0.6%) |
| food aversion                      | 2 | (0.6%) |
| alcohol use                        | 2 | (0.6%) |
| back pain                          | 2 | (0.6%) |
| body aches                         | 2 | (0.6%) |
| cognitive symptoms                 | 2 | (0.6%) |
| drug interaction                   | 2 | (0.6%) |
| excess skin                        | 2 | (0.6%) |
| flushing                           | 2 | (0.6%) |
| worsened pain                      | 2 | (0.6%) |
| increased food reward              | 1 | (0.3%) |
| smell sensitivity                  | 1 | (0.3%) |
| compulsive shopping                | 1 | (0.3%) |
| constipation                       | 1 | (0.3%) |
| skin laxity                        | 1 | (0.3%) |
| decreased libido                   | 1 | (0.3%) |
| shingles                           | 1 | (0.3%) |
| device or administration issue     | 1 | (0.3%) |
| sensory change                     | 1 | (0.3%) |
| sedation                           | 1 | (0.3%) |
| salivary gland issues              | 1 | (0.3%) |
| rhinitis                           | 1 | (0.3%) |
| dietary improvement                | 1 | (0.3%) |
| reduced motivation                 | 1 | (0.3%) |
| disordered eating                  | 1 | (0.3%) |

|                               |   |        |
|-------------------------------|---|--------|
| hypertension                  | 1 | (0.3%) |
| reduced inflammation          | 1 | (0.3%) |
| dry mouth                     | 1 | (0.3%) |
| reduced enjoyment of food     | 1 | (0.3%) |
| reduced efficacy              | 1 | (0.3%) |
| electrolyte imbalance         | 1 | (0.3%) |
| sleep-disordered breathing    | 1 | (0.3%) |
| stimulant sensitivity         | 1 | (0.3%) |
| raynaud phenomenon            | 1 | (0.3%) |
| increased alcohol sensitivity | 1 | (0.3%) |
| worsened inflammation         | 1 | (0.3%) |
| alcohol tolerance reduction   | 1 | (0.3%) |
| worsened cholesterol          | 1 | (0.3%) |
| worsened breathing            | 1 | (0.3%) |
| improved impulse control      | 1 | (0.3%) |
| altered alcohol response      | 1 | (0.3%) |
| worsened binge eating         | 1 | (0.3%) |
| improved mood                 | 1 | (0.3%) |
| increased alcohol desire      | 1 | (0.3%) |
| body composition change       | 1 | (0.3%) |
| body composition changes      | 1 | (0.3%) |
| water retention               | 1 | (0.3%) |
| body odor                     | 1 | (0.3%) |
| brain fog                     | 1 | (0.3%) |
| urticaria                     | 1 | (0.3%) |
| urine color change            | 1 | (0.3%) |
| breast imaging changes        | 1 | (0.3%) |
| chills                        | 1 | (0.3%) |
| cognitive issues              | 1 | (0.3%) |
| reduced alcohol effect        | 1 | (0.3%) |
| psychological wellbeing       | 1 | (0.3%) |
| increased impulsive spending  | 1 | (0.3%) |
| flu-like symptoms             | 1 | (0.3%) |
| low libido                    | 1 | (0.3%) |
| heart rate variability change | 1 | (0.3%) |
| food noise recurrence         | 1 | (0.3%) |
| gallbladder disease           | 1 | (0.3%) |
| leukopenia                    | 1 | (0.3%) |
| laboratory abnormality        | 1 | (0.3%) |
| lab abnormality               | 1 | (0.3%) |
| kidney pain                   | 1 | (0.3%) |

|                                |   |        |
|--------------------------------|---|--------|
| kidney concerns                | 1 | (0.3%) |
| joint pain                     | 1 | (0.3%) |
| jitteriness                    | 1 | (0.3%) |
| gout                           | 1 | (0.3%) |
| alcohol effects change         | 1 | (0.3%) |
| hair growth changes            | 1 | (0.3%) |
| injection discomfort           | 1 | (0.3%) |
| halitosis                      | 1 | (0.3%) |
| increased substance cravings   | 1 | (0.3%) |
| increased intraocular pressure | 1 | (0.3%) |
| increased inflammation         | 1 | (0.3%) |
| heart rate variability changes | 1 | (0.3%) |
| muscle cramps                  | 1 | (0.3%) |
| pruritus                       | 1 | (0.3%) |
| falls                          | 1 | (0.3%) |
| pots                           | 1 | (0.3%) |
| elevated intraocular pressure  | 1 | (0.3%) |
| palpitations                   | 1 | (0.3%) |
| improved blood pressure        | 1 | (0.3%) |
| elevated liver enzymes         | 1 | (0.3%) |
| high blood pressure            | 1 | (0.3%) |
| olfactory sensitivity          | 1 | (0.3%) |
| olfactory changes              | 1 | (0.3%) |
| no food noise reduction        | 1 | (0.3%) |
| no food noise control          | 1 | (0.3%) |
| excessive weight loss          | 1 | (0.3%) |
| neutropenia                    | 1 | (0.3%) |
| neuropathy                     | 1 | (0.3%) |
| neurological symptoms          | 1 | (0.3%) |
| nasal congestion               | 1 | (0.3%) |
| facial appearance changes      | 1 | (0.3%) |
| muscle twitching               | 1 | (0.3%) |
| muscle soreness                | 1 | (0.3%) |
| fainting                       | 1 | (0.3%) |
| pain                           | 1 | (0.3%) |

=== canonicalisedlabels that changed ===

DESIRED:

|                     |                               |     |
|---------------------|-------------------------------|-----|
| visualisation_group | visualisation_group_canonical |     |
| improved confidence | improved quality of life      | 145 |
| improved mood       | improved quality of life      | 212 |

|                         |                               |     |
|-------------------------|-------------------------------|-----|
| improved satiety        | reduced appetite              | 479 |
| portion control         | reduced appetite              | 633 |
| psychological wellbeing | improved quality of life      | 314 |
| ADVERSE:                |                               |     |
| visualisation_group     | visualisation_group_canonical |     |
| loss of appetite        | reduced appetite              | 526 |

```
In [ ]: #####
# Checking reta's canonicalised effects by prevalence.
#####
reta_ids = [a for a in has_both if _drug.get(a) == "reta"]
print(f"retatrutide authors (sex+age): {len(reta_ids)}\n")

def top_reta_effects(event, n=11):
    """The n most prevalent canonical effects among reta authors (sex+age).
    n=11 rather than 10 avoids splitting a tied point estimate at the 10th/11th."""
    reta_pool = effects_long[(effects_long["event"] == event) &
                             (effects_long["author_id"].isin(reta_ids))]
    prevs = [(eff, prevalence(eff, event, reta_ids))
              for eff in reta_pool["visualisation_group_canonical"].unique()]
    prevs.sort(key=lambda pair: pair[1], reverse=True) # most prevalent first
    return [eff for eff, _ in prevs[:n]]

for event in ["adverse", "desired"]:
    reta_pool = effects_long[(effects_long["event"] == event) &
                             (effects_long["author_id"].isin(reta_ids))]
    effects = reta_pool["visualisation_group_canonical"].value_counts().head(20).index
    rows = []
    for eff in effects:
        reporters = authors_reporting(eff, event, reta_ids)
        rate, ci_low, ci_high = wilson(reporters, len(reta_ids))
        rows.append(dict(effect=eff, reporters=reporters, rate=rate, lo=ci_low, hi=ci_high))
    df = pd.DataFrame(rows).sort_values("rate", ascending=False)
    df["prevalence"] = df.apply(
        lambda r: f"{r.rate*100:.1f}% [{r.lo*100:.0f}-{r.hi*100:.0f}] (n={int(r.reporters)})", axis=1)
    print(f"=== RETATRUTIDE {event.upper()} by prevalence (sex+age, canonical) ===")
    print(df[["effect", "prevalence"]].to_string(index=False), "\n")
```

retatrutide authors (sex+age): 104

=== RETATRUTIDE ADVERSE by prevalence (sex+age, canonical) ===

| effect                       | prevalence           |
|------------------------------|----------------------|
| gastrointestinal issues      | 58.7% [49-68] (n=61) |
| fatigue                      | 46.2% [37-56] (n=48) |
| no appetite suppression      | 30.8% [23-40] (n=32) |
| reduced appetite             | 27.9% [20-37] (n=29) |
| dose intolerance             | 24.0% [17-33] (n=25) |
| elevated heart rate          | 22.1% [15-31] (n=23) |
| sleep disturbance            | 21.2% [14-30] (n=22) |
| skin sensitivity             | 21.2% [14-30] (n=22) |
| weight gain                  | 18.3% [12-27] (n=19) |
| low mood                     | 17.3% [11-26] (n=18) |
| muscle loss                  | 17.3% [11-26] (n=18) |
| dehydration                  | 15.4% [10-24] (n=16) |
| headaches                    | 10.6% [6-18] (n=11)  |
| weight plateau               | 10.6% [6-18] (n=11)  |
| injection site inflammation  | 10.6% [6-18] (n=11)  |
| cold temperature sensitivity | 9.6% [5-17] (n=10)   |
| increased sugar appetite     | 4.8% [2-11] (n=5)    |
| sexual dysfunction           | 4.8% [2-11] (n=5)    |
| treatment failure            | 3.8% [2-9] (n=4)     |
| hypoglycemia                 | 3.8% [2-9] (n=4)     |

=== RETATRUTIDE DESIRED by prevalence (sex+age, canonical) ===

| effect                             | prevalence           |
|------------------------------------|----------------------|
| weight loss                        | 88.5% [81-93] (n=92) |
| reduced appetite                   | 71.2% [62-79] (n=74) |
| reduced food noise                 | 53.8% [44-63] (n=56) |
| improved quality of life           | 51.9% [42-61] (n=54) |
| improved impulse control           | 26.9% [19-36] (n=28) |
| dietary improvement                | 26.9% [19-36] (n=28) |
| maintained energy                  | 21.2% [14-30] (n=22) |
| relieved gastrointestinal symptoms | 21.2% [14-30] (n=22) |
| weight maintenance                 | 20.2% [14-29] (n=21) |
| improved blood glucose control     | 14.4% [9-22] (n=15)  |
| reduced inflammation               | 14.4% [9-22] (n=15)  |
| increased physical activity        | 13.5% [8-21] (n=14)  |
| improved fatigue                   | 11.5% [7-19] (n=12)  |
| improved blood pressure            | 10.6% [6-18] (n=11)  |



|                          |             |       |
|--------------------------|-------------|-------|
| reduced cholesterol      | 8.7% [5-16] | (n=9) |
| reduced alcohol cravings | 3.8% [2-9]  | (n=4) |
| sleep disturbance        | 3.8% [2-9]  | (n=4) |
| improved sleep           | 3.8% [2-9]  | (n=4) |
| improved liver function  | 1.9% [1-7]  | (n=2) |
| craving changes          | 1.0% [0-5]  | (n=1) |

```
In [ ]: #####
# Splits reta and pooled sema-family by PED use. #
#####
reta_all = [a for a in has_both if _drug.get(a) == "reta"]
sema_all = [a for a in has_both if _drug.get(a) in SEMA_FAMILY]      # pooled sema+oz+weg
reta_ped = [a for a in reta_all if uses_ped(a)]
reta_noped = [a for a in reta_all if not uses_ped(a)]
sema_ped = [a for a in sema_all if uses_ped(a)]
sema_noped = [a for a in sema_all if not uses_ped(a)]
print("sex+age PED split:")
print(f"  reta: {len(reta_all)} = {len(reta_ped)} PED + {len(reta_noped)} non-PED")
print(f"  sema: {len(sema_all)} = {len(sema_ped)} PED + {len(sema_noped)} non-PED")

def rate_str(effect, event, ids):
    """'rate [lo-hi]' percentage string for a subgroup, or '-' if empty."""
    reporters = authors_reporting(effect, event, ids); total = len(ids)
    if total == 0:
        return "-"
    rate, lo, hi = wilson(reporters, total)
    return f"{rate*100:.1f} [{lo*100:.0f}-{hi*100:.0f}]"

def deconfound(event, top_n=20):
    """Four-subgroup prevalence table; p/q test the DECONFOUNDED comparison
    (non-PED reta vs non-PED sema), BH-corrected across the effects shown."""
    pool = effects_long[(effects_long["event"] == event) &
                        (effects_long["author_id"].isin(has_both))]
    effects = pool["visualisation_group_canonical"].value_counts().head(top_n).index
    rows = []
    for eff in effects:
        # reported vs not, non-PED reta vs non-PED sema
        reta_k = authors_reporting(eff, event, reta_noped); reta_n = len(reta_noped)
        sema_k = authors_reporting(eff, event, sema_noped); sema_n = len(sema_noped)
        _, pval = fisher_exact([[reta_k, reta_n - reta_k], [sema_k, sema_n - sema_k]])
```

```

        rows.append(dict(effect=eff,
                        reta_PED    = rate_str(eff, event, reta_ped),
                        reta_nonPED= rate_str(eff, event, reta_noped),
                        sema_PED    = rate_str(eff, event, sema_ped),
                        sema_nonPED= rate_str(eff, event, sema_noped),
                        p=pval))
df = pd.DataFrame(rows)
df["q_BH"] = multipletests(df["p"], method="fdr_bh")[1]    #False discovery rate | Benjamini-Hochberg
df["sig"]   = (df["q_BH"] < 0.05).astype(int)
return df.sort_values("q_BH")                             #most significant effects sorted descending

for event in ["adverse", "desired"]:
    df = deconfound(event)
    out = df.copy()
    out["p"] = out["p"].map(lambda v: f"{v:.3f}"); out["q(BH)"] = out["q_BH"].map(lambda v: f"{v:.3f}")
    print(f"\n=== {event.upper()} - sex+age drug×PED (canonical; p/q = non-PED reta vs non-PED sema) ===")
    print(out[["effect", "reta_PED", "reta_nonPED", "sema_PED", "sema_nonPED", "p", "q(BH)", "sig"]].to_string(index=False))
    df.to_csv(f"{gpt_result_dir}/hasboth_PEDdeconfound_canonical_{event}.csv", index=False)

```

sex+age PED split:

reta: 104 = 38 PED + 66 non-PED

sema: 235 = 32 PED + 203 non-PED

=== ADVERSE – sex+age drug×PED (canonical; p/q = non-PED reta vs non-PED sema) ===

| effect                   | reta_PED     | reta_nonPED  | sema_PED     | sema_nonPED  | p     | q(BH) | sig |
|--------------------------|--------------|--------------|--------------|--------------|-------|-------|-----|
| elevated heart rate      | 21.1 [11-36] | 22.7 [14-34] | 3.1 [1-16]   | 3.0 [1-6]    | 0.000 | 0.000 | 1   |
| gastrointestinal issues  | 63.2 [47-77] | 56.1 [44-67] | 93.8 [80-98] | 77.3 [71-83] | 0.001 | 0.009 | 1   |
| skin sensitivity         | 21.1 [11-36] | 21.2 [13-33] | 9.4 [3-24]   | 5.9 [3-10]   | 0.001 | 0.009 | 1   |
| dysgeusia                | 10.5 [4-24]  | 3.0 [1-10]   | 31.2 [18-49] | 13.3 [9-19]  | 0.021 | 0.104 | 0   |
| headaches                | 13.2 [6-27]  | 9.1 [4-18]   | 28.1 [16-45] | 19.7 [15-26] | 0.059 | 0.235 | 0   |
| weight gain              | 28.9 [17-45] | 12.1 [6-22]  | 43.8 [28-61] | 21.7 [17-28] | 0.107 | 0.355 | 0   |
| body dysmorphia          | 5.3 [1-17]   | 4.5 [2-13]   | 31.2 [18-49] | 10.8 [7-16]  | 0.149 | 0.425 | 0   |
| increased sugar appetite | 2.6 [0-13]   | 6.1 [2-15]   | 25.0 [13-42] | 12.8 [9-18]  | 0.177 | 0.442 | 0   |
| weight plateau           | 10.5 [4-24]  | 10.6 [5-20]  | 25.0 [13-42] | 17.2 [13-23] | 0.243 | 0.503 | 0   |
| dose intolerance         | 28.9 [17-45] | 21.2 [13-33] | 25.0 [13-42] | 15.3 [11-21] | 0.260 | 0.503 | 0   |
| sleep disturbance        | 31.6 [19-47] | 15.2 [8-26]  | 15.6 [7-32]  | 10.3 [7-15]  | 0.277 | 0.503 | 0   |
| menstrual changes        | 2.6 [0-13]   | 4.5 [2-13]   | 12.5 [5-28]  | 9.4 [6-14]   | 0.303 | 0.505 | 0   |
| treatment failure        | 5.3 [1-17]   | 3.0 [1-10]   | 12.5 [5-28]  | 6.4 [4-11]   | 0.373 | 0.574 | 0   |
| muscle loss              | 39.5 [26-55] | 4.5 [2-13]   | 25.0 [13-42] | 7.4 [5-12]   | 0.575 | 0.639 | 0   |
| low mood                 | 13.2 [6-27]  | 19.7 [12-31] | 12.5 [5-28]  | 16.3 [12-22] | 0.573 | 0.639 | 0   |
| dehydration              | 26.3 [15-42] | 9.1 [4-18]   | 12.5 [5-28]  | 12.8 [9-18]  | 0.515 | 0.639 | 0   |
| reduced appetite         | 36.8 [23-53] | 22.7 [14-34] | 37.5 [23-55] | 27.1 [21-34] | 0.522 | 0.639 | 0   |
| orthostatic hypotension  | 0.0 [0-9]    | 1.5 [0-8]    | 0.0 [0-11]   | 1.0 [0-4]    | 0.572 | 0.639 | 0   |
| fatigue                  | 52.6 [37-68] | 42.4 [31-54] | 43.8 [28-61] | 40.4 [34-47] | 0.775 | 0.816 | 0   |
| no appetite suppression  | 34.2 [21-50] | 28.8 [19-41] | 46.9 [31-64] | 29.1 [23-36] | 1.000 | 1.000 | 0   |

=== DESIRED – sex+age drug×PED (canonical; p/q = non-PED reta vs non-PED sema) ===

| effect                             | reta_PED     | reta_nonPED  | sema_PED     | sema_nonPED  | p     | q(BH) | sig |
|------------------------------------|--------------|--------------|--------------|--------------|-------|-------|-----|
| weight loss                        | 94.7 [83-99] | 84.8 [74-92] | 96.9 [84-99] | 94.1 [90-97] | 0.035 | 0.693 | 0   |
| reduced alcohol use                | 0.0 [0-9]    | 1.5 [0-8]    | 0.0 [0-11]   | 4.4 [2-8]    | 0.459 | 0.807 | 0   |
| improved fatigue                   | 18.4 [9-33]  | 7.6 [3-17]   | 9.4 [3-24]   | 10.8 [7-16]  | 0.637 | 0.807 | 0   |
| reduced cholesterol                | 10.5 [4-24]  | 7.6 [3-17]   | 6.2 [2-20]   | 5.4 [3-9]    | 0.551 | 0.807 | 0   |
| improved blood pressure            | 7.9 [3-21]   | 12.1 [6-22]  | 12.5 [5-28]  | 9.9 [6-15]   | 0.644 | 0.807 | 0   |
| maintained energy                  | 28.9 [17-45] | 16.7 [10-27] | 6.2 [2-20]   | 10.3 [7-15]  | 0.190 | 0.807 | 0   |
| increased physical activity        | 18.4 [9-33]  | 10.6 [5-20]  | 43.8 [28-61] | 19.7 [15-26] | 0.097 | 0.807 | 0   |
| improved sleep                     | 7.9 [3-21]   | 1.5 [0-8]    | 6.2 [2-20]   | 3.9 [2-8]    | 0.461 | 0.807 | 0   |
| improved diabetes symptoms         | 0.0 [0-9]    | 0.0 [0-6]    | 3.1 [1-16]   | 3.9 [2-8]    | 0.206 | 0.807 | 0   |
| improved impulse control           | 39.5 [26-55] | 19.7 [12-31] | 43.8 [28-61] | 25.1 [20-32] | 0.409 | 0.807 | 0   |
| dietary improvement                | 28.9 [17-45] | 25.8 [17-37] | 50.0 [34-66] | 36.5 [30-43] | 0.134 | 0.807 | 0   |
| relieved gastrointestinal symptoms | 18.4 [9-33]  | 22.7 [14-34] | 34.4 [20-52] | 27.6 [22-34] | 0.521 | 0.807 | 0   |

|                                 |              |              |              |              |       |       |   |
|---------------------------------|--------------|--------------|--------------|--------------|-------|-------|---|
| reduced food noise              | 55.3 [40-70] | 53.0 [41-65] | 59.4 [42-74] | 46.8 [40-54] | 0.398 | 0.807 | 0 |
| improved quality of life        | 57.9 [42-72] | 48.5 [37-60] | 65.6 [48-80] | 55.7 [49-62] | 0.323 | 0.807 | 0 |
| reduced appetite                | 68.4 [53-81] | 72.7 [61-82] | 87.5 [72-95] | 69.5 [63-75] | 0.646 | 0.807 | 0 |
| weight maintenance              | 28.9 [17-45] | 15.2 [8-26]  | 43.8 [28-61] | 21.7 [17-28] | 0.292 | 0.807 | 0 |
| improved blood glucose control  | 10.5 [4-24]  | 16.7 [10-27] | 25.0 [13-42] | 19.7 [15-26] | 0.718 | 0.845 | 0 |
| reduced alcohol cravings        | 5.3 [1-17]   | 3.0 [1-10]   | 3.1 [1-16]   | 3.0 [1-6]    | 1.000 | 1.000 | 0 |
| improved relationship with food | 0.0 [0-9]    | 3.0 [1-10]   | 6.2 [2-20]   | 4.4 [2-8]    | 1.000 | 1.000 | 0 |
| reduced inflammation            | 13.2 [6-27]  | 15.2 [8-26]  | 31.2 [18-49] | 14.8 [11-20] | 1.000 | 1.000 | 0 |

```
In [ ]: #####
# Reta PED vs non-PED #
#####
print(f"retatrutide (sex+age): {len(reta_ped)} PED vs {len(reta_noped)} non-PED\n")
ped_driven = {}
for event in ["adverse", "desired"]:
    reta_pool = effects_long[(effects_long["event"] == event) &
                             (effects_long["author_id"].isin(reta_all))]
    effects = reta_pool["visualisation_group_canonical"].value_counts().head(25).index
    rows = []
    for eff in effects:
        ped_k = authors_reporting(eff, event, reta_ped); ped_rate, plo, phi = wilson(ped_k, len(reta_ped))
        noped_k = authors_reporting(eff, event, reta_noped); noped_rate, nlo, nhi = wilson(noped_k, len(reta_noped))
        _, pval = fisher_exact([[ped_k, len(reta_ped)-ped_k], [noped_k, len(reta_noped)-noped_k]])
        rows.append(dict(effect=eff, PED=ped_rate, PED_lo=plo, PED_hi=phi,
                        nonPED=noped_rate, nonPED_lo=nlo, nonPED_hi=nhi,
                        direction="higher in PED" if ped_rate > noped_rate else "higher in non-PED",
                        p=pval))
    df = pd.DataFrame(rows)
    df["q_BH"] = multipletests(df["p"], method="fdr_bh")[1]
    df["sig"] = (df["q_BH"] < 0.05).astype(int)
    df = df.sort_values("q_BH")
    out = df.copy()
    out["PED %"] = out.apply(lambda r: f"{r.PED*100:.1f} [{r.PED_lo*100:.0f}-{r.PED_hi*100:.0f}]", axis=1)
    out["non-PED %"] = out.apply(lambda r: f"{r.nonPED*100:.1f} [{r.nonPED_lo*100:.0f}-{r.nonPED_hi*100:.0f}]", axis=1)
    out["p"] = out["p"].map(lambda v: f"{v:.3f}"); out["q(BH)"] = out["q_BH"].map(lambda v: f"{v:.3f}")
    print(f"=== {event.upper()} - reta PED vs non-PED (sex+age, canonical, sorted by q) ===")
    print(out[["effect", "PED %", "non-PED %", "direction", "p", "q(BH)", "sig"]].to_string(index=False), "\n")
    ped_driven[event] = df[df["sig"] == 1]["effect"].tolist()
print("Effects differing by PED status WITHIN reta (BH q<0.05) = candidate artefacts:")
print(f"  adverse: {ped_driven['adverse']}\n  desired: {ped_driven['desired']}")
```

retatrutide (sex+age): 38 PED vs 66 non-PED

=== ADVERSE – reta PED vs non-PED (sex+age, canonical, sorted by q) ===

|  | effect                         | PED %        | non-PED %    | direction         | p     | q(BH) | sig |
|--|--------------------------------|--------------|--------------|-------------------|-------|-------|-----|
|  | muscle loss                    | 39.5 [26–55] | 4.5 [2–13]   | higher in PED     | 0.000 | 0.000 | 1   |
|  | reduced libido                 | 10.5 [4–24]  | 0.0 [0–6]    | higher in PED     | 0.016 | 0.201 | 0   |
|  | dehydration                    | 26.3 [15–42] | 9.1 [4–18]   | higher in PED     | 0.025 | 0.210 | 0   |
|  | weight gain                    | 28.9 [17–45] | 12.1 [6–22]  | higher in PED     | 0.039 | 0.242 | 0   |
|  | sexual dysfunction             | 10.5 [4–24]  | 1.5 [0–8]    | higher in PED     | 0.058 | 0.292 | 0   |
|  | sleep disturbance              | 31.6 [19–47] | 15.2 [8–26]  | higher in PED     | 0.079 | 0.330 | 0   |
|  | cold temperature sensitivity   | 15.8 [7–30]  | 6.1 [2–15]   | higher in PED     | 0.165 | 0.479 | 0   |
|  | reduced appetite               | 36.8 [23–53] | 22.7 [14–34] | higher in PED     | 0.173 | 0.479 | 0   |
|  | increased food noise           | 7.9 [3–21]   | 1.5 [0–8]    | higher in PED     | 0.137 | 0.479 | 0   |
|  | injection site inflammation    | 15.8 [7–30]  | 7.6 [3–17]   | higher in PED     | 0.204 | 0.510 | 0   |
|  | dose intolerance               | 28.9 [17–45] | 21.2 [13–33] | higher in PED     | 0.475 | 0.869 | 0   |
|  | low mood                       | 13.2 [6–27]  | 19.7 [12–31] | higher in non-PED | 0.436 | 0.869 | 0   |
|  | treatment failure              | 5.3 [1–17]   | 3.0 [1–10]   | higher in PED     | 0.622 | 0.869 | 0   |
|  | no appetite suppression        | 34.2 [21–50] | 28.8 [19–41] | higher in PED     | 0.660 | 0.869 | 0   |
|  | headaches                      | 13.2 [6–27]  | 9.1 [4–18]   | higher in PED     | 0.525 | 0.869 | 0   |
|  | fatigue                        | 52.6 [37–68] | 42.4 [31–54] | higher in PED     | 0.414 | 0.869 | 0   |
|  | hypoglycemia                   | 5.3 [1–17]   | 3.0 [1–10]   | higher in PED     | 0.622 | 0.869 | 0   |
|  | increased sugar appetite       | 2.6 [0–13]   | 6.1 [2–15]   | higher in non-PED | 0.650 | 0.869 | 0   |
|  | gastrointestinal issues        | 63.2 [47–77] | 56.1 [44–67] | higher in PED     | 0.539 | 0.869 | 0   |
|  | elevated heart rate            | 21.1 [11–36] | 22.7 [14–34] | higher in non-PED | 1.000 | 1.000 | 0   |
|  | other                          | 7.9 [3–21]   | 7.6 [3–17]   | higher in PED     | 1.000 | 1.000 | 0   |
|  | weight plateau                 | 10.5 [4–24]  | 10.6 [5–20]  | higher in non-PED | 1.000 | 1.000 | 0   |
|  | worsened blood glucose control | 0.0 [0–9]    | 1.5 [0–8]    | higher in non-PED | 1.000 | 1.000 | 0   |
|  | skin sensitivity               | 21.1 [11–36] | 21.2 [13–33] | higher in non-PED | 1.000 | 1.000 | 0   |
|  | anhedonia                      | 2.6 [0–13]   | 1.5 [0–8]    | higher in PED     | 1.000 | 1.000 | 0   |

=== DESIRED – reta PED vs non-PED (sex+age, canonical, sorted by q) ===

|  | effect                   | PED %        | non-PED %    | direction     | p     | q(BH) | sig |
|--|--------------------------|--------------|--------------|---------------|-------|-------|-----|
|  | improved sleep           | 7.9 [3–21]   | 1.5 [0–8]    | higher in PED | 0.137 | 0.686 | 0   |
|  | improved impulse control | 39.5 [26–55] | 19.7 [12–31] | higher in PED | 0.039 | 0.686 | 0   |
|  | sleep disturbance        | 7.9 [3–21]   | 1.5 [0–8]    | higher in PED | 0.137 | 0.686 | 0   |
|  | improved fatigue         | 18.4 [9–33]  | 7.6 [3–17]   | higher in PED | 0.117 | 0.686 | 0   |
|  | weight maintenance       | 28.9 [17–45] | 15.2 [8–26]  | higher in PED | 0.128 | 0.686 | 0   |
|  | weight loss              | 94.7 [83–99] | 84.8 [74–92] | higher in PED | 0.203 | 0.756 | 0   |
|  | maintained energy        | 28.9 [17–45] | 16.7 [10–27] | higher in PED | 0.212 | 0.756 | 0   |
|  | improved liver enzymes   | 2.6 [0–13]   | 0.0 [0–6]    | higher in PED | 0.365 | 0.843 | 0   |
|  | craving changes          | 2.6 [0–13]   | 0.0 [0–6]    | higher in PED | 0.365 | 0.843 | 0   |

|                                    |              |              |                   |       |       |   |
|------------------------------------|--------------|--------------|-------------------|-------|-------|---|
| increased physical activity        | 18.4 [9-33]  | 10.6 [5-20]  | higher in PED     | 0.371 | 0.843 | 0 |
| other                              | 0.0 [0-9]    | 4.5 [2-13]   | higher in non-PED | 0.298 | 0.843 | 0 |
| improved quality of life           | 57.9 [42-72] | 48.5 [37-60] | higher in PED     | 0.417 | 0.869 | 0 |
| reduced cholesterol                | 10.5 [4-24]  | 7.6 [3-17]   | higher in PED     | 0.721 | 1.000 | 0 |
| improved sleep quality             | 2.6 [0-13]   | 1.5 [0-8]    | higher in PED     | 1.000 | 1.000 | 0 |
| improved blood pressure            | 7.9 [3-21]   | 12.1 [6-22]  | higher in non-PED | 0.742 | 1.000 | 0 |
| relieved gastrointestinal symptoms | 18.4 [9-33]  | 22.7 [14-34] | higher in non-PED | 0.804 | 1.000 | 0 |
| reduced alcohol cravings           | 5.3 [1-17]   | 3.0 [1-10]   | higher in PED     | 0.622 | 1.000 | 0 |
| improved liver function            | 2.6 [0-13]   | 1.5 [0-8]    | higher in PED     | 1.000 | 1.000 | 0 |
| dietary improvement                | 28.9 [17-45] | 25.8 [17-37] | higher in PED     | 0.819 | 1.000 | 0 |
| reduced food noise                 | 55.3 [40-70] | 53.0 [41-65] | higher in PED     | 0.841 | 1.000 | 0 |
| treatment continuation             | 2.6 [0-13]   | 1.5 [0-8]    | higher in PED     | 1.000 | 1.000 | 0 |
| reduced appetite                   | 68.4 [53-81] | 72.7 [61-82] | higher in non-PED | 0.659 | 1.000 | 0 |
| reduced cravings                   | 0.0 [0-9]    | 1.5 [0-8]    | higher in non-PED | 1.000 | 1.000 | 0 |
| reduced inflammation               | 13.2 [6-27]  | 15.2 [8-26]  | higher in non-PED | 1.000 | 1.000 | 0 |
| improved blood glucose control     | 10.5 [4-24]  | 16.7 [10-27] | higher in non-PED | 0.564 | 1.000 | 0 |

Effects differing by PED status WITHIN reta (BH  $q < 0.05$ ) = candidate artefacts:

adverse: ['muscle loss']

desired: []

```
In [ ]: #####
# DEMOGRAPHICS DASHBOARD (visual summary of the recovery EDA + cohort choice) #
#####

dash_fields = ["age", "sex", "height", "BMI", "race/ethnicity"]
field_have = {f: {a for a in analysis_set_ids if FIELDS[f].get(a) is not None} for f in dash_fields}
field_pct = [100*len(field_have[f])/N for f in dash_fields]
field_n = [len(field_have[f]) for f in dash_fields]
per_drug = {f: [sum(1 for a in field_have[f] if _drug.get(a) == dg) for dg in DRUG_ORDER]
            for f in dash_fields}

dash_cohorts = ["injection\nroute-conf.", "sex", "age", "sex+age", "sex+age+\nheight+BMI"]
dash_cohort_n = [N, len(has_sex), len(has_age), len(has_both), len(full_demo)]

fig = plt.figure(figsize=(16, 10))
gs = GridSpec(2, 3, figure=fig, hspace=0.35, wspace=0.3)

# (a) recovery rate per field
ax1 = fig.add_subplot(gs[0, 0])
bars = ax1.barh(dash_fields, field_pct, color=["#4daf4a" if p > 25 else "#e41a1c" for p in field_pct], alpha=0.8)
for bar, n, pct in zip(bars, field_n, field_pct):
```

```

    ax1.text(pct+0.25, bar.get_y()+bar.get_height()/2, f"{pct:.1f}% (n={n})", va="center", fontsize=5.5)
ax1.set_xlim(0, 40); ax1.set_xlabel("% of cohort with field recorded"); ax1.invert_yaxis()
ax1.set_title("Demographic recovery rate", fontweight="bold", fontsize=11)

# (b) cohort size by completeness
ax2 = fig.add_subplot(gs[0, 1])
bars = ax2.bar(range(len(dash_cohorts)), dash_cohort_n, color="#377eb8", alpha=0.8)
for bar, n in zip(bars, dash_cohort_n):
    ax2.text(bar.get_x()+bar.get_width()/2, n+20, f"{n}", ha="center", fontsize=9, fontweight="bold")
ax2.set_xticks(range(len(dash_cohorts))); ax2.set_xticklabels(dash_cohorts, fontsize=8)
ax2.set_ylabel("authors"); ax2.set_title("Cohort size by demographic completeness", fontweight="bold", fontsize=11)

# (c) recovery per drug group
ax3 = fig.add_subplot(gs[0, 2])
x = np.arange(len(dash_fields)); w = 0.2
for i, dg in enumerate(DRUG_ORDER):
    ax3.bar(x + (i-1.5)*w, [per_drug[f][i] for f in dash_fields], w,
            color=DRUG_COLOUR[dg], label=dg.upper(), alpha=0.85)
ax3.set_xticks(x); ax3.set_xticklabels(dash_fields, fontsize=8, rotation=20, ha="right")
ax3.set_ylabel("authors with field"); ax3.set_title("Recovery by drug group", fontweight="bold", fontsize=11)
ax3.legend(fontsize=8, ncol=2)

# (d) age distribution (sex+age)
ax4 = fig.add_subplot(gs[1, 0])
ages = [age_map[a] for a in has_both if age_map.get(a) is not None]
ax4.hist(ages, bins=15, color="#984ea3", alpha=0.75, edgecolor="white")
ax4.axvline(np.median(ages), color="black", ls="--", lw=1.5, label=f"median {np.median(ages):.0f}")
ax4.set_xlabel("age (years)"); ax4.set_ylabel("authors")
ax4.set_title("Age distribution (sex+age)", fontweight="bold", fontsize=11); ax4.legend(fontsize=8)

# (e) baseline BMI distribution (sex+age) with WHO category thresholds
ax5 = fig.add_subplot(gs[1, 1])
bmis = [bmi_map[a] for a in has_both if bmi_map.get(a) is not None]
ax5.hist(bmis, bins=15, color="#ff7f00", alpha=0.75, edgecolor="white")
for thr, lab in [(18.5, "underweight"), (25, "normal"), (30, "overweight"), (35, "obese")]:
    ax5.axvline(thr, color="grey", ls=":", lw=1)
    ax5.text(thr-0.4, ax5.get_ylim()[1]*0.9, lab, fontsize=7, rotation=90, va="top", ha="right")
ax5.axvline(np.median(bmis), color="black", ls="--", lw=1.5, label=f"median {np.median(bmis):.1f}")
ax5.set_xlabel("BMI"); ax5.set_ylabel("authors")
ax5.set_title("Baseline BMI distribution (sex+age)", fontweight="bold", fontsize=11); ax5.legend(fontsize=8)

```

```

# (f) sex composition per drug (sex+age)
ax6 = fig.add_subplot(gs[1, 2])
female = [sum(1 for a in has_both if _drug.get(a)==dg and sex_map.get(a=="female") for dg in DRUG_ORDER]
male    = [sum(1 for a in has_both if _drug.get(a)==dg and sex_map.get(a=="male")    for dg in DRUG_ORDER]
ax6.bar(DRUG_ORDER, female, color="#c2185b", alpha=0.8, label="female")
ax6.bar(DRUG_ORDER, male, bottom=female, color="#1565c0", alpha=0.8, label="male")
for i, dg in enumerate(DRUG_ORDER):
    tot = female[i]+male[i]
    if tot: ax6.text(i, tot+1, f"{100*male[i]/tot:.0f}% M", ha="center", fontsize=8)
ax6.set_ylabel("authors"); ax6.set_xticklabels([d.upper() for d in DRUG_ORDER])
ax6.set_title("Sex composition by drug (sex+age)", fontweight="bold", fontsize=11); ax6.legend(fontsize=8)

for label, ax in zip("abcdef", [ax1, ax2, ax3, ax4, ax5, ax6]):
    ax.text(-0.08, 1.08, f"({label})", transform=ax.transAxes,
           fontsize=11, fontweight="bold", va="top", ha="right")
fig.suptitle("GLP-1 Reddit cohort – demographic recovery & composition",
            fontsize=15, fontweight="bold", y=0.98)
plt.savefig(f"{gpt_result_dir}/demographics_dashboard.png", dpi=140, bbox_inches="tight")
plt.show()

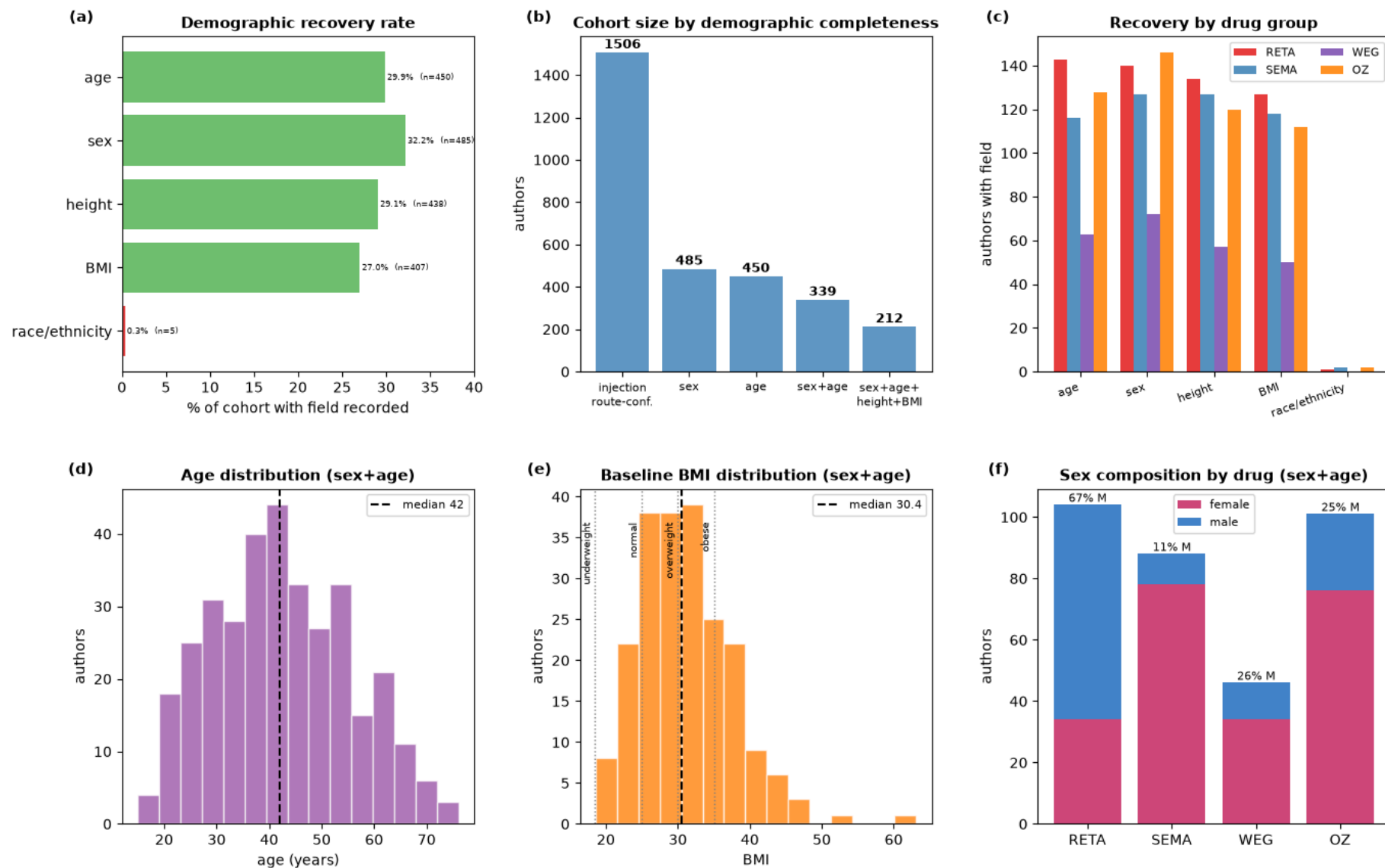
```

/var/folders/5l/fhws2zt56lg1ns3d7llkfmkw0000gn/T/ipykernel\_32450/3952683660.py:71: UserWarning: set\_ticklabels() should only be used with a fixed number of ticks, i.e. after set\_ticks() or using a FixedLocator. Otherwise, ticks may be mislabeled.

```
ax6.set_ylabel("authors"); ax6.set_xticklabels([d.upper() for d in DRUG_ORDER])
```



## GLP-1 Reddit cohort — demographic recovery & composition



```
In [ ]: #####
#BUBBLE HEATMAP: reta vs sema across demographic subgroups #
#####
```

```

ADVERSE_6 = ["gastrointestinal issues", "dose intolerance", "elevated heart rate",
             "skin sensitivity", "weight gain", "muscle loss"]
DESIRED_6 = ["weight loss", "relieved gastrointestinal symptoms", "reduced appetite",
             "reduced food noise", "improved quality of life", "improved impulse control"]
COLUMNS = [(e, "adverse") for e in ADVERSE_6] + [(e, "desired") for e in DESIRED_6]

age_median = np.median([age_map[a] for a in has_both if age_map.get(a) is not None])
bmi_median = np.median([bmi_map[a] for a in has_both if bmi_map.get(a) is not None])
ROWS = ["All", "Males", "Females",
        f"Age<{age_median:.0f}", f"Age≥{age_median:.0f}",
        f"BMI<{bmi_median:.1f}", f"BMI≥{bmi_median:.1f}"]

def subgroup_members(name):
    """sex+age author-ids belonging to a named demographic subgroup row."""
    if name == "All": return list(has_both)
    if name == "Males": return [a for a in has_both if sex_map.get(a) == "male"]
    if name == "Females": return [a for a in has_both if sex_map.get(a) == "female"]
    if name.startswith("Age<"): return [a for a in has_both if age_map.get(a) is not None and age_map[a] < age_med
    if name.startswith("Age≥"): return [a for a in has_both if age_map.get(a) is not None and age_map[a] >= age_med
    if name.startswith("BMI<"): return [a for a in has_both if bmi_map.get(a) is not None and bmi_map[a] < bmi_med
    if name.startswith("BMI≥"): return [a for a in has_both if bmi_map.get(a) is not None and bmi_map[a] >= bmi_med
    return []

def cell_stats(effect, event, members):
    """Per subgroup cell: reta-sema rate difference, pooled rate, Fisher p."""
    reta = [a for a in members if _drug.get(a) == "reta"]
    sema = [a for a in members if _drug.get(a) in SEMA_FAMILY]
    if not reta or not sema:
        return None
    reta_k = authors_reporting(effect, event, reta); reta_n = len(reta)
    sema_k = authors_reporting(effect, event, sema); sema_n = len(sema)
    _, p = fisher_exact([[reta_k, reta_n-reta_k], [sema_k, sema_n-sema_k]])
    return dict(diff=reta_k/reta_n - sema_k/sema_n, pooled=(reta_k+sema_k)/(reta_n+sema_n), p=p)

grid = {}; pvals_flat = []
for r, row in enumerate(ROWS):
    members = subgroup_members(row)
    for c, (effect, event) in enumerate(COLUMNS):
        st = cell_stats(effect, event, members)
        grid[(r, c)] = st
        if st is not None:

```

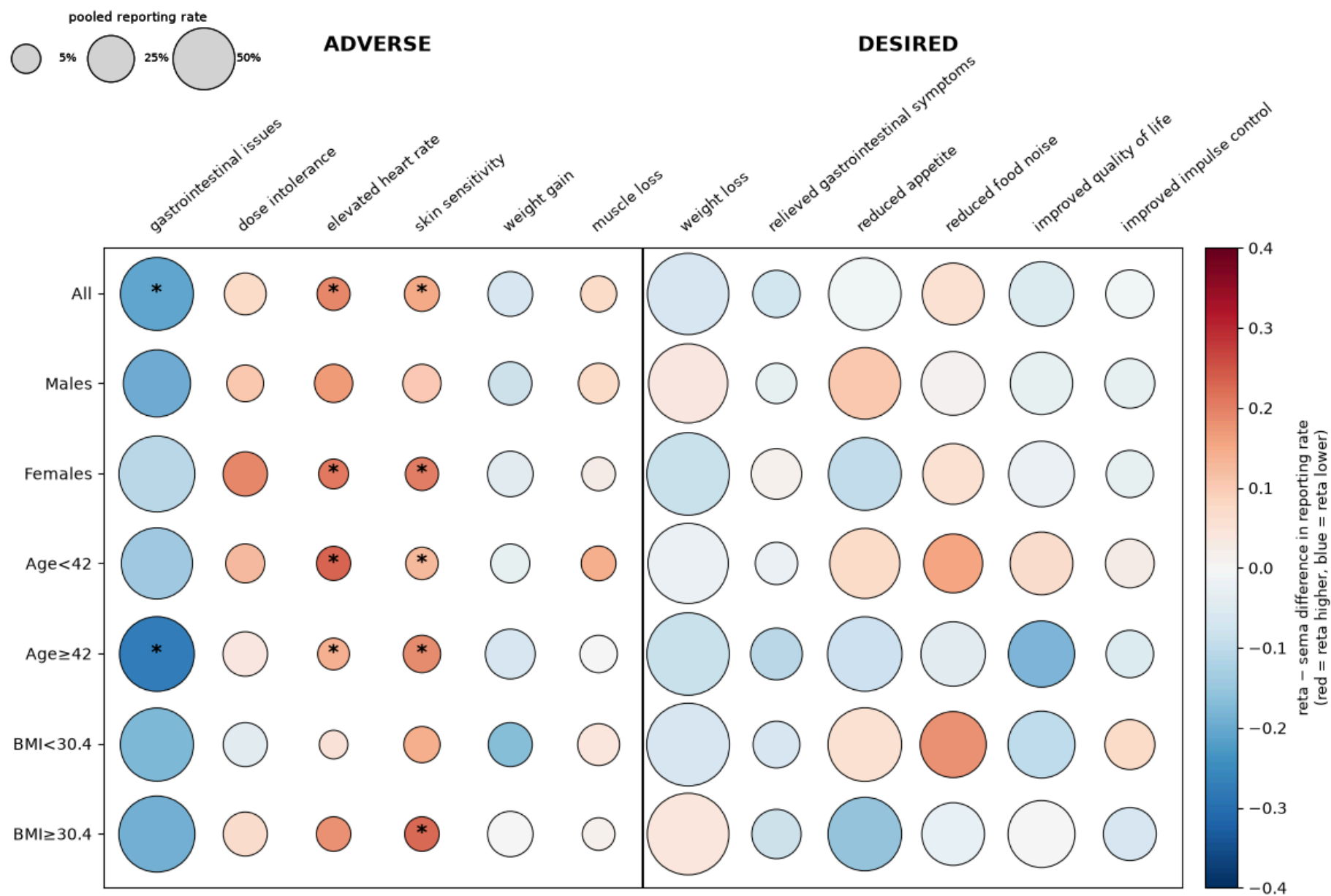
```

        pvals_flat.append(((r, c), st["p"]))
qmap = dict(zip([k for k, _ in pvals_flat],
                multipletests([p for _, p in pvals_flat], method="fdr_bh")[1])) # BH across all cells

fig, ax = plt.subplots(figsize=(15, 9))
norm = TwoSlopeNorm(vmin=-0.4, vcenter=0, vmax=0.4); cmap = plt.cm.RdBu_r
for (r, c), st in grid.items():
    if st is None:
        ax.add_patch(plt.Rectangle((c-0.5, r-0.5), 1, 1, color="#e0e0e0", zorder=0)); continue
    ax.scatter(c, r, s=200+2600*st["pooled"], color=cmap(norm(st["diff"])), # size ~ pooled rate
              edgecolors="black", linewidths=0.7, zorder=2)
    if qmap.get((r, c), 1) < 0.05:
        ax.text(c, r, "*", ha="center", va="center", fontsize=13, fontweight="bold", zorder=3)
ax.set_yticks(range(len(ROWS))); ax.set_yticklabels(ROWS, fontsize=10); ax.set_ylim(len(ROWS)-0.4, -0.5)
ax.set_xticks(range(len(COLUMNS)))
ax.set_xticklabels([e for e, _ in COLUMNS], rotation=40, ha="left", rotation_mode="anchor", fontsize=9)
ax.xaxis.tick_top(); ax.xaxis.set_label_position("top"); ax.tick_params(axis="x", length=0, pad=8)
ax.set_xlim(-0.6, len(COLUMNS)-0.4)
ax.axvline(len(ADVERSE_6)-0.5, color="black", lw=1.5) # adverse and desired effects divider
ax.text((len(ADVERSE_6)-1)/2, 1.3, "ADVERSE", ha="center", va="bottom", fontweight="bold",
        fontsize=13, transform=ax.get_xaxis_transform())
ax.text((len(ADVERSE_6)+(len(DESIRED_6)-1)/2), 1.3, "DESIRED", ha="center", va="bottom", fontweight="bold",
        fontsize=13, transform=ax.get_xaxis_transform())
sm = ScalarMappable(norm=norm, cmap=cmap); sm.set_array([])
cbar = fig.colorbar(sm, ax=ax, fraction=0.03, pad=0.02)
cbar.set_label("reta - sema difference in reporting rate\n(red = reta higher, blue = reta lower)", fontsize=9)
size_handles = [plt.scatter([], [], s=200+2600*rate, color="lightgrey", edgecolors="black",
                            linewidths=0.8, label=f"{int(rate*100)}%") for rate in [0.05, 0.25, 0.5]]
size_legend = ax.legend(handles=size_handles, title="pooled reporting rate",
                        loc="upper left", bbox_to_anchor=(-0.1, 1.4),
                        fontsize=7, title_fontsize=8, labelspace=2.5,
                        borderpad=1.2, handletextpad=2, frameon=False,
                        ncol=3)
ax.add_artist(size_legend); ax.grid(False)
plt.subplots_adjust(top=0.68, left=0.12, right=0.78, bottom=0.06)
fig.suptitle("Retatrutide vs semaglutide-family effect differences across demographic subgroups",
            fontsize=13, fontweight="bold", y=0.97)
plt.savefig(f"{gpt_result_dir}/bubble_heatmap_subgroups.png", dpi=140)
plt.show()

```

# Retatrutide vs semaglutide-family effect differences across demographic subgroups



```

In [ ]: # #####
# PED vs nonPED AE+DE HEATMAP #
#####

subgroups = {"reta-PED": reta_ped, "reta-nonPED": reta_noped,
             "sema-PED": sema_ped, "sema-nonPED": sema_noped}
subgroup_sizes = {name: len(ids) for name, ids in subgroups.items()}

def build_panel(effects, event):
    """Prevalence grid (effects x 4 subgroups) + BH q of the deconfounded test."""
    table = pd.DataFrame({name: [prevalence(e, event, ids) for e in effects]
                          for name, ids in subgroups.items()}, index=effects)

    pvals = []
    for eff in effects:
        reta_k = authors_reporting(eff, event, reta_noped)
        sema_k = authors_reporting(eff, event, sema_noped)
        _, p = fisher_exact([[reta_k, len(reta_noped)-reta_k], [sema_k, len(sema_noped)-sema_k]])
        pvals.append(p)
    return table, multipletests(pvals, method="fdr_bh")[1]

adv_table, adv_q = build_panel(top_reta_effects("adverse", 11), "adverse")
des_table, des_q = build_panel(top_reta_effects("desired", 11), "desired")

def draw_panel(ax, table, qvals, cmap, title, vmax=0.8):
    im = ax.imshow(table.values, aspect="auto", cmap=cmap, vmin=0, vmax=vmax)
    ax.axvline(1.5, color="black", lw=2) # reta and pooled sema divider
    ax.set_xticks(range(len(subgroups)))
    ax.set_xticklabels([f"{n}\n(n={subgroup_sizes[n]})" for n in subgroups], fontsize=8)
    ax.set_yticks(range(len(table.index))); ax.set_yticklabels(table.index, fontsize=8.5)
    reta_col = list(subgroups).index("reta-nonPED")
    for row in range(len(table.index)):
        for col in range(len(subgroups)):
            value = table.values[row, col]
            if np.isnan(value): continue
            star = " *" if (col == reta_col and qvals[row] < 0.05) else ""
            ax.text(col, row, f"{value*100:.0f}%{star}", ha="center", va="center",
                    fontsize=9, color="black" if value < vmax*0.8 else "white")
    ax.set_title(title, fontsize=11, fontweight="bold")
    return im

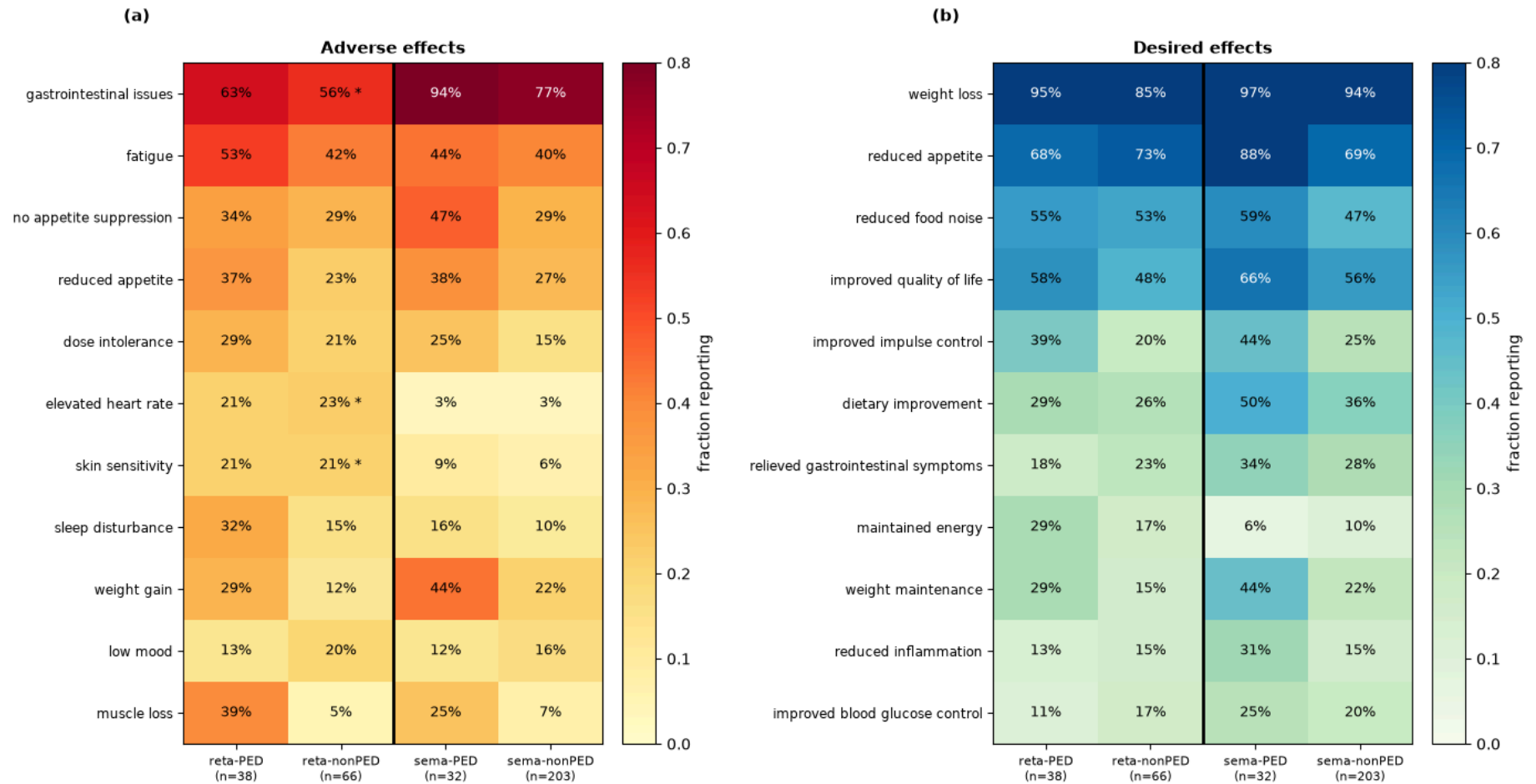
```

```

fig, (ax_adv, ax_des) = plt.subplots(1, 2, figsize=(15, 8))
plt.subplots_adjust(wspace=0.7)
im_adv = draw_panel(ax_adv, adv_table, adv_q, "YlOrRd", "Adverse effects")
im_des = draw_panel(ax_des, des_table, des_q, "GnBu", "Desired effects")
fig.colorbar(im_adv, ax=ax_adv, fraction=0.08, pad=0.04, label="fraction reporting")
fig.colorbar(im_des, ax=ax_des, fraction=0.08, pad=0.04, label="fraction reporting")
for label, ax in zip("ab", [ax_adv, ax_des]):
    ax.text(-0.08, 1.08, f"({label})", transform=ax.transAxes,
           fontsize=11, fontweight="bold", va="top", ha="right")
fig.suptitle("Effect prevalence by PED status × drug", fontsize=11, fontweight="bold")
plt.savefig(f"{gpt_result_dir}/ped_deconfound_heatmap_top11.png", dpi=140, bbox_inches="tight")
plt.show()

```

Effect prevalence by PED status x drug



```
In [ ]: #####
# EVENT FREQUENCY: PED vs non-PED (sema pooled + by drug) #
#####
def author_event_counts(aid):
    """Per-author reported effect-event counts: total, adverse, desired."""
    p = _by.get(aid) or {}; adverse = desired = 0
    for e in (p.get("timeline_events") or []):
        if isinstance(e, dict):
            adverse += len([x for x in (e.get("adverse_effects") or []) if isinstance(x, dict)])
            desired += len([x for x in (e.get("desired_effects") or []) if isinstance(x, dict)])
```

```

    return {"total": adverse + desired, "adverse": adverse, "desired": desired}

freq = pd.DataFrame([{"author_id": a, "drug": _drug.get(a), "ped": uses_ped(a), **author_event_counts(a)}
                     for a in has_both])

def med_iqr(values):
    """'median (IQR lo-hi)' string for a set of counts."""
    values = np.asarray([v for v in values if v is not None and not np.isnan(v)])
    return "-" if len(values) == 0 else f"{np.median(values):.1f} ({np.percentile(values,25):.1f}-{np.percentile(va

def mwu_p(a, b):
    """Mann-Whitney U two-sided p (skewed counts); nan if either group empty."""
    a = [v for v in a if v is not None and not np.isnan(v)]; b = [v for v in b if v is not None and not np.isnan(v)
    return np.nan if not a or not b else mannwhitneyu(a, b, alternative="two-sided")[1]

def comparison_table(subset, label):
    """PED vs non-PED median(IQR) + MWU p for total/adverse/desired counts."""
    ped = subset[subset["ped"]]; noped = subset[~subset["ped"]]
    rows = [{"metric": m, "PED med (IQR)": med_iqr(ped[m]), "nonPED med (IQR)": med_iqr(noped[m]),
            "MWU p": mwu_p(ped[m].values, noped[m].values)} for m in ["total", "adverse", "desired"]]
    df = pd.DataFrame(rows)
    df["MWU p"] = df["MWU p"].map(lambda v: f"{v:.3f}" if not np.isnan(v) else "-")
    df["sig"] = [( "*" if p != "-" and float(p) < 0.05 else "" ) for p in df["MWU p"]]
    df.insert(0, "cohort", f"{label} (PED={len(ped)}, nonPED={len(noped)})")
    return df

tables = [comparison_table(freq, "POOLED (all drugs)")]
for drug in DRUG_ORDER:
    sub = freq[freq["drug"] == drug]
    if sub["ped"].any() and (~sub["ped"]).any():
        tables.append(comparison_table(sub, drug))
print("Event frequency per author – PED vs non-PED (sex+age cohort)\n")
print(pd.concat(tables, ignore_index=True).to_string(index=False))

```



# Event frequency per author – PED vs non-PED (sex+age cohort)

|                                         | cohort | metric  | PED med (IQR)    | nonPED med (IQR) | MWU p | sig |
|-----------------------------------------|--------|---------|------------------|------------------|-------|-----|
| POOLED (all drugs) (PED=70, nonPED=269) |        | total   | 29.5 (14.2–74.8) | 20.0 (10.0–37.0) | 0.001 | *   |
| POOLED (all drugs) (PED=70, nonPED=269) |        | adverse | 15.0 (4.2–29.5)  | 9.0 (4.0–17.0)   | 0.005 | *   |
| POOLED (all drugs) (PED=70, nonPED=269) |        | desired | 16.5 (9.0–35.2)  | 11.0 (5.0–22.0)  | 0.002 | *   |
| reta (PED=38, nonPED=66)                |        | total   | 23.0 (10.5–47.0) | 13.5 (8.0–28.0)  | 0.033 | *   |
| reta (PED=38, nonPED=66)                |        | adverse | 8.5 (3.0–25.2)   | 6.0 (2.0–14.0)   | 0.072 |     |
| reta (PED=38, nonPED=66)                |        | desired | 12.5 (5.0–27.2)  | 7.0 (4.0–15.5)   | 0.022 | *   |
| sema (PED=15, nonPED=73)                |        | total   | 46.0 (24.5–81.0) | 23.0 (14.0–44.0) | 0.031 | *   |
| sema (PED=15, nonPED=73)                |        | adverse | 18.0 (7.5–31.5)  | 10.0 (5.0–19.0)  | 0.075 |     |
| sema (PED=15, nonPED=73)                |        | desired | 22.0 (13.5–39.0) | 13.0 (6.0–27.0)  | 0.149 |     |
| weg (PED=2, nonPED=44)                  |        | total   | 61.5 (52.8–70.2) | 14.5 (8.0–27.0)  | 0.038 | *   |
| weg (PED=2, nonPED=44)                  |        | adverse | 27.5 (24.2–30.8) | 6.0 (2.8–12.8)   | 0.049 | *   |
| weg (PED=2, nonPED=44)                  |        | desired | 34.0 (28.5–39.5) | 8.5 (4.0–14.0)   | 0.043 | *   |
| oz (PED=15, nonPED=86)                  |        | total   | 48.0 (20.5–91.5) | 25.0 (15.0–45.2) | 0.059 |     |
| oz (PED=15, nonPED=86)                  |        | adverse | 24.0 (6.5–36.0)  | 11.0 (5.0–18.8)  | 0.061 |     |
| oz (PED=15, nonPED=86)                  |        | desired | 19.0 (11.5–52.0) | 13.5 (6.2–26.5)  | 0.076 |     |

```
In [ ]: #####
# SWIMMER PLOTS (for report: 6 AEs + 6 DEs) #
# Author-level temporal reconstruction.      #
#####

# vertical offset: adverse markers above the lane, desired below
OFFSET = 0.18

STYLE_6 = {
    # adverse (above the lane) – reds
    ("gastrointestinal issues", "adverse"): ("^", "#b2182b"),
    ("dose intolerance", "adverse"):         ("v", "#d6604d"),
    ("elevated heart rate", "adverse"):       ("X", "#d73027"),
    ("skin sensitivity", "adverse"):           ("P", "#fc8d59"),
    ("weight gain", "adverse"):                ("s", "#8c510a"),
    ("muscle loss", "adverse"):                ("h", "#7f0000"),
    # desired (below the lane) – greens
    ("weight loss", "desired"):                ("o", "#1a9850"),
    ("relieved gastrointestinal symptoms", "desired"): ("^", "#00806a"),
    ("reduced appetite", "desired"):           ("D", "#4daf4a"),
    ("reduced food noise", "desired"):         ("s", "#006837"),
    ("improved quality of life", "desired"):   ("h", "#66c2a4"),
```

```

    ("improved impulse control", "desired"):          ("P", "#35978f"),
}

def author_journey(aid, hl_adverse, hl_desired):
    """Reconstruct one author's journey, keeping only the highlighted effects.
    For each timeline event with a date, calculate the number of weeks since the
    author's FIRST dated event (week = days_since_first / 7), then count a hit
    for every adverse or desired effect from CANONICAL labels in the highlight
    set.

    Returns a dict:
        "duration"      : weeks from the first to the last dated event (journey length)
        "hits"          : list of (week, canonical_effect, event_type) for highlighted effects
        "adverse_count" : number of highlighted ADVERSE hits (used by the min-adverse filter)
        "desired_count" : number of highlighted DESIRED hits (used by the min-desired filter)
    Returns None if the author has no dated events."""
    p = _by.get(aid) or {}
    events = [e for e in (p.get("timeline_events") or []) if isinstance(e, dict) and e.get("date")]
    if not events:
        return None
    dates = pd.to_datetime([e["date"] for e in events], errors="coerce")
    first = dates.min()
    if pd.isna(first):
        return None
    hits = []; adverse_count = desired_count = 0; weeks_all = []
    for e, d in zip(events, dates):
        if pd.isna(d):
            continue
        week = (d - first).days / 7.0
        weeks_all.append(week)
        for it in (e.get("adverse_effects") or []):
            if isinstance(it, dict):
                name = canonical_label(it.get("visualisation_group") or "", "adverse")
                if name in hl_adverse:
                    hits.append((week, name, "adverse")); adverse_count += 1
        for it in (e.get("desired_effects") or []):
            if isinstance(it, dict):
                name = canonical_label(it.get("visualisation_group") or "", "desired")
                if name in hl_desired:
                    hits.append((week, name, "desired")); desired_count += 1
    if not weeks_all:

```

```

        return None
    return {"duration": max(weeks_all), "hits": hits,
            "adverse_count": adverse_count, "desired_count": desired_count}

def draw_swimmer(drug, style, hl_adverse, hl_desired, per_drug, min_wk, max_wk,
                 min_des=0, title_extra="", fname=""):
    """One swimmer plot per drug. Lane bar black=PED / drug-colour=non-PED.
    Selects the longest qualifying journeys within min_wk, max_wk range with at
    least min_des highlighted desired events, up to per_drug authors."""
    journeys = []
    for a in [x for x in has_both if _drug.get(x) == drug]:
        j = author_journey(a, hl_adverse, hl_desired)
        if j and min_wk <= j["duration"] <= max_wk and j["desired_count"] >= min_des:
            journeys.append((a, j))
    journeys.sort(key=lambda t: t[1]["duration"], reverse=True)
    journeys = journeys[:per_drug]
    if not journeys:
        print(f"{DRUG_NAME[drug]}: no qualifying authors"); return

    adv_handles = [Line2D([0],[0], marker=mk, color="w", markerfacecolor=c, markeredgecolor=c,
                           linestyle="None", markersize=9, label=e)
                    for (e, ev), (mk, c) in style.items() if ev == "adverse"]
    des_handles = [Line2D([0],[0], marker=mk, color="w", markerfacecolor=c, markeredgecolor=c,
                           linestyle="None", markersize=9, label=e)
                    for (e, ev), (mk, c) in style.items() if ev == "desired"]

    fig, ax = plt.subplots(figsize=(14, max(4, 0.5*len(journeys))))
    for lane, (aid, j) in enumerate(journeys):
        is_ped = uses_ped(aid)
        lane_colour = "black" if is_ped else DRUG_COLOUR[drug]
        ax.barh(lane, j["duration"], height=0.06, color=lane_colour,
                 alpha=0.8 if is_ped else 0.5, zorder=1)
        ax.plot([0, j["duration"]], [lane+OFFSET]*2, color="#eeeeee", lw=0.6, zorder=1) # adverse row
        ax.plot([0, j["duration"]], [lane-OFFSET]*2, color="#eeeeee", lw=0.6, zorder=1) # desired row
        for week, effect, event in j["hits"]:
            marker, colour = style[(effect, event)]
            y = lane + OFFSET if event == "adverse" else lane - OFFSET
            ax.scatter(week, y, marker=marker, color=colour, s=52, zorder=3,
                       edgecolors="white", linewidths=0.4, clip_on=False)
    ax.set_yticks(range(len(journeys))); ax.set_yticklabels([a[:12] for a, _ in journeys], fontsize=7)
    max_dur = max(j["duration"] for _, j in journeys)

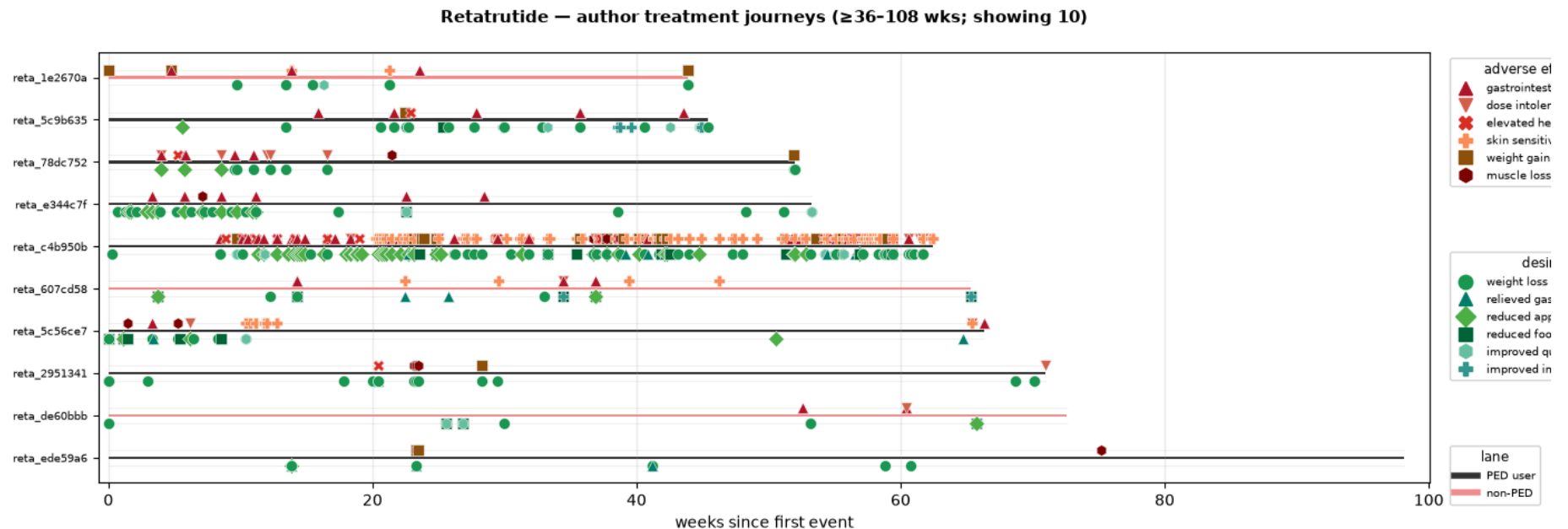
```

```

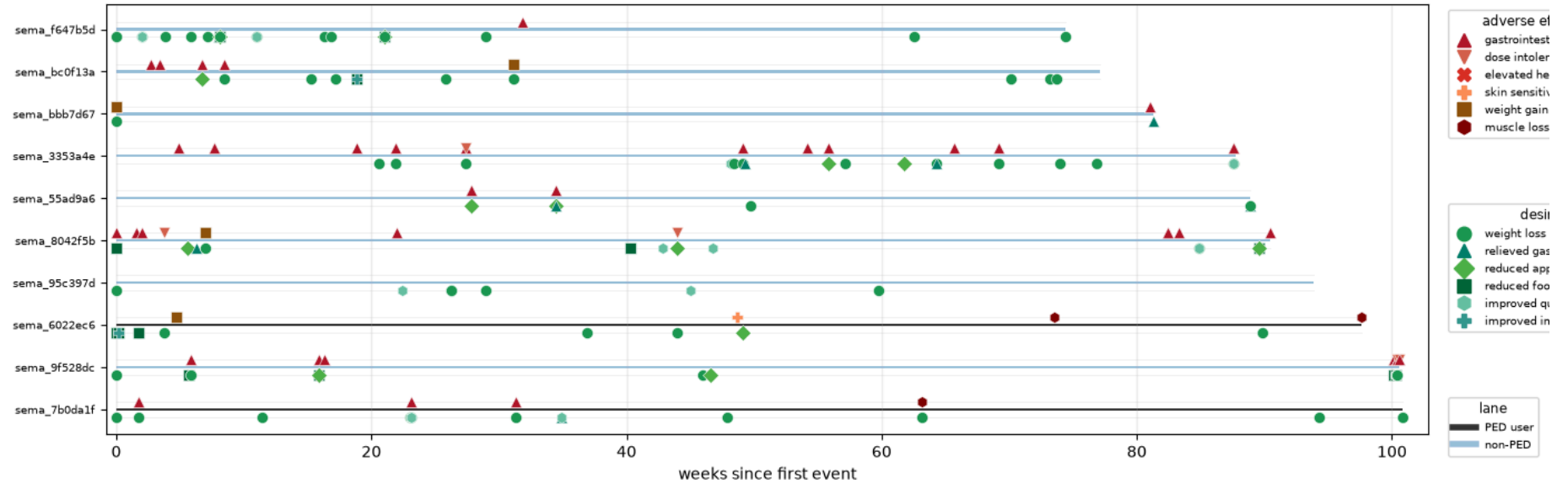
ax.set_xlim(-max_dur*0.008, max_dur*1.02) # small left margin so week=0 markers aren't clipped
ax.set_ylim(-0.6, len(journeys)-0.4)
ax.set_xlabel("weeks since first event")
ax.set_title(f"{DRUG_NAME[drug]} — author treatment journeys ({title_extra}; showing {len(journeys)})\n",
             fontsize=11, fontweight="bold", color="black")
ax.grid(axis="x", alpha=0.2)
lane_handles = [Line2D([0],[0], color="black", lw=4, alpha=0.8, label="PED user"),
                Line2D([0],[0], color=DRUG_COLOUR[drug], lw=4, alpha=0.5, label="non-PED")]
leg_adv = ax.legend(handles=adv_handles, title="adverse effect", loc="upper left",
                   bbox_to_anchor=(1.01, 1.0), fontsize=7.5, title_fontsize=8.5); ax.add_artist(leg_adv)
leg_des = ax.legend(handles=des_handles, title="desired effect", loc="upper left",
                   bbox_to_anchor=(1.01, 0.55), fontsize=7.5, title_fontsize=8.5); ax.add_artist(leg_des)
ax.legend(handles=lane_handles, title="lane", loc="upper left",
          bbox_to_anchor=(1.01, 0.10), fontsize=7.5, title_fontsize=8.5)
plt.tight_layout()
plt.savefig(f"{gpt_result_dir}/{fname}_{drug}.png", dpi=140, bbox_inches="tight")
plt.show()

for drug in DRUG_ORDER:
    draw_swimmer(drug, STYLE_6, set(ADVERSE_6), set(DESIRED_6), per_drug=10,
                 min_wk=36, max_wk=108, title_extra="≥36-108 wks", fname="swimmer_main")

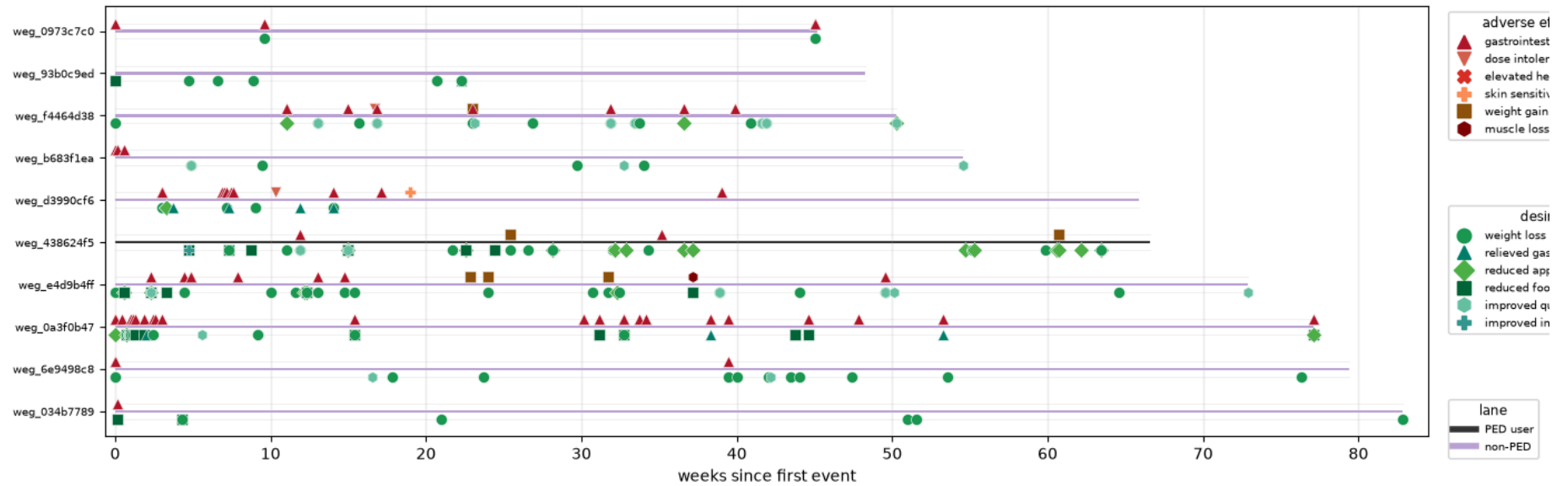
```

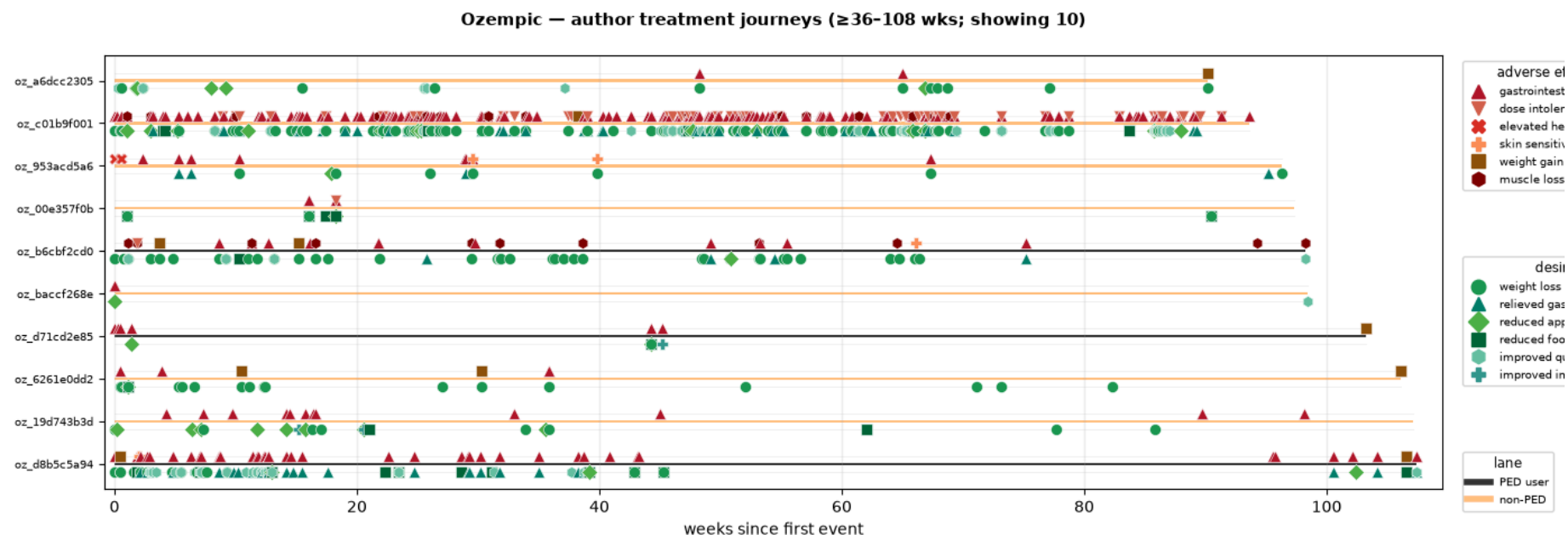


Semaglutide — author treatment journeys (≥36-108 wks; showing 10)



Wegovy — author treatment journeys (≥36-108 wks; showing 10)





```
In [ ]: #####
# Stacked swimmer #
#####

# panel order: (a) reta, (b) sema, (c) weg, (d) oz
panel_drugs = ["reta", "sema", "weg", "oz"]
panel_labels = ["a", "b", "c", "d"]
panel_paths = [f"{gpt_result_dir}/swimmer_main_{dg}.png" for dg in panel_drugs]

# load images and size the stacked canvas to their combined aspect ratio
images = [mpimg.imread(p) for p in panel_paths]
# each image is roughly the same width; total height = sum of heights
heights = [im.shape[0] for im in images]
widths = [im.shape[1] for im in images]
fig_w = 15
fig_h = fig_w * (sum(heights) / max(widths))

fig, axes = plt.subplots(4, 1, figsize=(fig_w, fig_h))
for ax, im, label in zip(axes, images, panel_labels):
    ax.imshow(im)
```

```

    ax.axis("off")
    # panel label in the top-left corner of each image
    ax.text(0.01, 0.97, f"({label})", transform=ax.transAxes,
            fontsize=8, fontweight="bold", va="top", ha="left")

plt.subplots_adjust(hspace=0.03, left=0.005, right=0.995, top=0.995, bottom=0.005)
plt.savefig(f"{gpt_result_dir}/swimmer_stacked_abcd.png", dpi=150, bbox_inches="tight")
plt.show()
print("saved stacked figure -> swimmer_stacked_abcd.png")

```

```

In [ ]: #####
# EXTENDED SWIMMER PLOTS (11 AEs + 11 DEs) #
#####

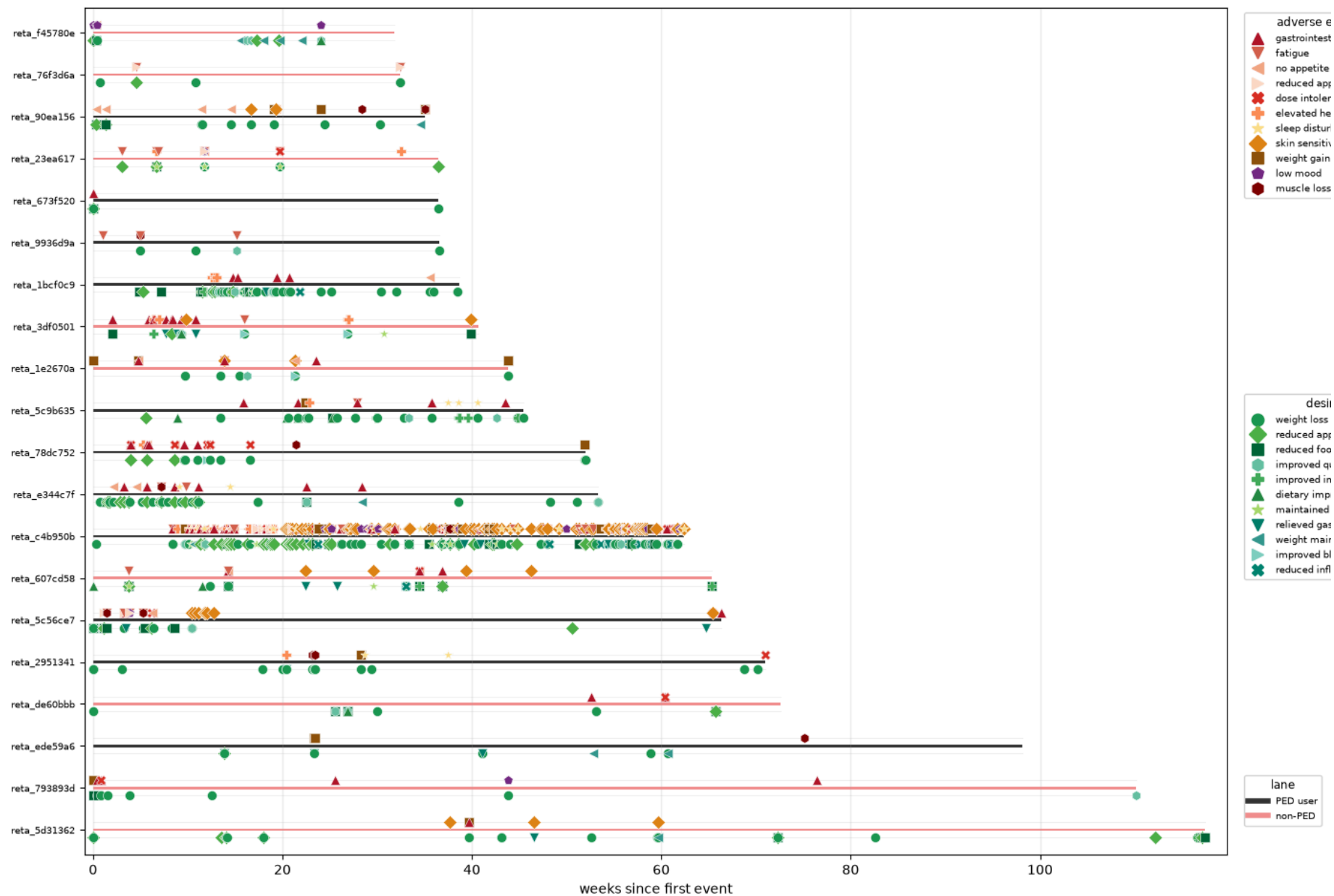
HL_ADVERSE_11 = set(top_reta_effects("adverse", 11))
HL_DESIRE_11 = set(top_reta_effects("desired", 11))
STYLE_11 = {
    ("gastrointestinal issues", "adverse"): ("^", "#b2182b"),
    ("fatigue", "adverse"): ("v", "#d6604d"),
    ("no appetite suppression", "adverse"): ("<", "#f4a582"),
    ("reduced appetite", "adverse"): (">", "#fddbc7"),
    ("dose intolerance", "adverse"): ("X", "#d73027"),
    ("elevated heart rate", "adverse"): ("P", "#fc8d59"),
    ("sleep disturbance", "adverse"): ("*", "#fee08b"),
    ("skin sensitivity", "adverse"): ("D", "#e08214"),
    ("weight gain", "adverse"): ("s", "#8c510a"),
    ("low mood", "adverse"): ("p", "#762a83"),
    ("muscle loss", "adverse"): ("h", "#7f0000"),
    ("weight loss", "desired"): ("o", "#1a9850"),
    ("reduced appetite", "desired"): ("D", "#4daf4a"),
    ("reduced food noise", "desired"): ("s", "#006837"),
    ("improved quality of life", "desired"): ("h", "#66c2a4"),
    ("improved impulse control", "desired"): ("P", "#41ab5d"),
    ("dietary improvement", "desired"): ("^", "#238b45"),
    ("maintained energy", "desired"): ("*", "#a6d96a"),
    ("relieved gastrointestinal symptoms", "desired"): ("v", "#00806a"),
    ("weight maintenance", "desired"): ("<", "#35978f"),
    ("improved blood glucose control", "desired"): (">", "#80cdc1"),
    ("reduced inflammation", "desired"): ("X", "#018571"),
}

```

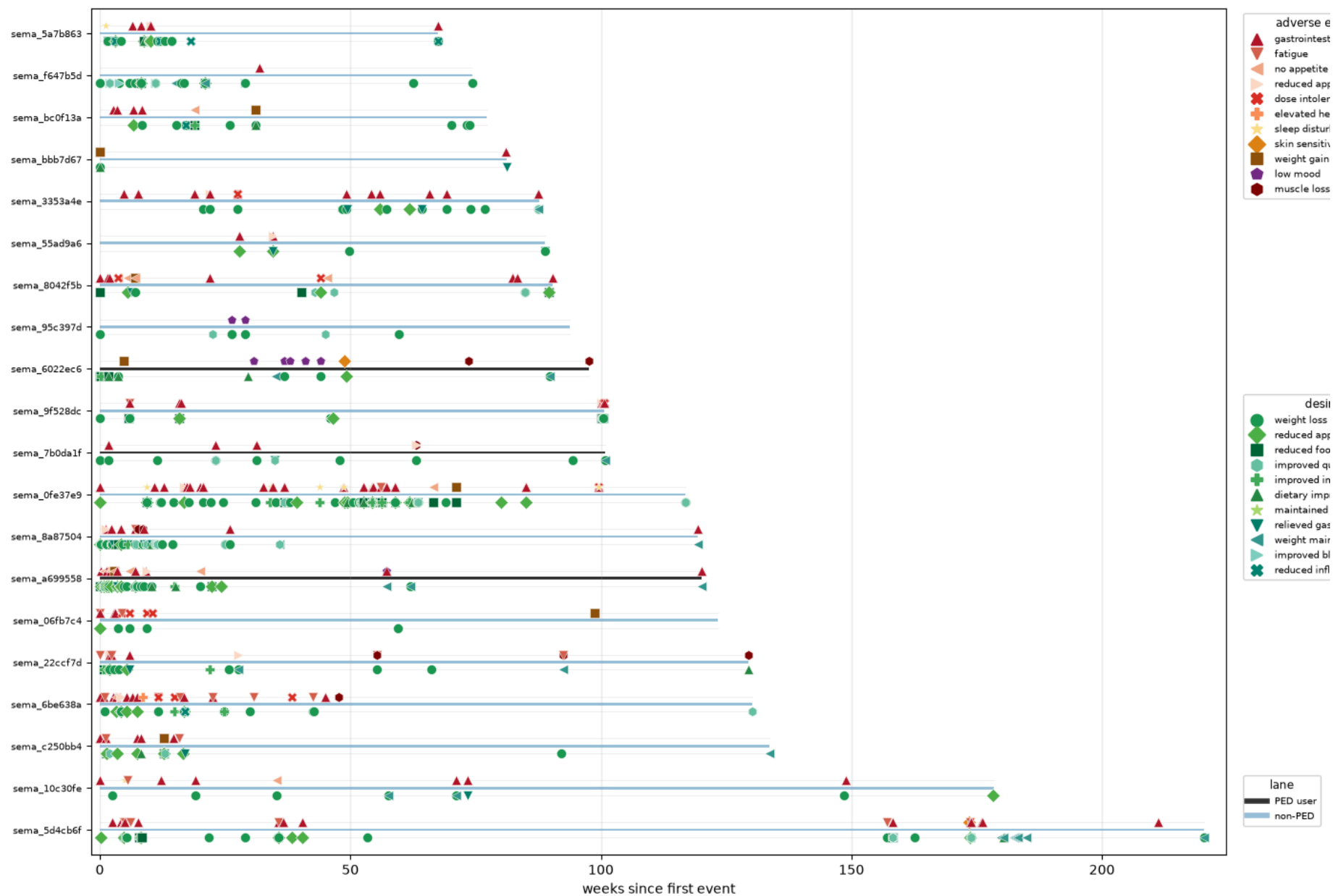
```
for drug in DRUG_ORDER:
    draw_swimmer(drug, STYLE_11, HL_ADVERSE_11, HL_DESIRE_11, per_drug=20,
                  min_wk=12, max_wk=10_000, min_des=3,
                  title_extra="sex+age, ≥12 wks & ≥3 desired", fname="swimmer_appendix")
```



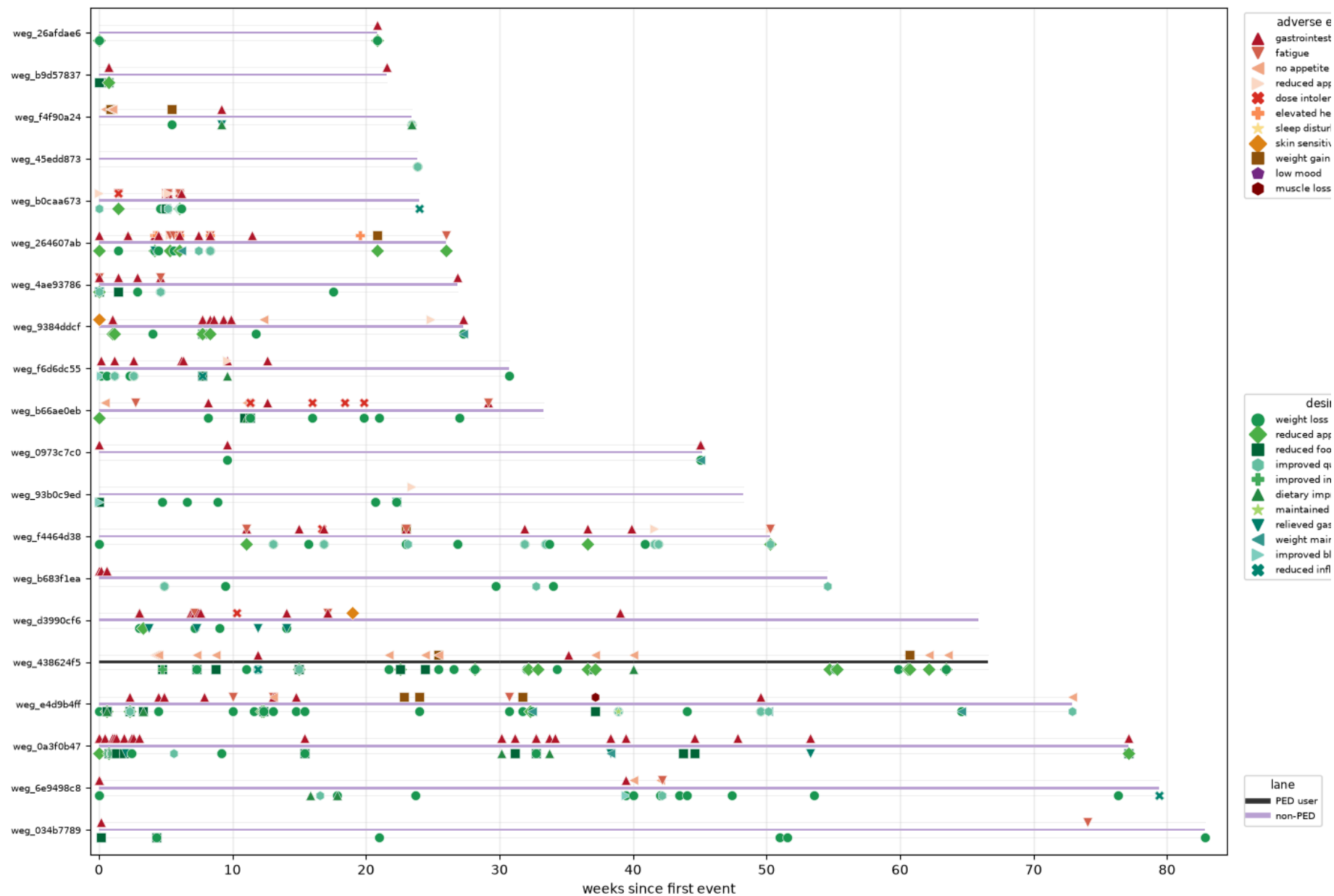
Retatrutide — author treatment journeys (sex+age, ≥12 wks & ≥3 desired; showing 20)



Semaglutide — author treatment journeys (sex+age,  $\geq 12$  wks &  $\geq 3$  desired; showing 20)



Wegovy — author treatment journeys (sex+age, ≥12 wks & ≥3 desired; showing 20)



Ozempic — author treatment journeys (sex+age,  $\geq 12$  wks &  $\geq 3$  desired; showing 20)

